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### Data Communication CS601 Lecture No 1

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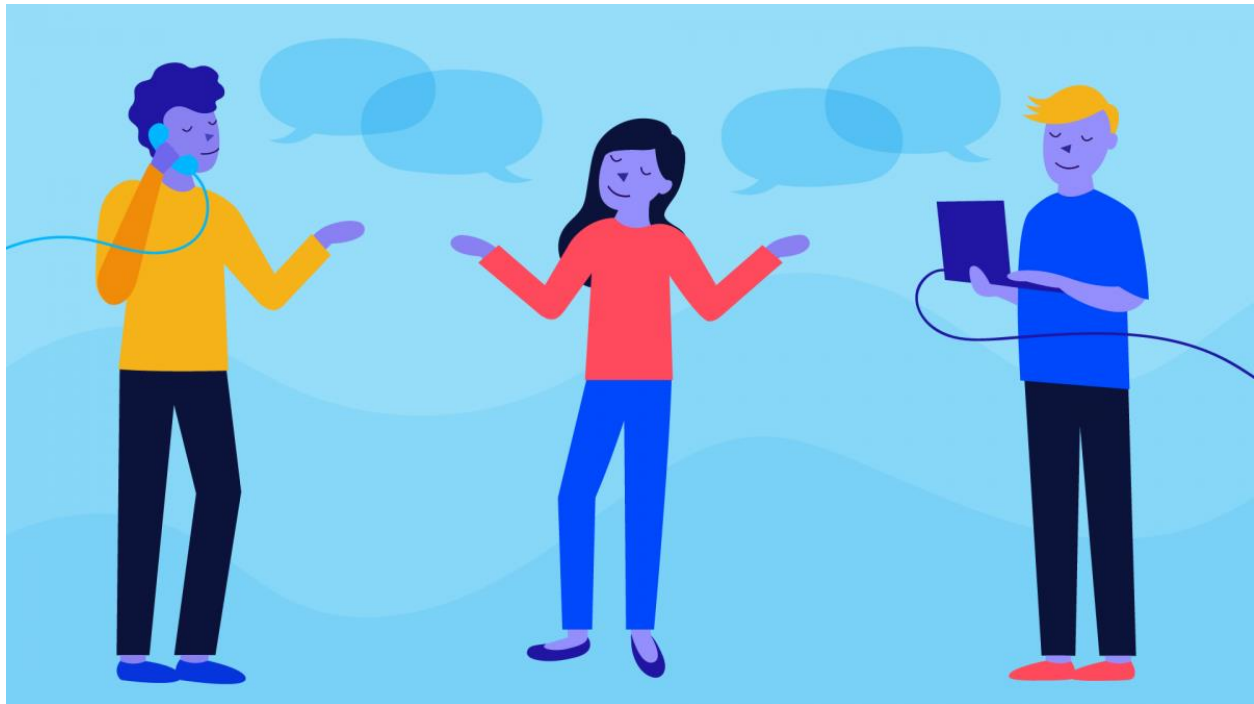


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## Communication

A process by which information is exchanged. Communication is simply the act of transferring information from one place, person or group to another.



## Telecommunications

Telecommunications are **the means of electronic transmission of information over distances**. The information may be in the form of voice telephone calls, data, text, images, or video. Today, telecommunications are used to organize more or less remote computer systems into telecommunications networks.

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### Data communications

Exchange of data between two devices via some form of transmission media. Data communications (DC) is **the process of using computing and communication technologies to transfer data from one place to another, or between participating parties.**

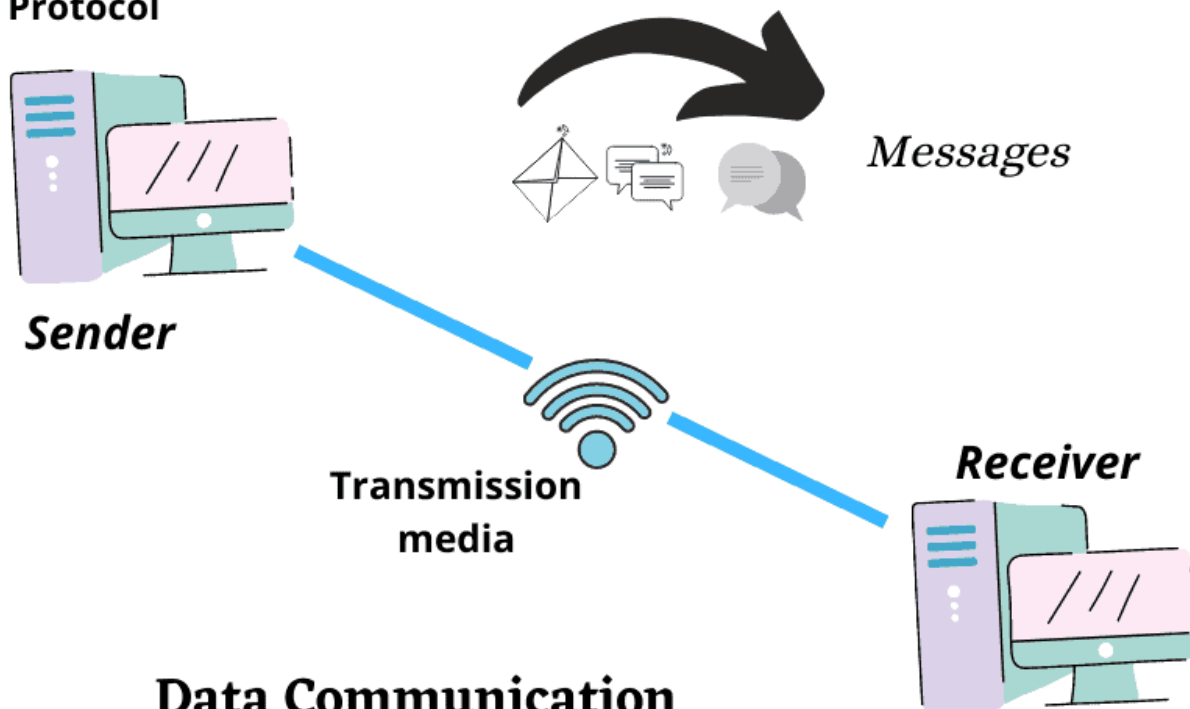
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### Protocol



### Data Communication

### Characteristics of a Data Communication System

- **Delivery**
- **Accuracy**
- **Timeliness**
- **Jitter**



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- **Delivery** means sending the data to the correct destination.
- **Accuracy** means reception of data with no errors.
- **Timeliness** means delivering/transmitting the data within a time interval.
- **Jitter** refers to the variation in the packet arrival of time.

**I. Delivery-** The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

**2. Accuracy-** The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

**3. Timeliness-** The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.

**4. Jitter-** Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

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### Forms of Information

- **Text**
- **Numbers**
- **Images**
- **Audio**
- **Video**

♦ **Text: ASCII:** 7-bit pattern (128 different symbols) Extended ASCII: 8-bit pattern  
Unicode: 32 bits pattern, which is definitely enough to represent any symbol in the world.

♦ **Numbers:** represented by bit pattern (binary number)

♦ **Images:** represented by matrix of pixels (picture element), small dot. The size of pixel represents the resolution.

♦ **Audio:** represent sound by continuous (analog) signal

♦ **Video:** can be analog or digital signal

### 1. Text-

In data communications, text is represented as a bit pattern, a sequence of bits (Os or Is). Different sets of bit patterns have been designed to represent text symbols. Each set is called a code, and the process of representing symbols is called coding.

### 2. Numbers-

Numbers are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify mathematical operations.

### 3. Images-

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Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the resolution. For example, an image can be divided into 1000 pixels or 10,000 pixels.

### 4. Audio-

Audio refers to the recording or broadcasting of sound or music. Audio is by nature different from text, numbers, or images. It is continuous, not discrete. Even when we use a microphone to change voice or music to an electric signal, we create a continuous signal.

### 5. Video-

Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion. Again, we can change video to a digital or an analog signal.

### Data Flow between two devices:

- **Simplex**
- **Half-Duplex**
- **Full-Duplex**

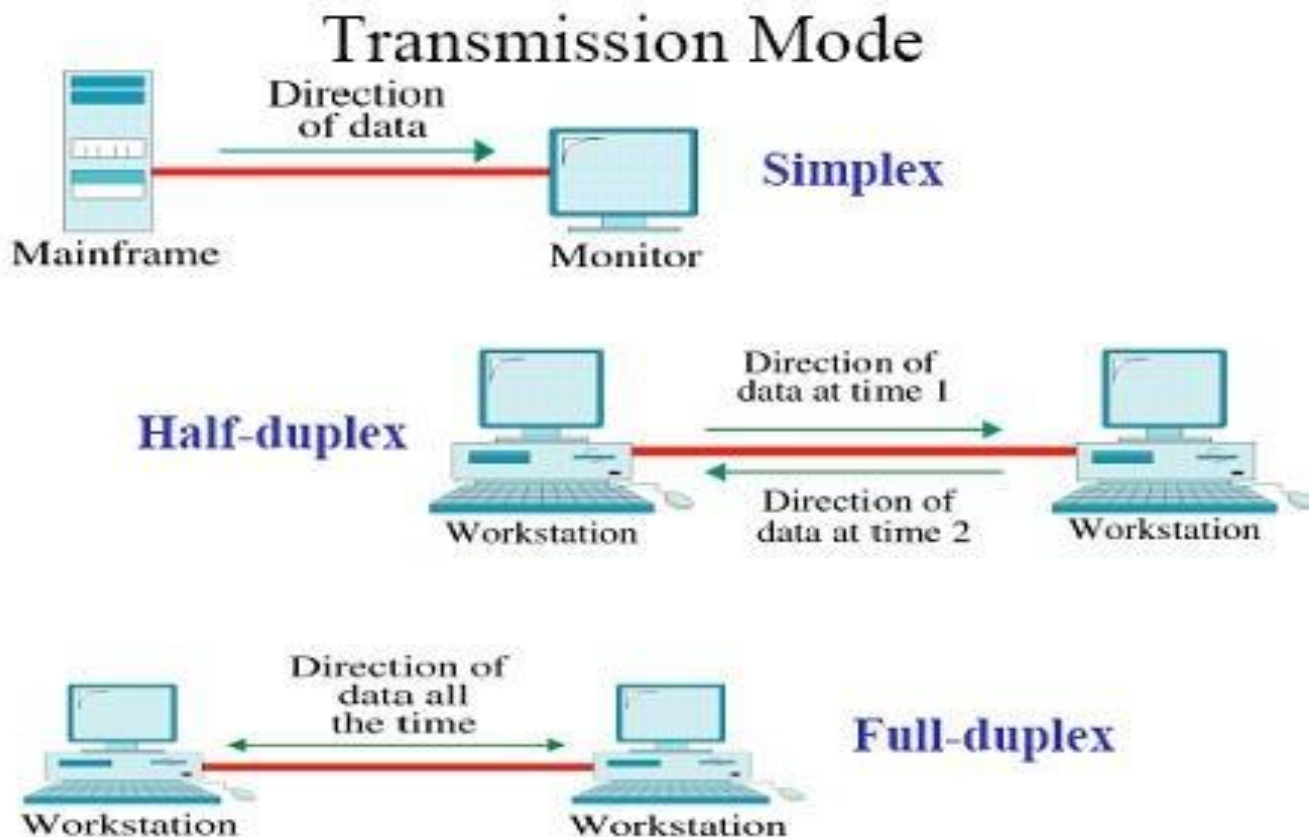
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### 1. Simplex:

In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; Keyboards and traditional monitors are examples of simplex devices. the other can only receive which can be represented in the following figure.

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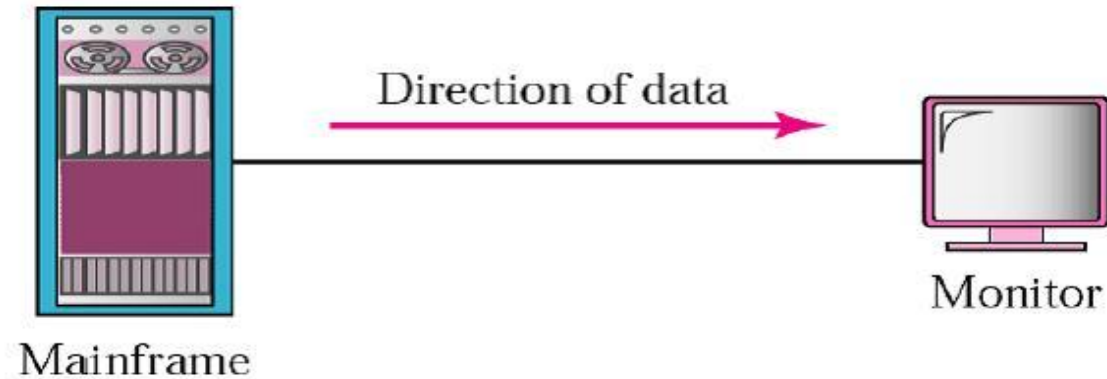
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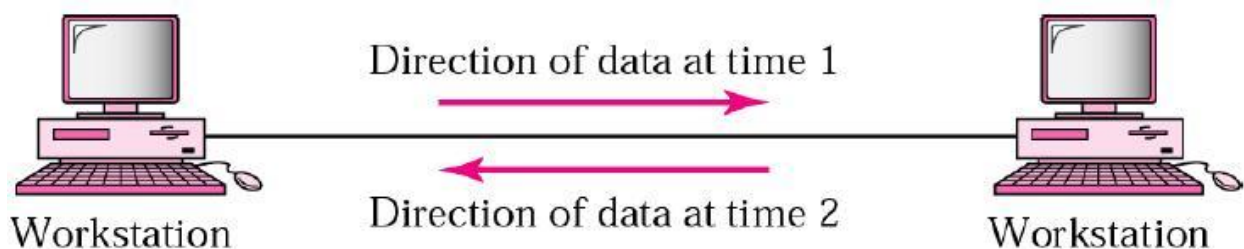
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## 2. Half-Duplex:

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa which will represent in the following figure.



he half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait.

## 3. Full-Duplex:

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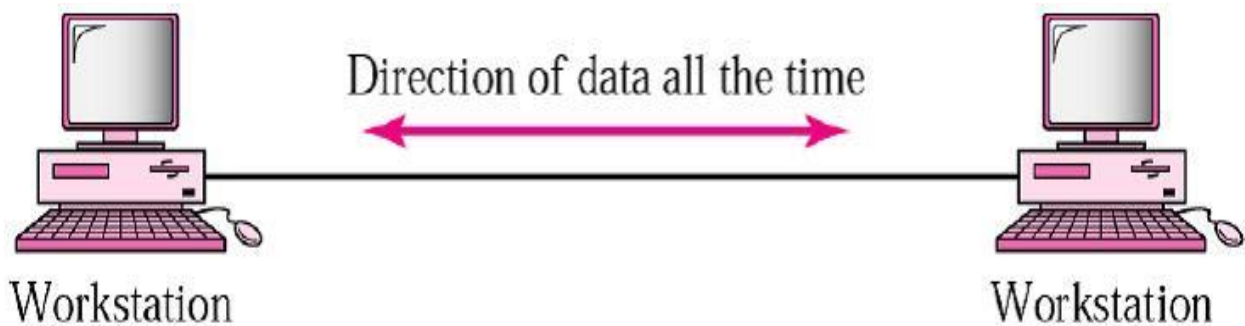
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In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously as shown in the following figure.



### Network:

Interconnection of a set of devices capable of communication



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A data network is a **system designed to transfer data from one network access point to one other or more network access points via data switching, transmission lines, and system controls**. Data networks consist of communication systems such as circuit switches, leased lines, and packet switching networks.

## Throughput

Throughput is **the actual amount of data that is successfully sent/received over the communication link**. Throughput is presented as kbps, Mbps or Gbps, and can differ from bandwidth due to a range of technical issues, including latency, packet loss, jitter and more.



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### Delay

Network delay is a design and performance characteristic of a telecommunications network. It **specifies the latency for a bit of data to travel across the network from one communication endpoint to another.**

### Reliability

In computer networking, a reliable protocol is **a communication protocol that notifies the sender whether or not the delivery of data to intended recipients was successful.**

### Physical Network Attributes

- Link
- Type of Connection
  - ✓ Point-to-Point
  - ✓ Multipoint



Link



a. Point-to-point

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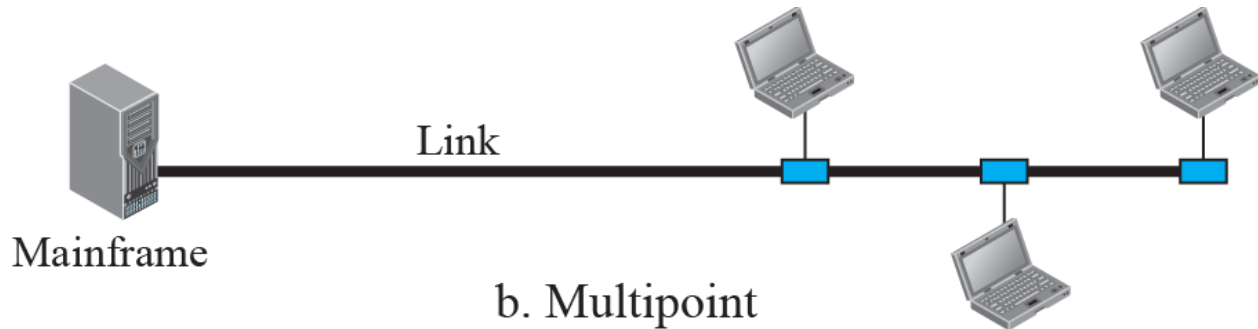
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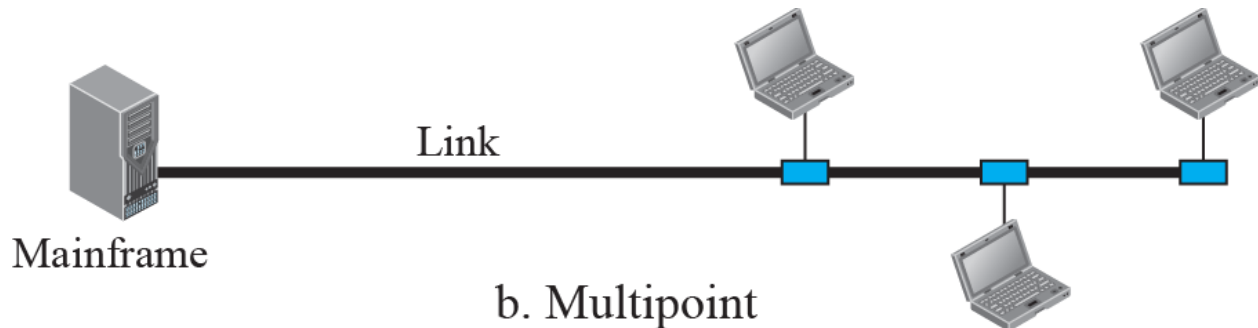
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### Physical Network Attributes

- Link
- Type of Connection
  - ✓ Point-to-Point
  - ✓ Multipoint



a. Point-to-point



b. Multipoint

### Physical Topologies

- Physical Layout of Network
- Links + Nodes = Topology
- Physical Topologies:
  - ✓ Mesh

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✓ **Star**

✓ **Bus**

✓ **Ring**

### Mesh topology

In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. A fully connected mesh network therefore has  $[n(n-1) / 2]$  physical channels to link it devices.

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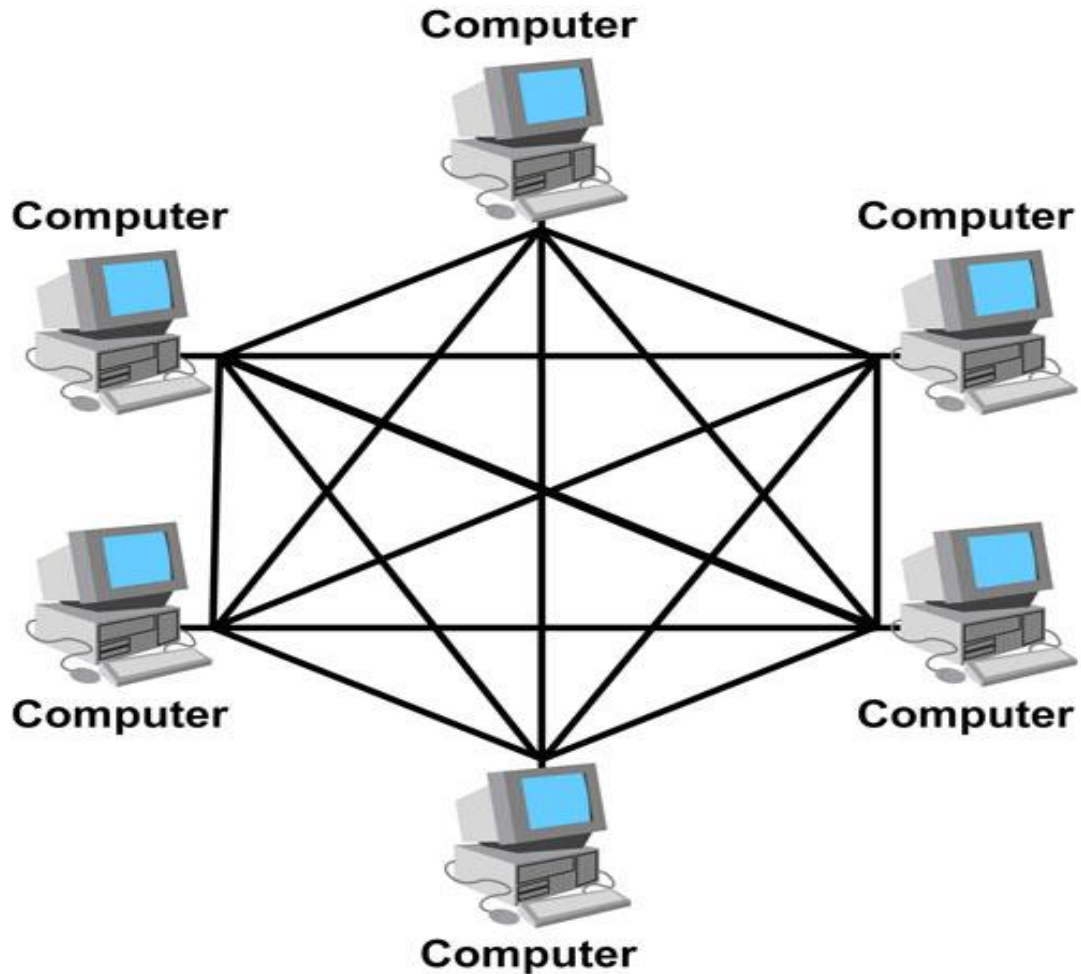
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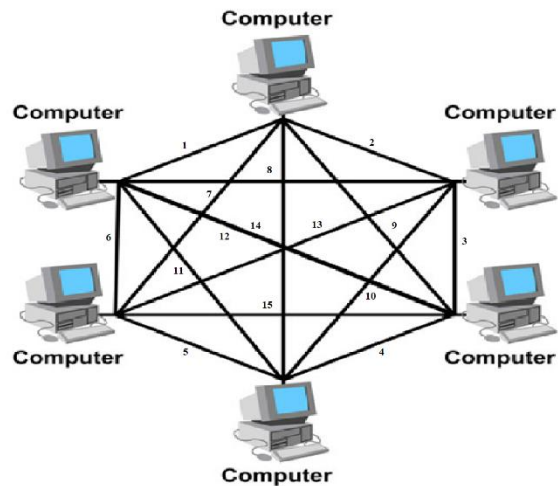
**A fully connected mesh network therefore has  $[n(n-1) / 2]$  physical channels to link its devices.**

$$n(n-1) / 2$$

$$6(6-1) / 2$$

$$6(5) / 2$$

$$30 / 2 = 15 \text{ Links}$$



**A mesh offers several advantages over other network topologies:**

1. The use of dedicated links guarantees that each connection can carry its data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
2. A mesh topology is robust; if one link becomes unusable, it does not affect the entire system.
3. Security where every message sent travels along a dedicated line, only the intended recipient sees it.
4. Point-to-point links make fault identification and fault isolation easy.

**The main disadvantages of a mesh topology are:**



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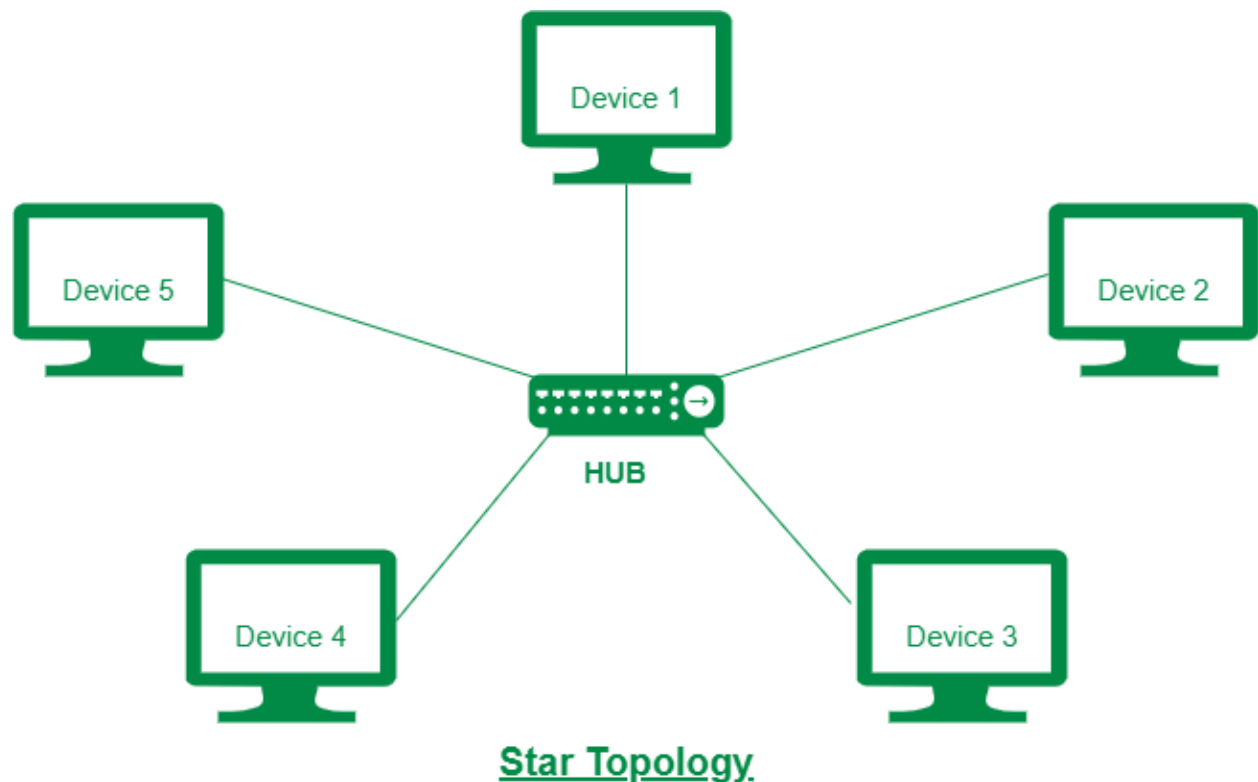


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1. Difficult installation and reconfiguration because every device must be connected to every other device.
2. The bulk of the wiring can be greater than the available space can accommodate.
3. The hardware required to connect each link can be expensive.

## Star Topology

In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not linked to each other. Unlike a mesh topology, a star topology does not allow direct traffic between devices.



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### The advantages of star topology are:

1. Less expensive than a mesh topology.
2. Easy to install.
3. Robustness. If one link fails, only that link is affected.
4. Easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

### The disadvantages of the star topology are:

1. Any fault occur to the central controller will drop down the entire network.
2. Although the star needs less cabling than mesh, it still needs more cables than ring and tree topology.

### Bus Topology

The preceding examples all describe point-to-point configurations. A bus topology on the other hand, is multipoint. One long cable act as a backbone to link all the devices in the network.

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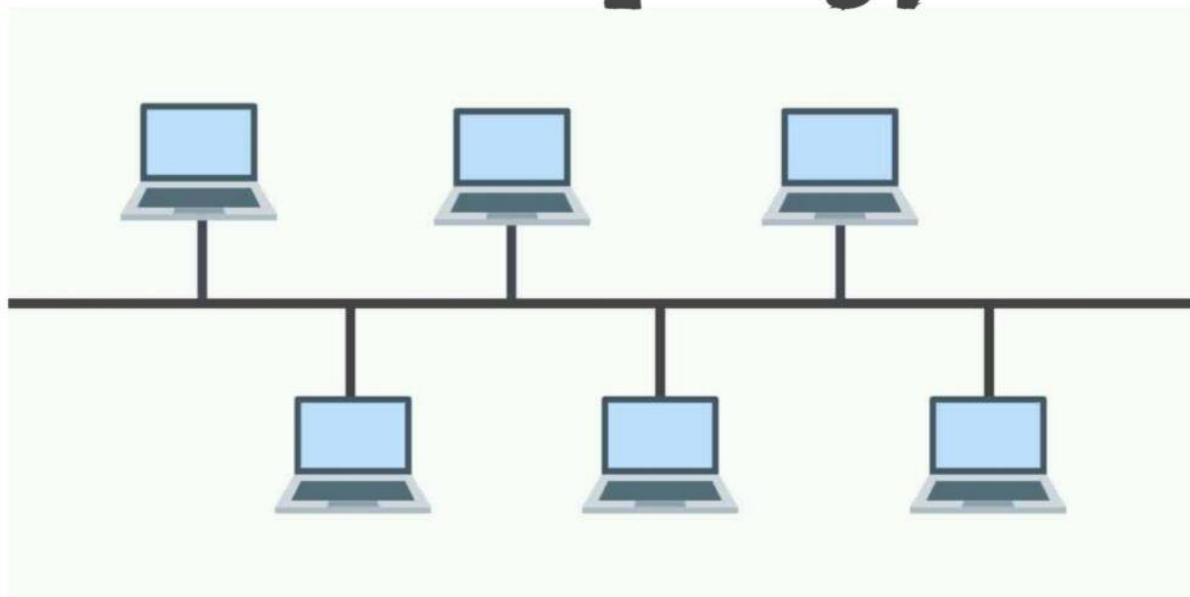
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## Bus Topology



### The advantages of a bus topology are:

1. Ease of installation.
2. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths.
3. Use less cabling than mesh, star, and tree topologies. Terminator Terminator Tap Tap 11

### While the disadvantages are:

1. Difficult reconfiguration and fault isolation.
2. A fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

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3. Adding new devices may therefore require modification or replacement of the backbone.

### Ring Topology

In a ring topology, each device has a dedicated point-to-point line configuration only with the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater.

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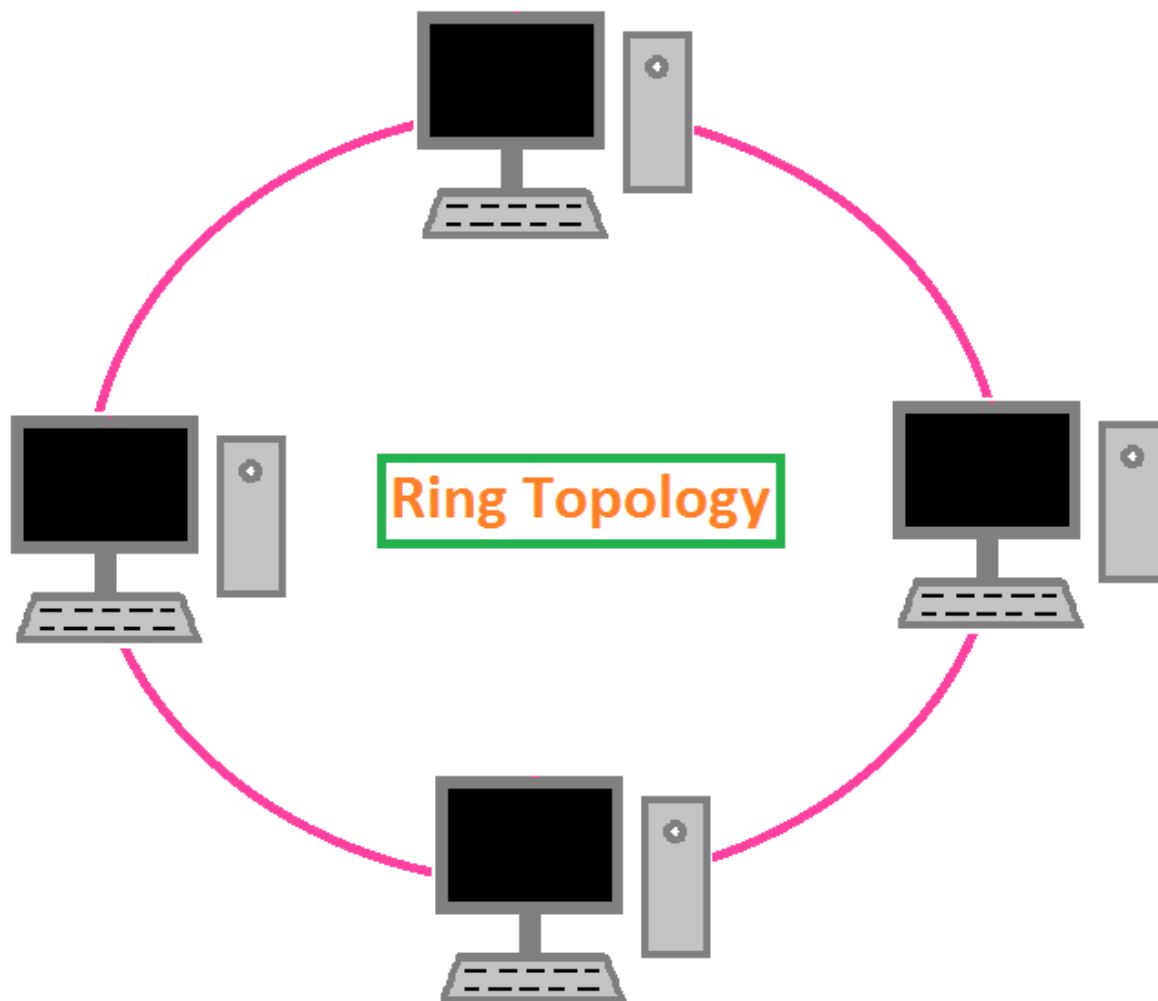
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### The advantages of ring topology are:

1. Relatively easy to install and reconfigure.
2. Addition or deletion of a device requires moving only two connections.

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3. Fault isolation is simplified.

**The disadvantage** of the ring topology is that a break in the ring (such as a disabled station) can disable the entire network, and it is one way data flow.

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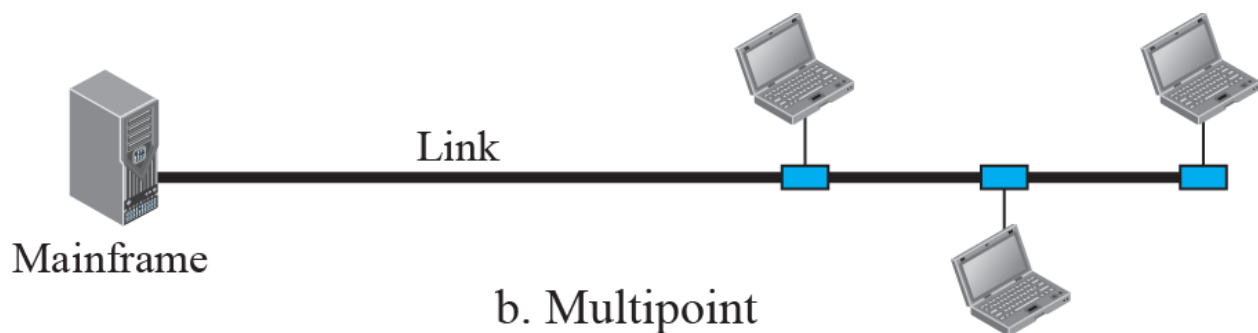
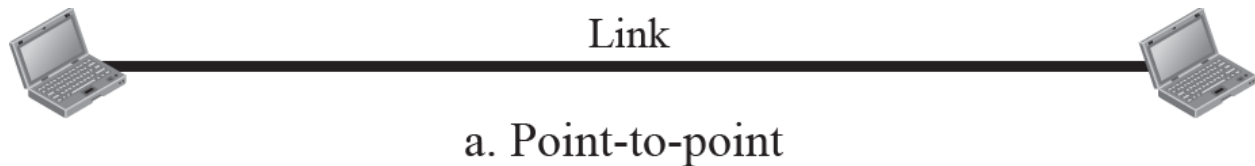


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### Physical Network Attributes

- Link
- Type of Connection
  - ✓ Point-to-Point
  - ✓ Multipoint



### Physical Topologies

- Physical Layout of Network
- Links + Nodes = Topology



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- **Physical Topologies:**

- ✓ **Mesh**
- ✓ **Star**
- ✓ **Bus**
- ✓ **Ring**

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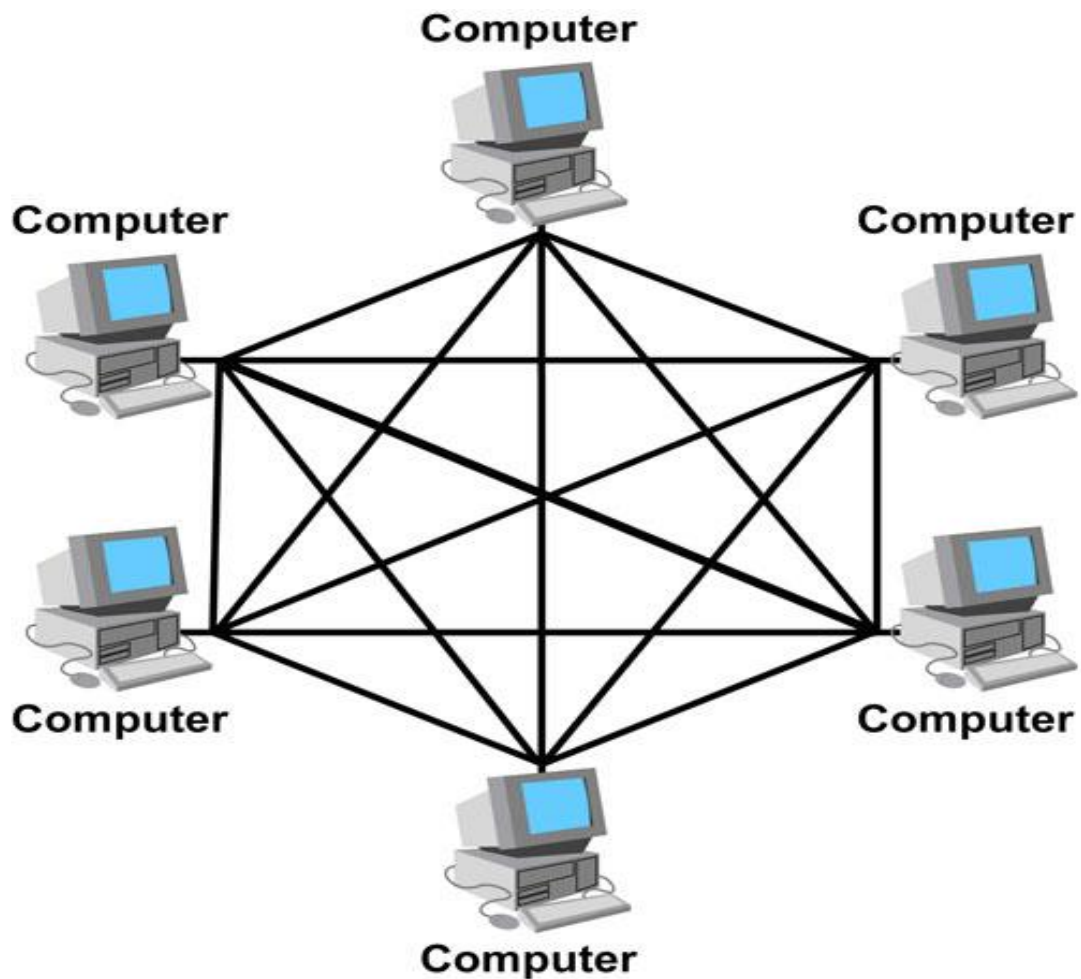
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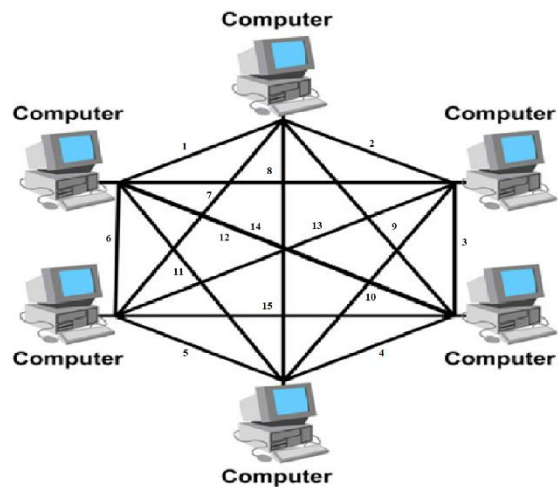
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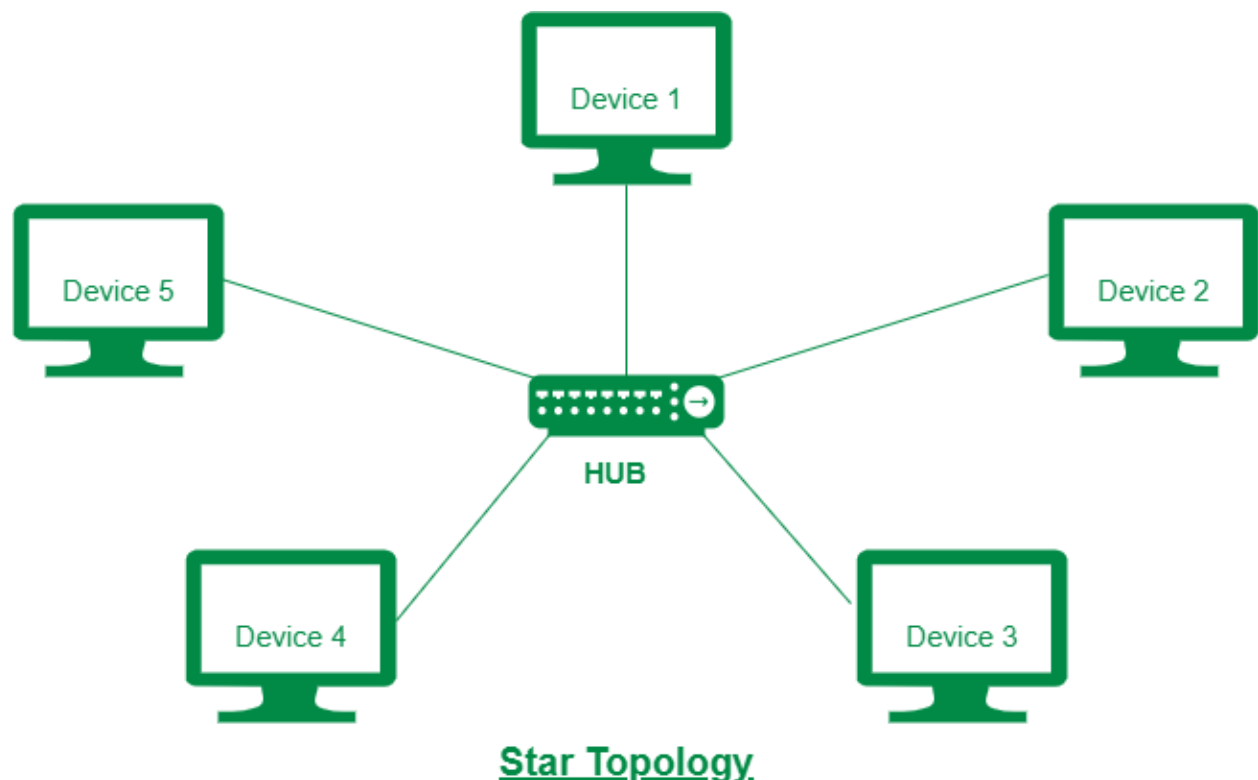
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### The main disadvantages of a mesh topology are:

1. Difficult installation and reconfiguration because every device must be connected to every other device.
2. The bulk of the wiring can be greater than the available space can accommodate.
3. The hardware required to connect each link can be expensive.

### Star Topology

In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not linked to each other. Unlike a mesh topology, a star topology does not allow direct traffic between devices.



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### The advantages of star topology are:

1. Less expensive than a mesh topology.
2. Easy to install.
3. Robustness. If one link fails, only that link is affected.
4. Easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

### The disadvantages of the star topology are:

1. Any fault occur to the central controller will drop down the entire network.
2. Although the star needs less cabling than mesh, it still needs more cables than ring and tree topology.

### Bus Topology

The preceding examples all describe point-to-point configurations. A bus topology on the other hand, is multipoint. One long cable act as a backbone to link all the devices in the network.

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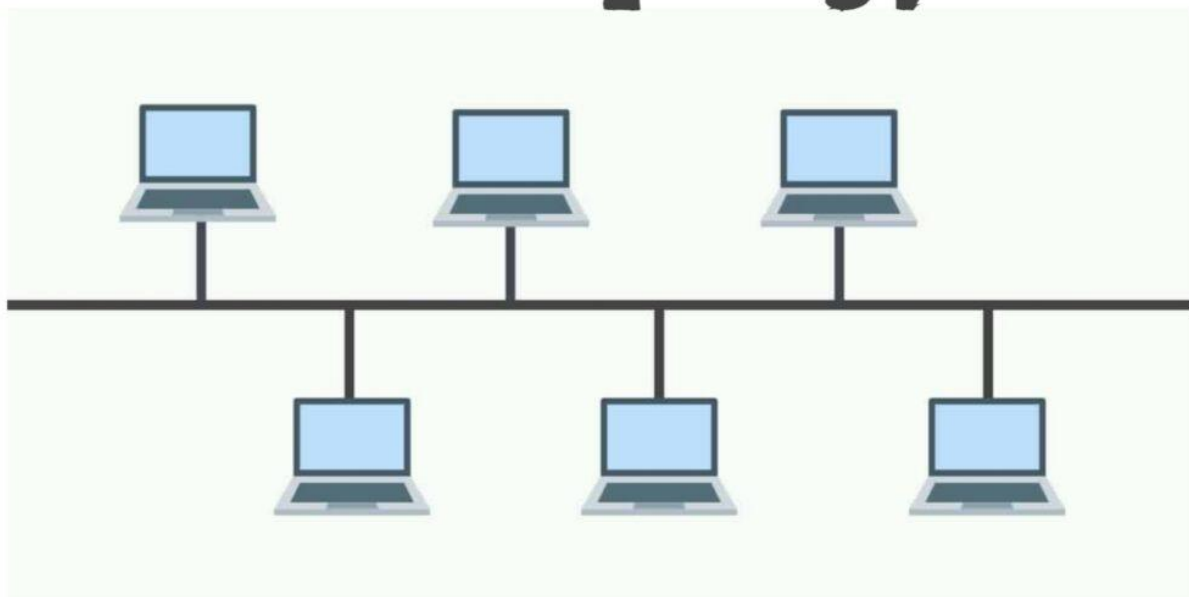
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## Bus Topology



### The advantages of a bus topology are:

1. Ease of installation.
2. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths.
3. Use less cabling than mesh, star, and tree topologies. Terminator Terminator Tap Tap 11

### While the disadvantages are:

1. Difficult reconfiguration and fault isolation.

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2. A fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

3. Adding new devices may therefore require modification or replacement of the backbone.

### Ring Topology

In a ring topology, each device has a dedicated point-to-point line configuration only with the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater.

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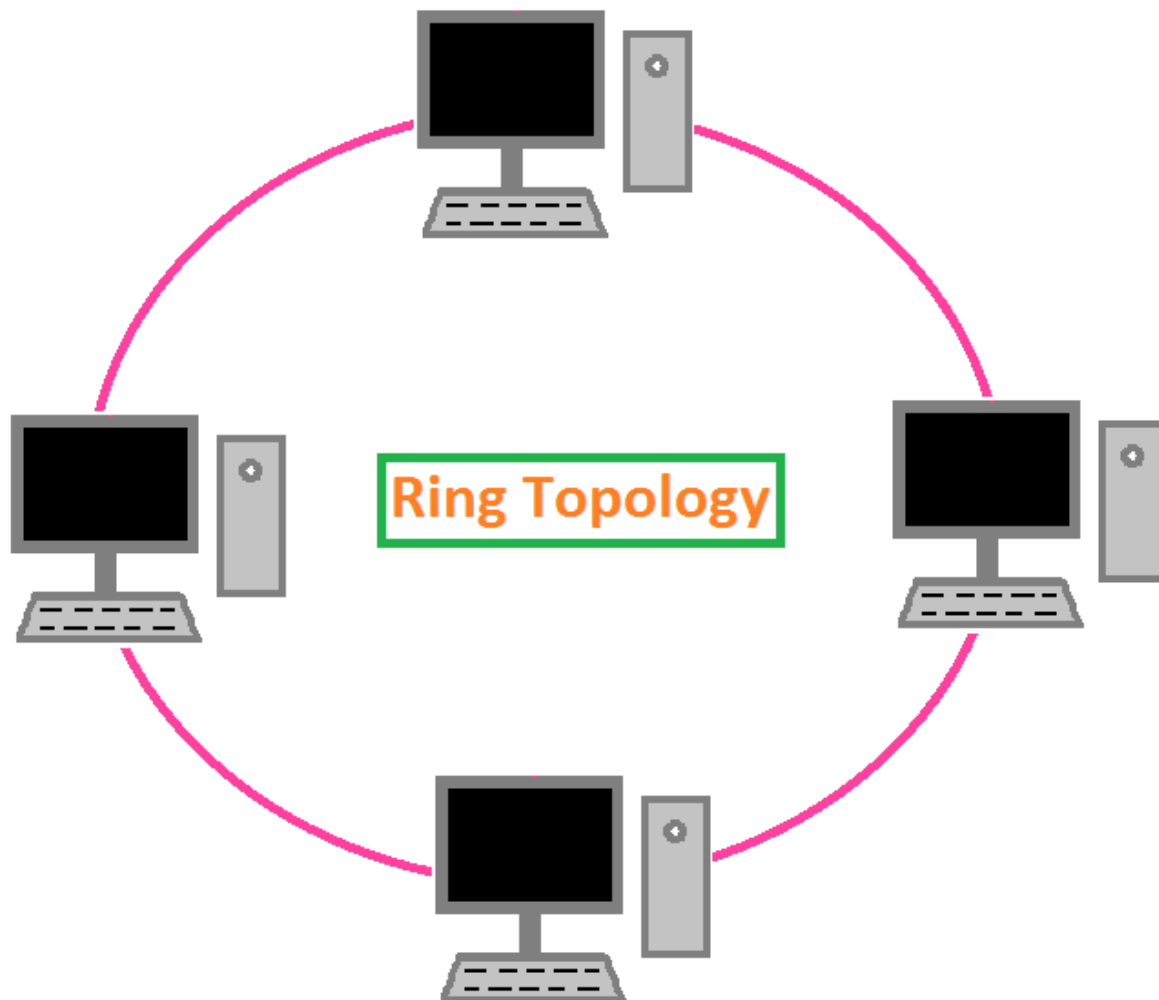
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### The advantages of ring topology are:

1. Relatively easy to install and reconfigure.
2. Addition or deletion of a device requires moving only two connections.



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3. Fault isolation is simplified.

**The disadvantage** of the ring topology is that a break in the ring (such as a disabled station) can disable the entire network, and it is one way data flow.

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### The Internet

- An internet (note the lowercase i) is two or more networks that can communicate with each other

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- **The Internet (uppercase I), and is composed of thousands of interconnected networks.**

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## Internet History

- **Telegraph and Telephone networks, before 1960:**
  - ✓ **Constant-rate communication only**
  - ✓ **ARPANET- Packet Switched**
- **Birth of the Internet & TCP/IP**
- **MILNET**
- **CSNET**
- **NSFNET**
- **Internet Today**

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### What is ARPANET and why is it important?

The Advanced Research Projects Agency (ARPA), an arm of the U.S. Defense Department, funded the development of the Advanced Research Projects Agency Network (ARPANET) in the late 1960s. Its initial purpose was **to link computers at Pentagon-funded research institutions over telephone lines**.

### Birth of the Internet & TCP/IP

**January 1, 1983** is considered the official birthday of the Internet. Prior to this, the various computer networks did not have a standard way to communicate with each other. A new communications protocol was established called Transfer Control Protocol/Internetwork Protocol (TCP/IP).

### MILNET

MILNET was **the name given to a network that split off from ARPANET in 1983 to create an internetwork designated for use by the U.S. Department of Defense**. MILNET later became part of the DoD Defense Network (DDN). MILNET is also known as military net.

### Who developed Csnets?

Short for Computer Science Network, CSNET is a computer network that was developed by the **U.S. National Science Foundation** in the early 1980s.

### NSFNET

**The National Science Foundation Network (NSFNET)** was a program of coordinated, evolving projects sponsored by the National Science Foundation (NSF) from 1985 to 1995 to promote advanced research and education networking in the United States.

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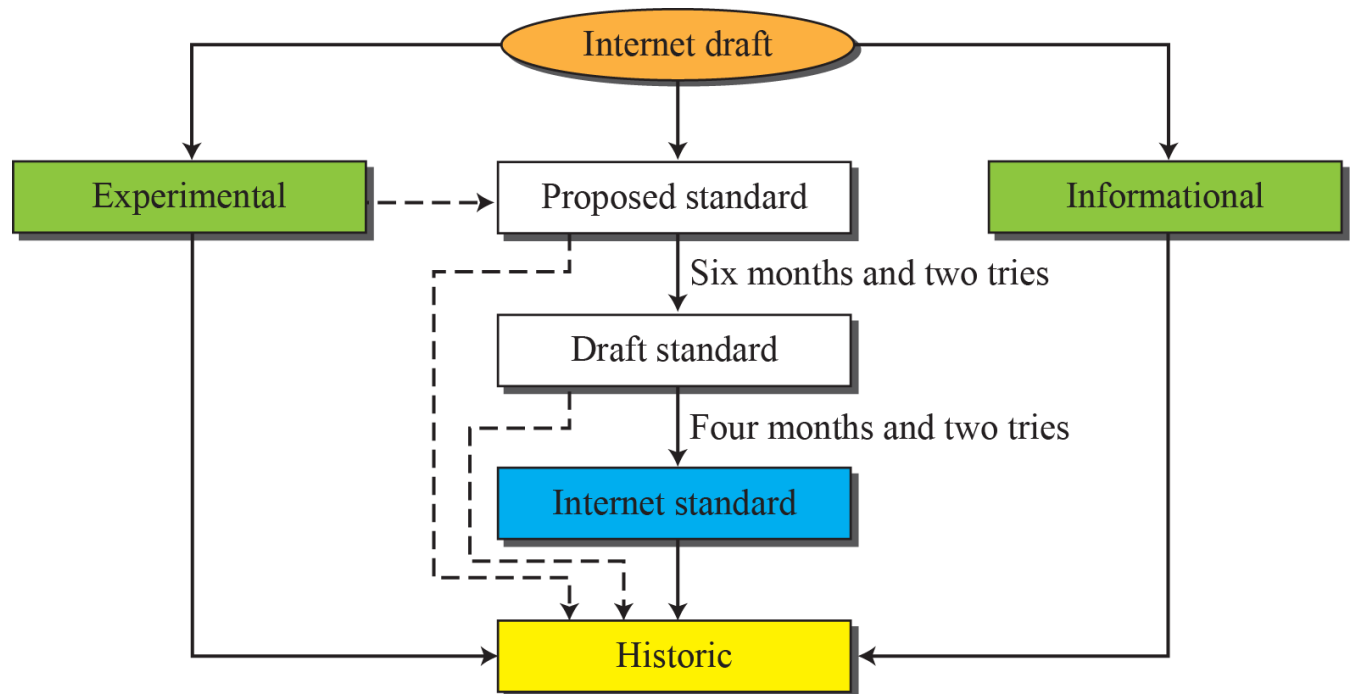
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### Internet Standards

- Internet draft
- Request for Comments (RFC)
  - ✓ Proposed Standard
  - ✓ Draft Standard
  - ✓ Internet Standard
  - ✓ Historic
  - ✓ Experimental
  - ✓ Informational



### Protocol Layering - Introduction

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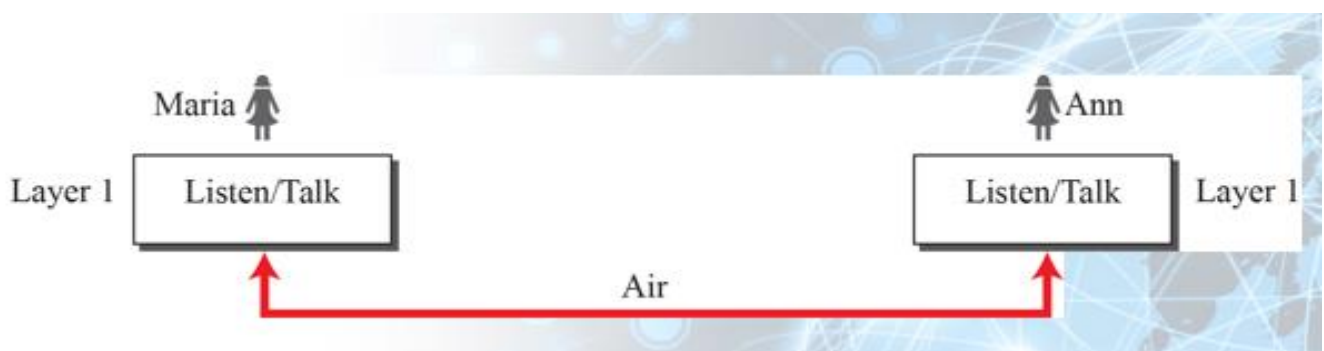
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- **Protocol**

Rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively

- **Protocol Layering**

**Simple Communication**: only one simple protocol



**Complex Communication**, we need a protocol at each layer, or Protocol Layering

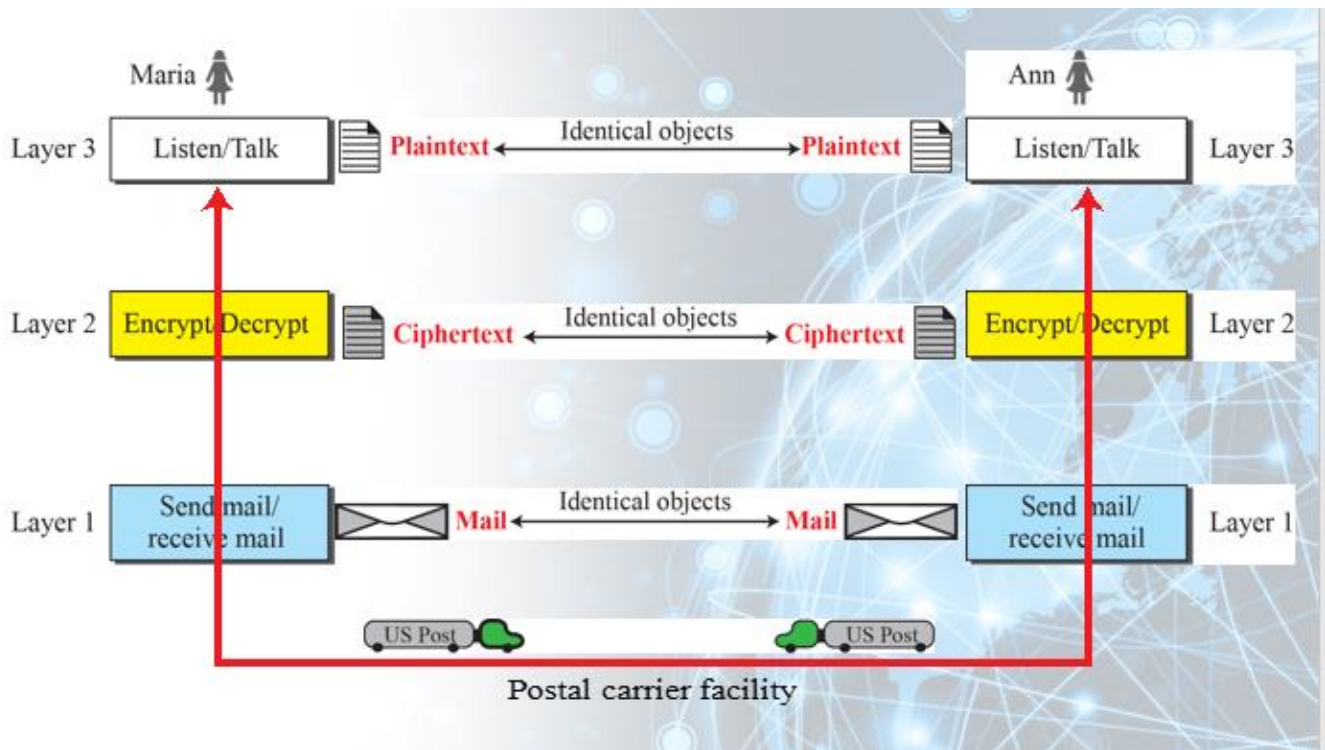


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## Protocol Layering - Advantages and Disadvantages

### Advantages

- ✓ Modularity
- ✓ Separation of Service & Implementation
- ✓ Reduced Complexity & Cost

### Disadvantages

- None Really!



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### Protocol Layering - Principles

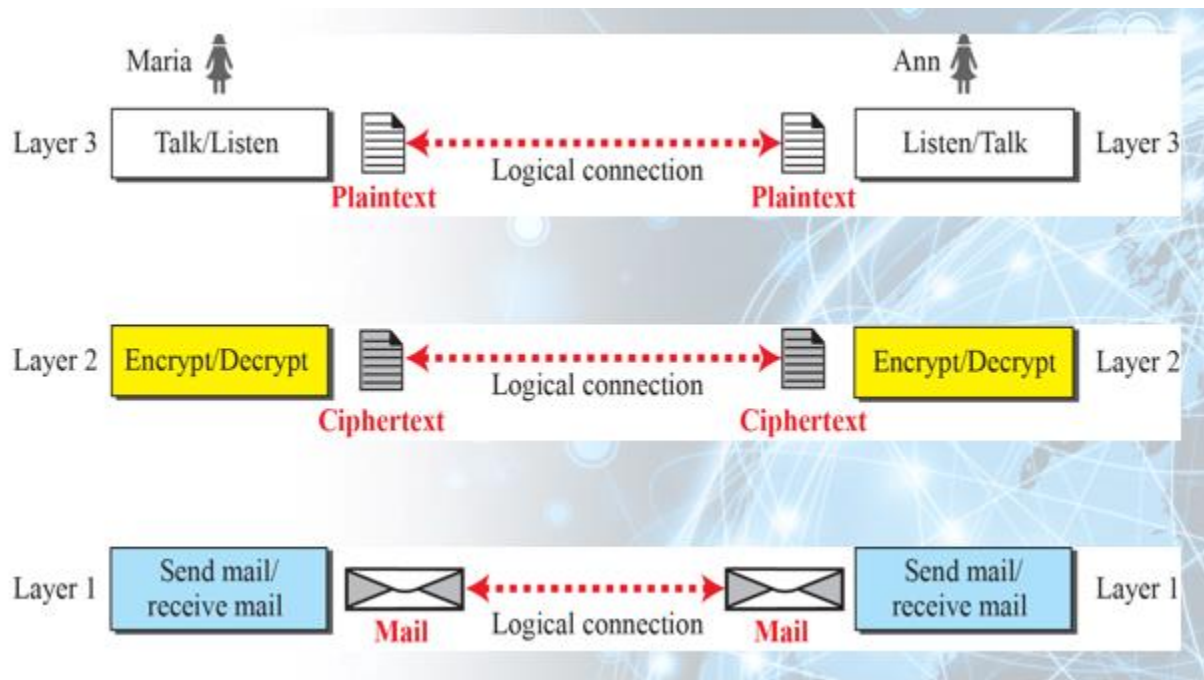
#### • Two Principles

- ✓ Bidirectional Communication → Each Layer performs two opposite tasks in each direction
- ✓ Two objects under each layer at both sites should be identical

### Protocol Layering - Logical Connections

#### • Logical Connections

- ✓ Imaginary connection between each layer



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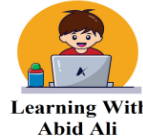
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### TCP/IP Protocol Suite

TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a **suite of communication protocols used to interconnect network devices on the internet**. TCP/IP is also used as a communications protocol in a private computer network (an intranet or extranet).

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# TCP/IP Protocol Suite

## History

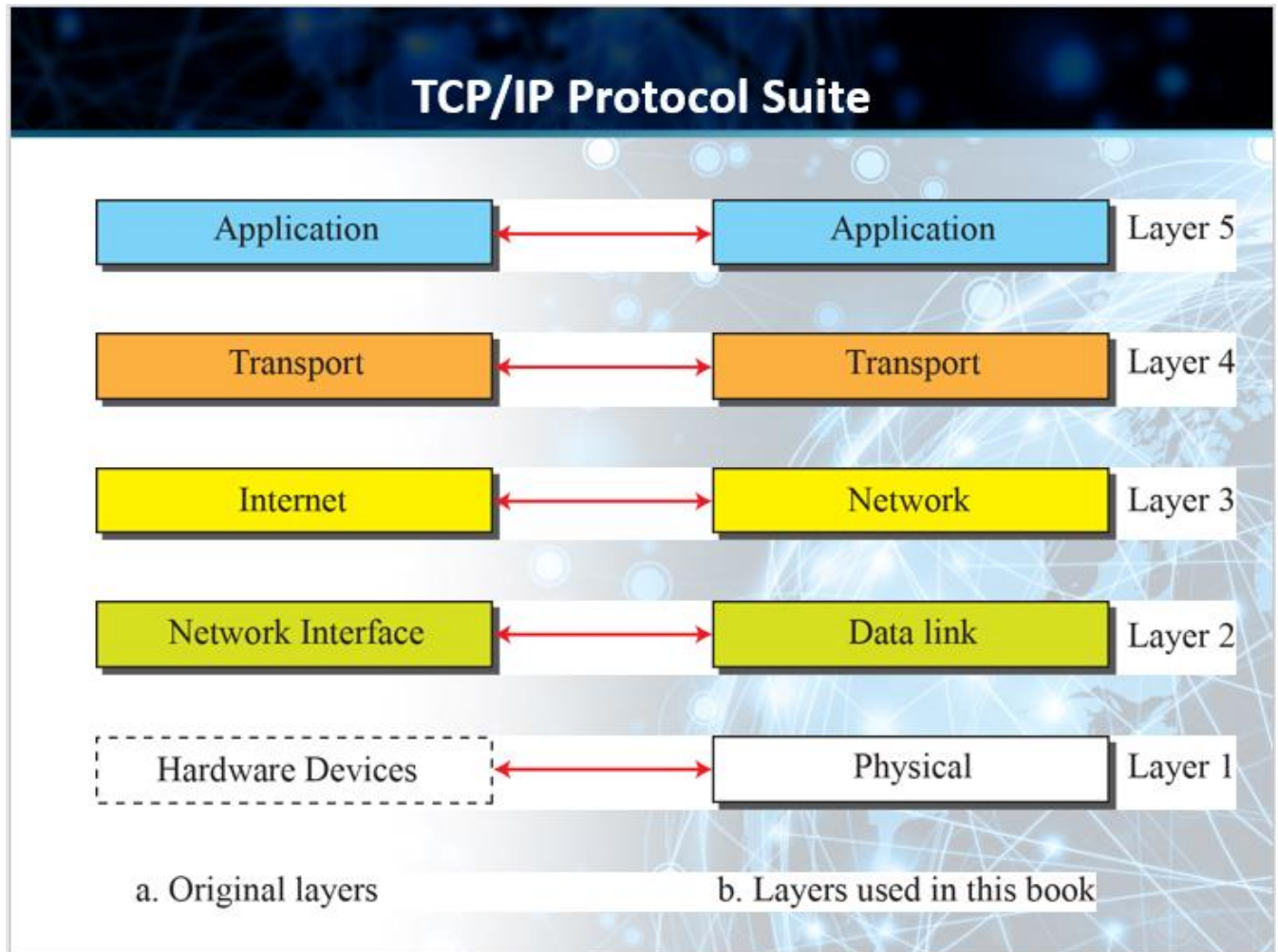
- The Internet Protocol Suite (commonly known as TCP/IP Suite or the TCP/IP Model) is the set of computer network communications protocols and a description framework used for the Internet and other similar networks.
- The TCP/IP Model was created in the 1970s by DARPA, an agency of the United States Department of Defense (DOD). It evolved from ARPANET, which was the world's first wide area network and a predecessor of the Internet.

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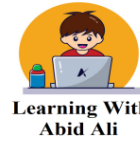


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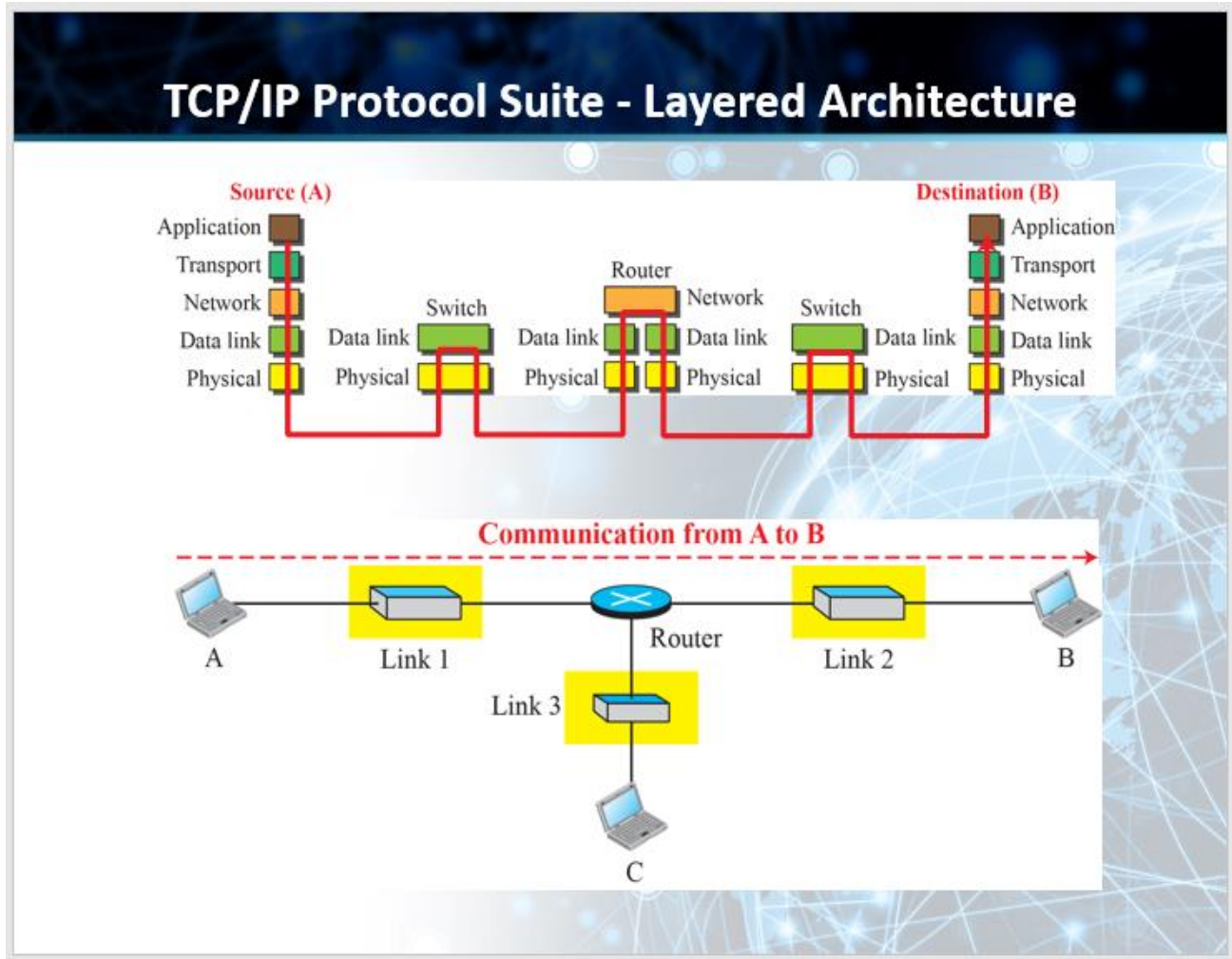


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- **Logical Connections between TCP/IP Layers**

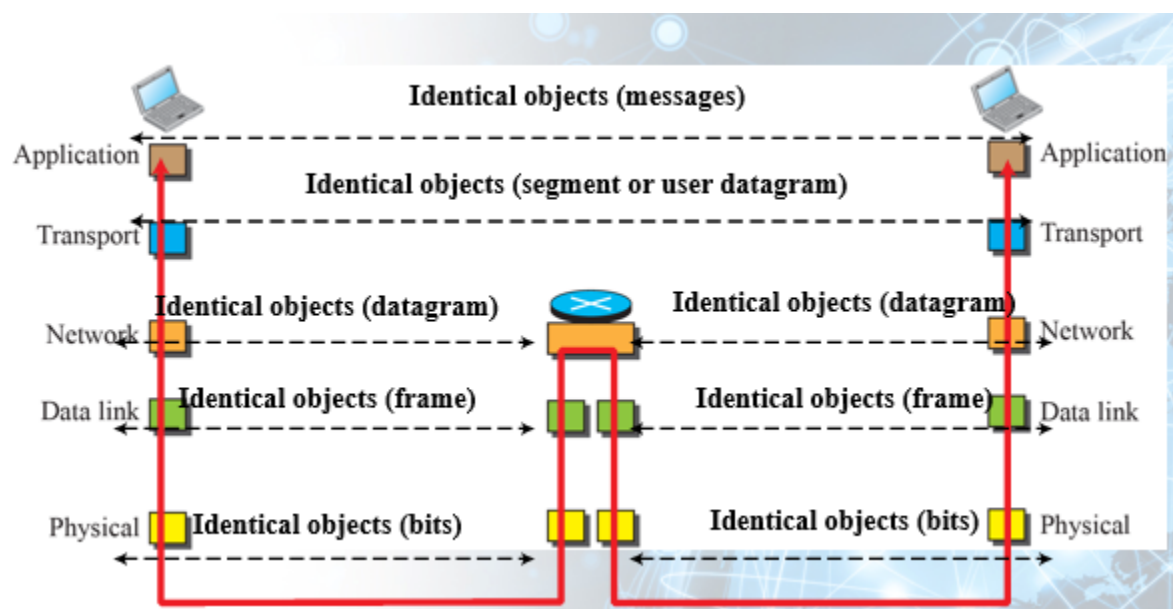
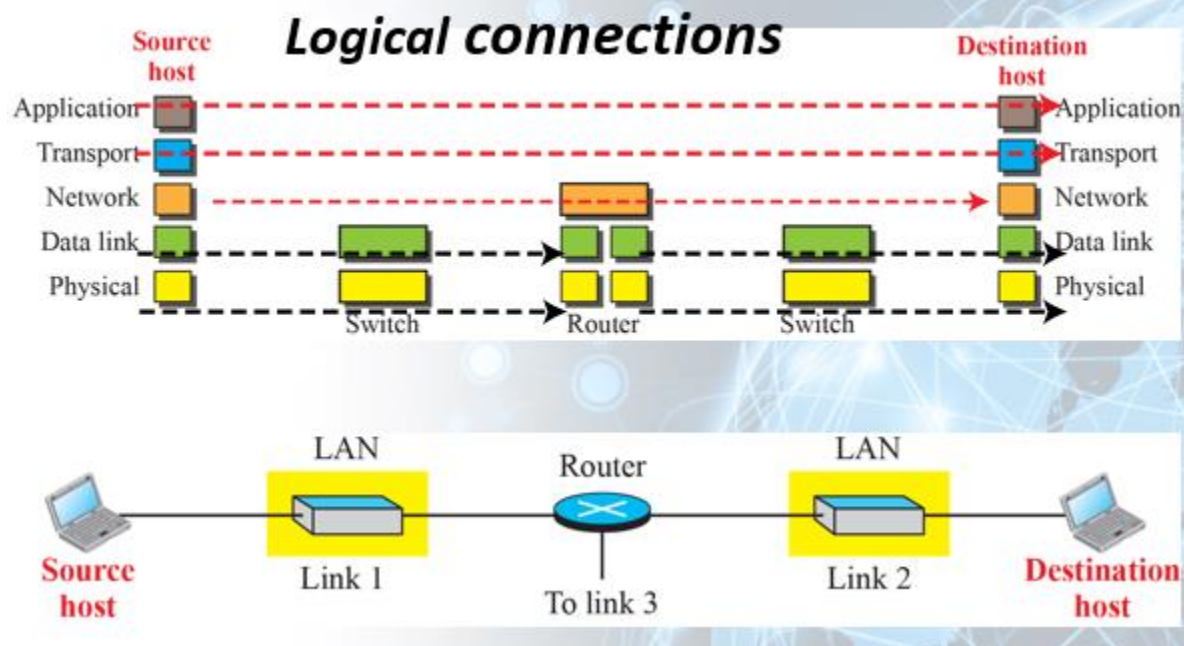


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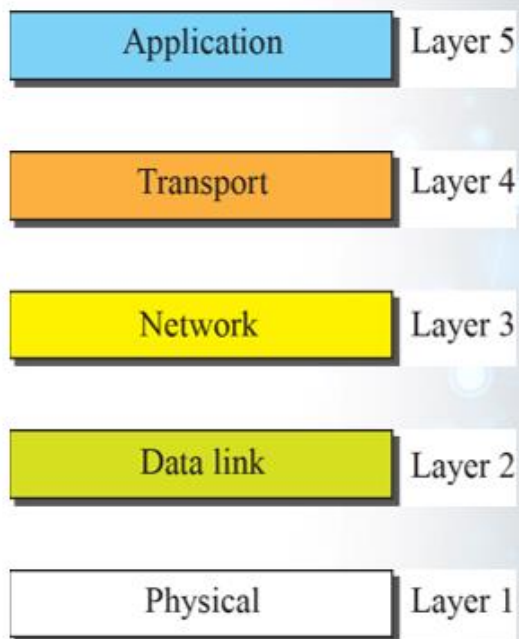
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### TCP/IP Protocol Suite – Layer Description

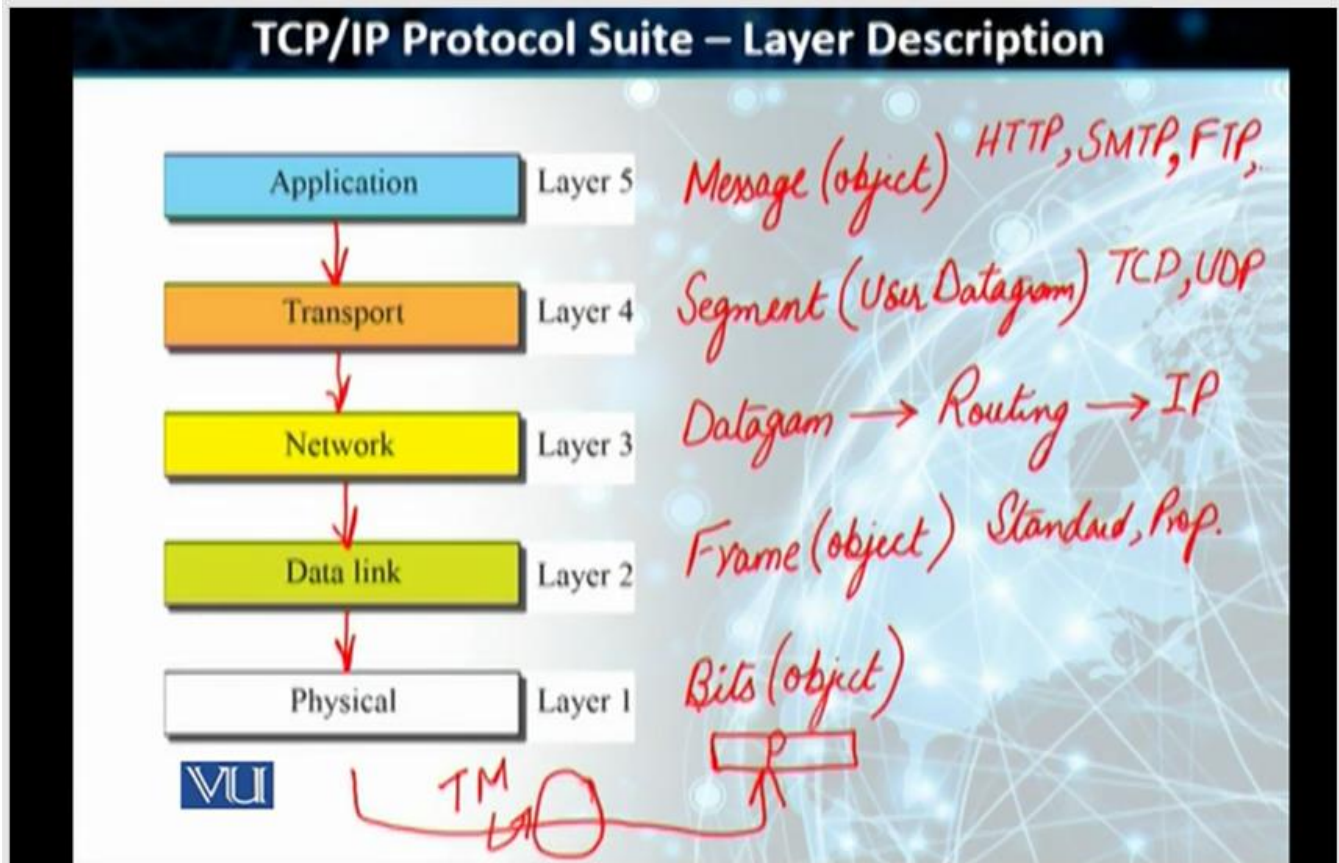


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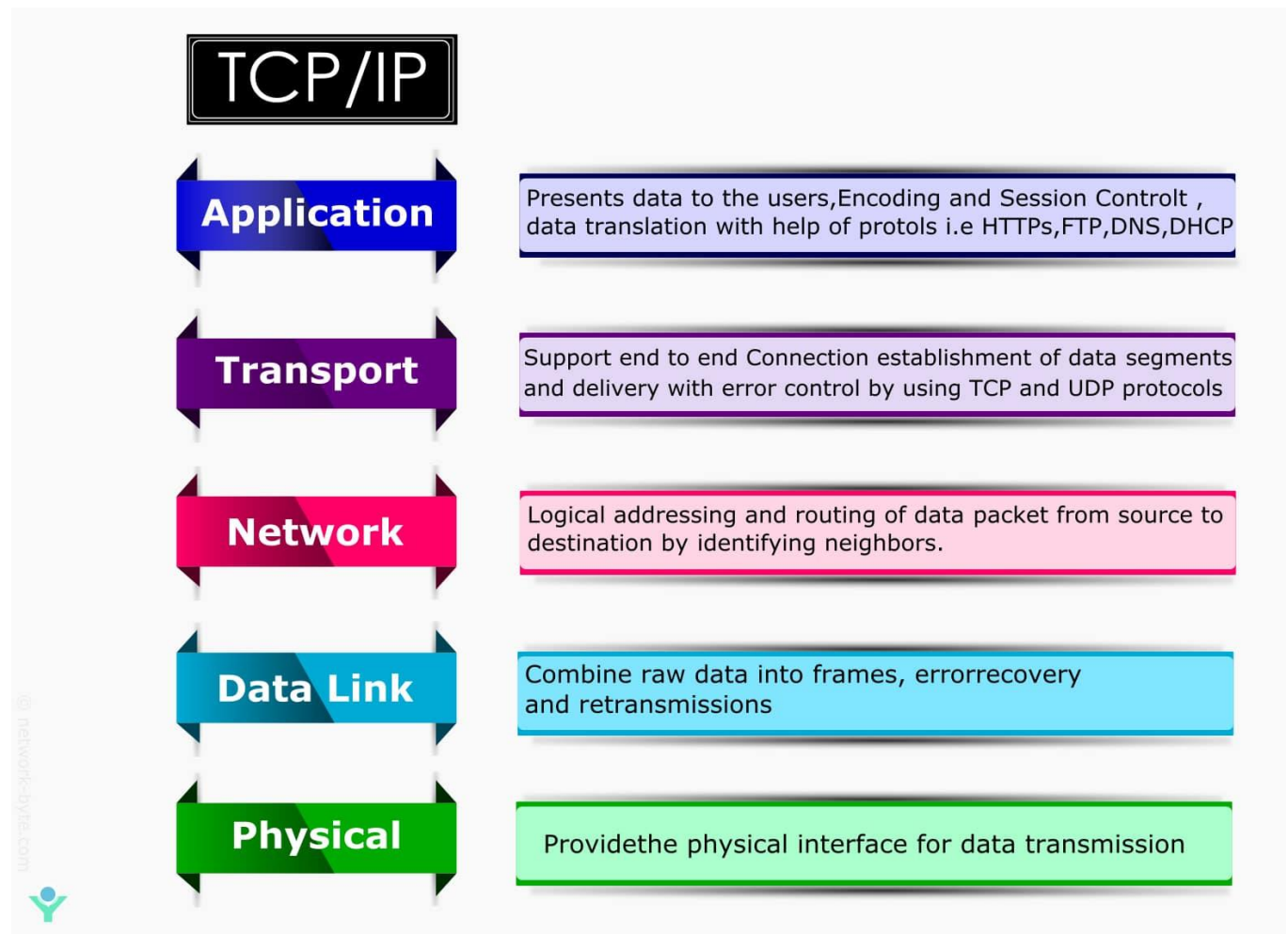


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### What is encapsulation and Decapsulation?

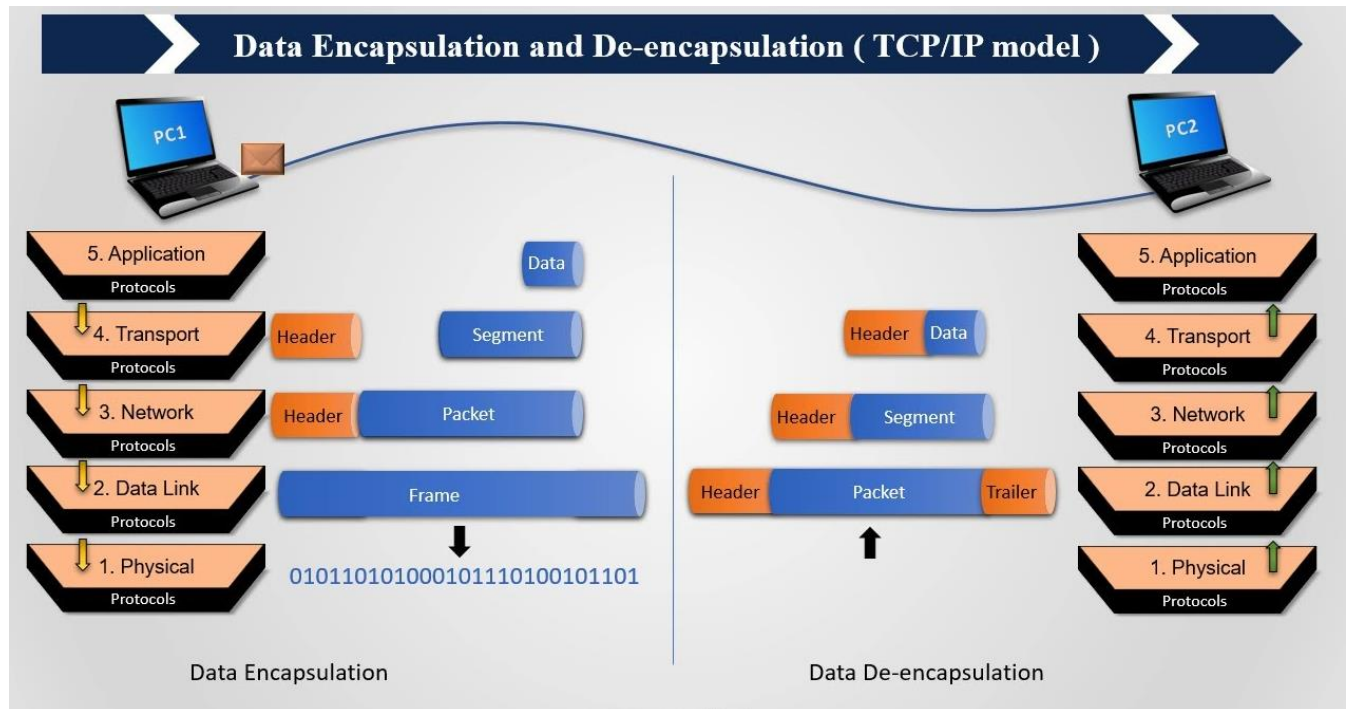
Encapsulation adds information to a packet as it travels to its destination. Decapsulation reverses the process by removing the info, so a destination device can read the original data.

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### Encapsulation:

Encapsulation refers to attaching new information in the Application Layer data as it is passed onto next layers in the [TCP/IP](#) model. This additional information basically divided into two parts, Header and Trailer. These are elements attached in order to make the transmission smoother, on each layer a PDU (Protocol Data Unit) is generated. The concept of Encapsulations can be summarized in the screenshot attached ahead.

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## ENCAPSULATION



### **Decapsulation:**

Decapsulation refers to the removal of all these additional information and extraction of originally existing data, and this process continues till the last [layer](#) i.e., the Application Layer. This process removes, fragments of distinct information in each layer as it approaches that layer. Here is the pictorial representation of the whole process.

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### DECAPSULATION

HEADER

DATA

TRAILER

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### The Open System Interconnection (OSI) Model

Open Systems Interconnection (OSI) – a framework for transmitting messages between any two entities in a network. It defines seven layers of functions: Physical Layer – is at the electrical and mechanical level; referring to the hardware and physical wires/devices required to transport the data.

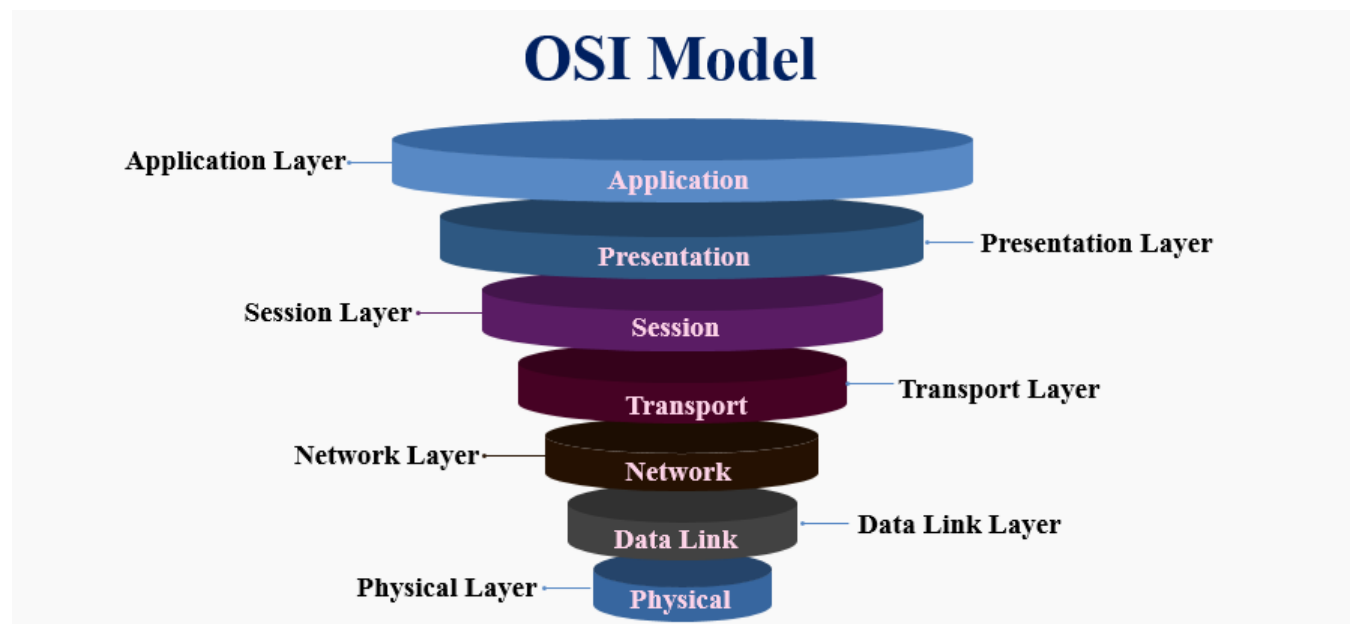
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- **International Organization for Standardization (ISO)**
- **ISO established in 1947**
- **Close to three-fourths of countries represented**
- **Introduced OSI Model in late 1970s**
- **OSI: a 7-Layer Model**



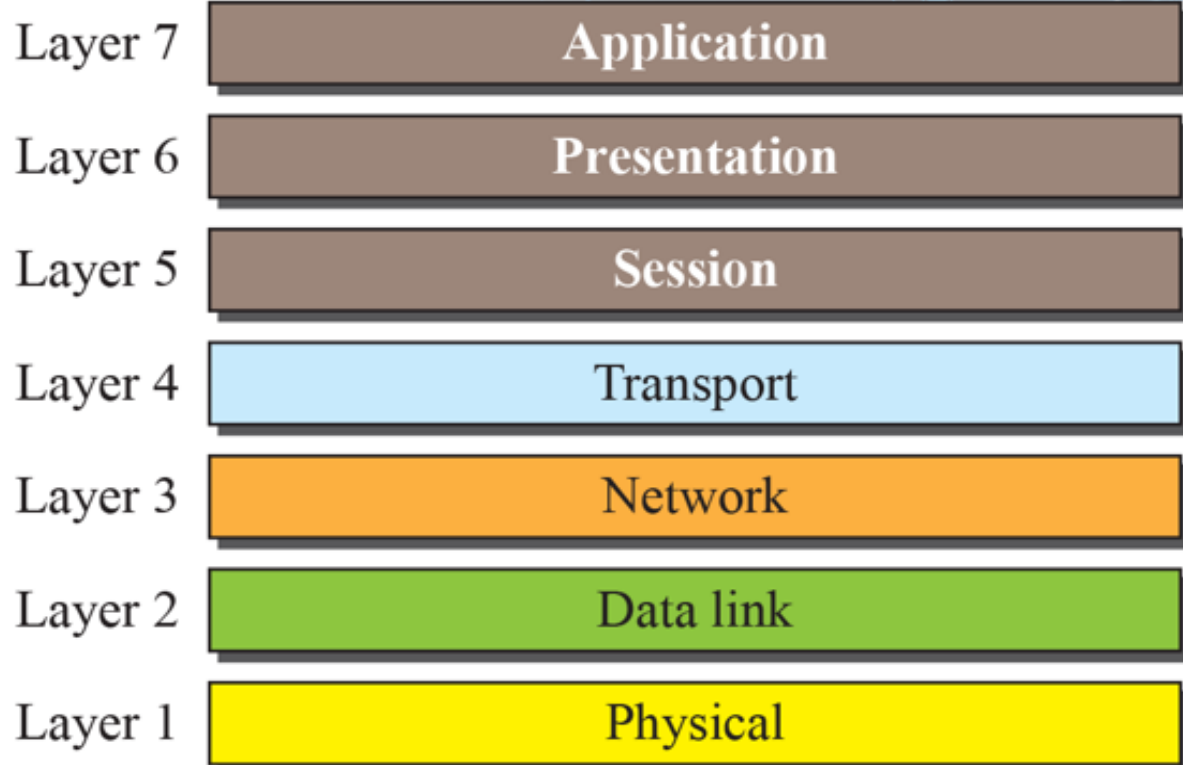
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### The Open System Interconnection (OSI) Model



### Lack of OSI Model's Success

- Three reasons OSI did not replace TCP/IP:
  - ✓ OSI was completed when TCP/IP was fully in place
  - ✓ Some layers in OSI not fully defined
  - ✓ Performance of TCP/IP better than that of OSI



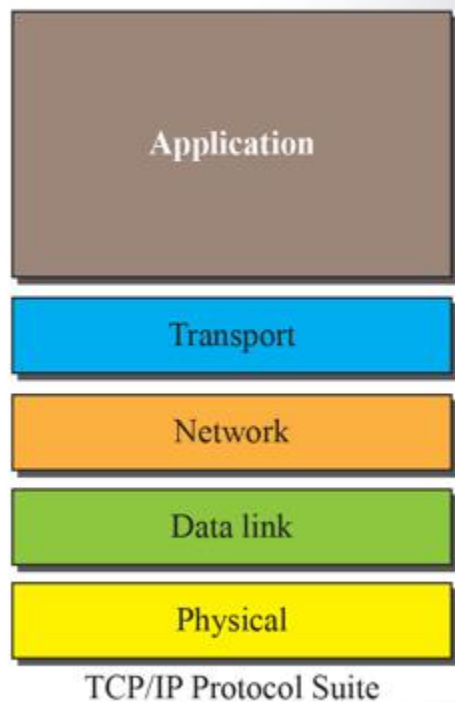
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### Data Communication versus Computer Networks



- Analog & Digital Transmission
- Transmission Media
- Switching
- Error Detection and Correction
- Media Access and Data Link Control
- Wired and Wireless LANs



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## Physical Layers

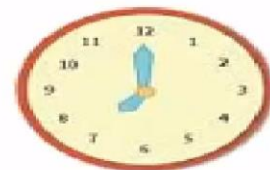
The data is travel from one network to another network through Physical Layer in the Form of Analog & Digital Data.

**Analog Data → Continuous**

Analog Clock, Human Voice



Analog Signal



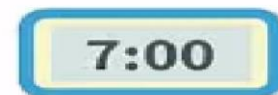
Analog clock

**Digital Data → Discrete**

Digital Clock, Data Store in Computer



Digital Signal



Digital clock

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### Analog & Digital Signals

- Signals represent Data
- Signals → Analog or Digital
- Analog Signal → Infinite Levels of Intensity over time
- Digital Signal → Limited number of defined values

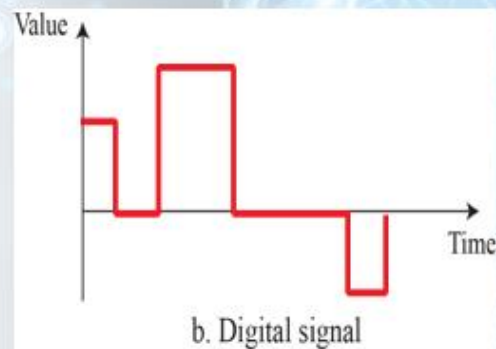
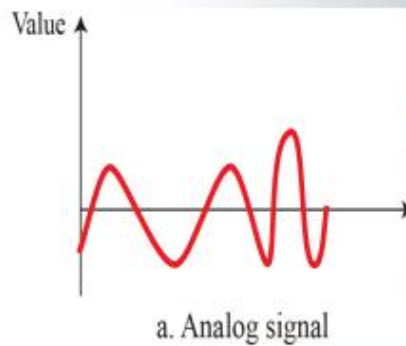
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### Analog & Digital Signals



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### Analog & Digital Data

- Data → Analog or Digital
- Analog Data → Continuous
- Digital Data → Discrete
- Examples: Analog Clock vs. Digital Clock
- Human voice vs. Data in Computer



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### Periodic & Non-periodic Signals

- Analog/Digital Signal → Periodic or Non-periodic
- Periodic Signal → Pattern
- Period and Cycle
- Non-Periodic → No Pattern
- Periodic ANALOG Signals and Non-periodic DIGITAL Signals

### Periodic & Non-periodic Signals

A signal is said to be periodic signal if it has a definite pattern and repeats itself at a regular interval of time. Whereas, the signal which does not at the regular interval of time is known as an aperiodic signal or non-periodic signal

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### Periodic Signals

A Periodic Signals is a signal that complete pattern in a certain period of time. (Cycle).

**Simple Periodic Analog Signal - Sine Wave**

**Composite Periodic Analog Signal - Multiple Sine Wave**

### Non-Periodic Signals

A Non-Periodic Signals is a signal that changes its pattern in each cycle.

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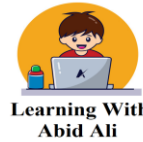
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### A Simple Periodic Analog Signal, A Sine Wave

A **simple periodic analog signal**, a sine wave, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves. A sine wave can be represented by three parameters: the peak amplitude, the frequency, and the phase. These three parameters fully describe a sine wave.

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## Periodic Signal – Sine Wave

There are three main attribute of a Sine Wave

### 1. Amplitude

### 2. Period / Frequency

### 3. Phase

#### 1. Amplitude (Key Attribute):

It is equal to the vertical distance from a given point on the wave form to the horizontal axis

#### 2. Period / Frequency (Second Attribute):

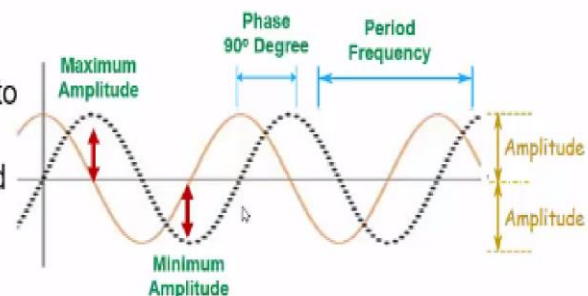
**Period:** Amount of time (in seconds) a signal need to complete one cycle.

**Frequency:** Number of cycles completed in one second

**Unit of Period:** Period is expressed in seconds

#### 3. Phase

The Phase Shift is how far the function is shifted horizontally from the usual position.



Periodic Analog Signal



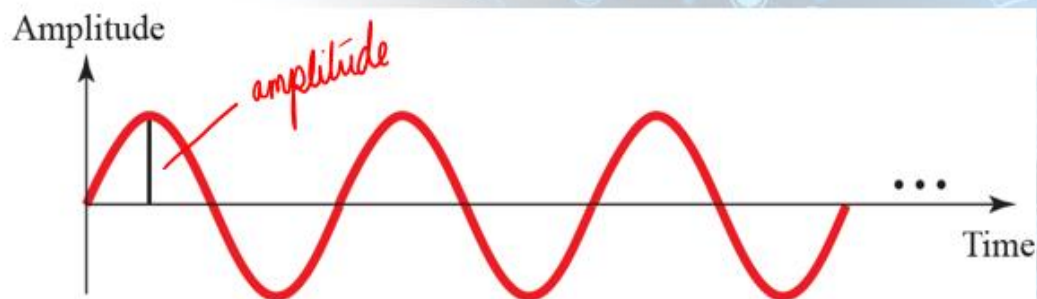
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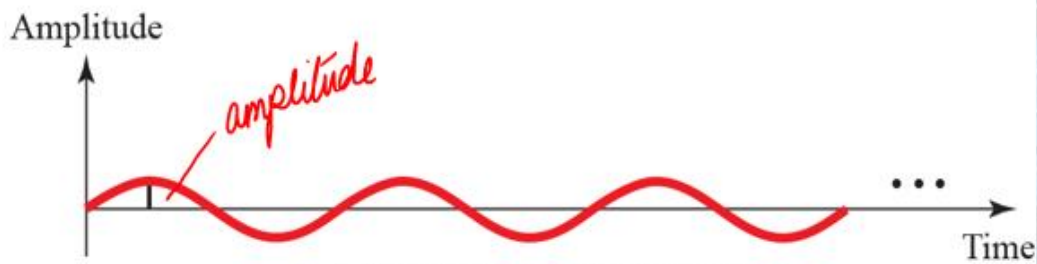


<https://www.youtube.com/watch?v=-0kr2sG8JUQ&list=PLAUocDaOe9K6nDHAXKa-7fHsIFAKBPE8q&index=9>

### Sine Wave – Peak Amplitude



a. A signal with high peak amplitude



b. A signal with low peak amplitude

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### Sine Wave –Frequency

- Period (T) → Amount of time required to complete 1 cycle
- Frequency (f) → No. of Periods in 1 sec
- $f = 1/T$  or  $T = 1/f$

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## Period / Frequency Examples

- ❑ A Sine wave has a frequency of 6 Hz.  
What is its period?
- ❑ A Sine wave completes one cycle in 4 seconds. What is its frequency?

- ❑ **Period: Formula:  $T = 1 / f$**
- ❑ **Frequency: Formula:  $f = 1 / T$**

| Period                  |              | Frequency       |              |
|-------------------------|--------------|-----------------|--------------|
| Unit                    | Equivalent   | Unit            | Equivalent   |
| Seconds (s)             | 1 s          | Hertz (Hz)      | 1 Hz         |
| Milliseconds (ms)       | $10^{-3}$ s  | Kilohertz (kHz) | $10^3$ Hz    |
| Microseconds ( $\mu$ s) | $10^{-6}$ s  | Megahertz (MHz) | $10^6$ Hz    |
| Nanoseconds (ns)        | $10^{-9}$ s  | Gigahertz (GHz) | $10^9$ Hz    |
| Picoseconds (ps)        | $10^{-12}$ s | Terahertz (THz) | $10^{12}$ Hz |

Pause (s)

## SOLUTION

- ❑ **Period: Formula:  $T = 1 / f$**   
 $= 1 / 6$   
 $= 0.17 \text{ Second}$
- ❑ **Frequency: Formula:  $f = 1 / T$**   
 $= 1 / 4$   
 $= 0.25 \text{ Hz}$

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### Sine Wave –Frequency

| Period                  |                | Frequency       |                |
|-------------------------|----------------|-----------------|----------------|
| Unit                    | Equivalent     | Unit            | Equivalent     |
| Seconds (s) ✓           | 1 s            | Hertz (Hz) ✓    | 1 Hz           |
| Milliseconds (ms)       | $10^{-3}$ s ✓  | Kilohertz (kHz) | $10^3$ Hz ✓    |
| Microseconds ( $\mu$ s) | $10^{-6}$ s ✓  | Megahertz (MHz) | $10^6$ Hz ✓    |
| Nanoseconds (ns)        | $10^{-9}$ s ✓  | Gigahertz (GHz) | $10^9$ Hz ✓    |
| Picoseconds (ps)        | $10^{-12}$ s ✓ | Terahertz (THz) | $10^{12}$ Hz ✓ |

### Example

The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ sec} \\ = 16.6 \text{ msec}$$



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### Example

The period of a signal is 100 ms. What is its frequency in kilohertz?

$$T = 100 \text{ ms} \Rightarrow 100 \times 10^{-3} \text{ sec} = 10^{-1} \text{ sec}$$

$$f = \frac{1}{T} = \frac{1}{10^{-1}} = 10 \text{ Hz}$$

$$10 \times 10^{-3} \text{ kHz}$$

$$= 10^{-2} \text{ kHz}$$

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### Phase (or Phase Shift)

- Position of waveform relative to time 0
- Phase describes the amount of shift of the wave
- Indicates start of the first cycle

### Phase

$$360^\circ = 2\pi \text{ radians}$$

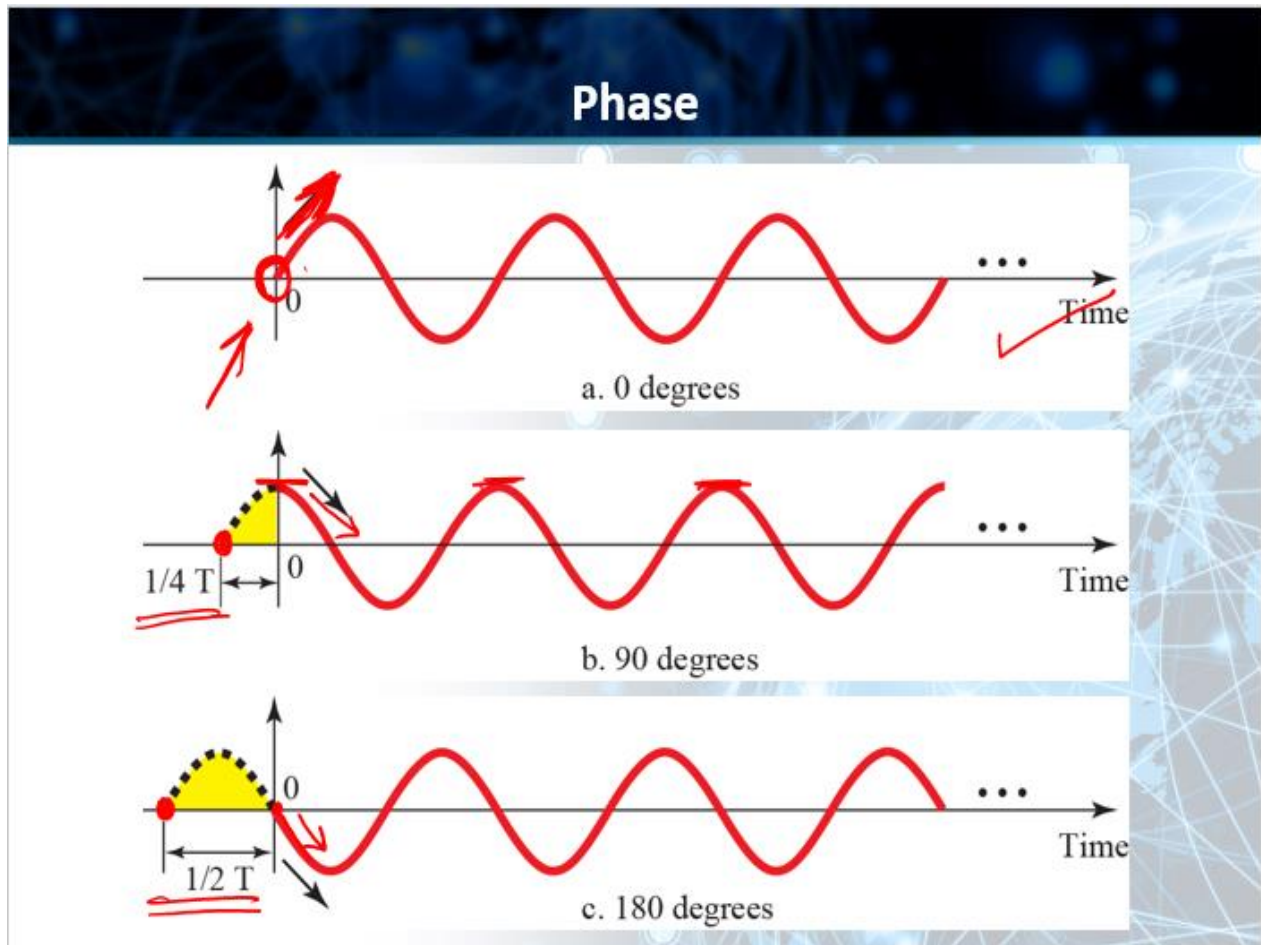
$$1^\circ = \frac{2\pi}{360} \text{ rad}$$

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## SOLUTION

□ Phase:

$$= 1 / 6 \times 360$$

$$= 360 / 6$$

$$= 60^{\circ} \text{ Degree}$$

## Phase Example

A sine wave is offset 1/6 of a cycle with respect to time zero. What is its phase in radius?

□ One Cycle = 360 Degrees

$$(1^{\circ} = 2 \pi / 360^{\circ} \text{ radians})$$

$$= 60 \times 2 \pi / 360$$

$$= 60 \times (2 \times (3.146 / 360))$$

$$= 60 \times (2 \times 0.0087)$$

$$= 60 \times 0.0174$$

$$= 1.04 \text{ Radius}$$

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## EXERCISE

- ☐ A Sine wave has a frequency of 60 Hz. What is its period?
- ☐ A Sine wave completes one cycle in 100 seconds. What is its frequency?
- ☐ A sine wave is offset  $\frac{1}{4}$  of a cycle with respect to time zero. What is its phase in radius?

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### Wavelength

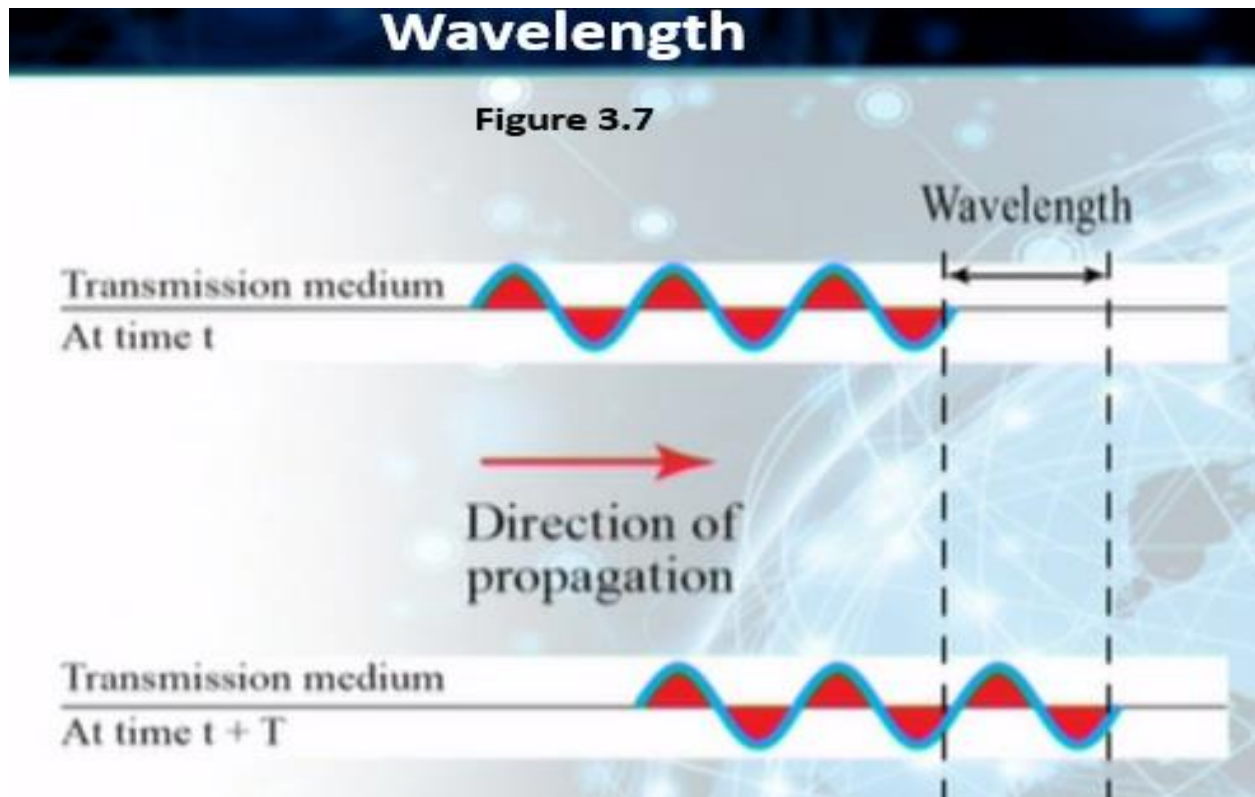
**Wavelength is another characteristic of a signal traveling through a transmission medium. Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium (see Figure 3.7).**

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### Wavelength

Prop. Speed =  $c$  = Light (speed)

frequency =  $f$

Wavelength =  $\lambda$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/sec}}{f}$$

$$\lambda = \frac{3 \times 10^8}{4 \times 10^{14}}$$





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$$\lambda = 0.75 \times 10^{-6} \text{ m}$$
$$= \underline{\underline{0.75 \mu\text{m}}}$$

### Composite Signals

So far, we have focused on simple sine waves. Simple sine waves have many applications in daily life. We can send a single sine wave to carry electric energy from one place to another. For example, the power company sends a single sine wave with a frequency of 60 Hz to distribute electric energy to houses and businesses.



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### Composite Signals

- **Single Sine Wave can only carry limited information**
- **Composite Signal is made up of multiple simple sine waves**
- **Can be periodic or non-periodic**

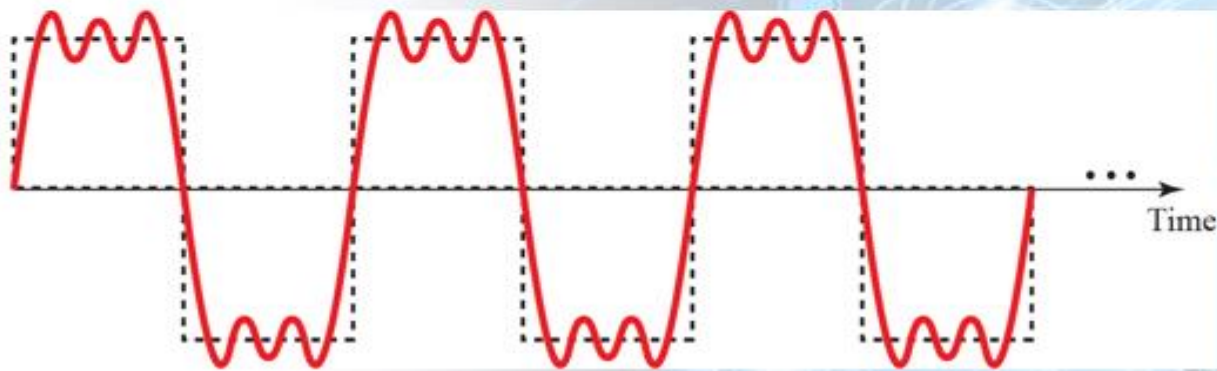
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### A Composite Periodic Signal



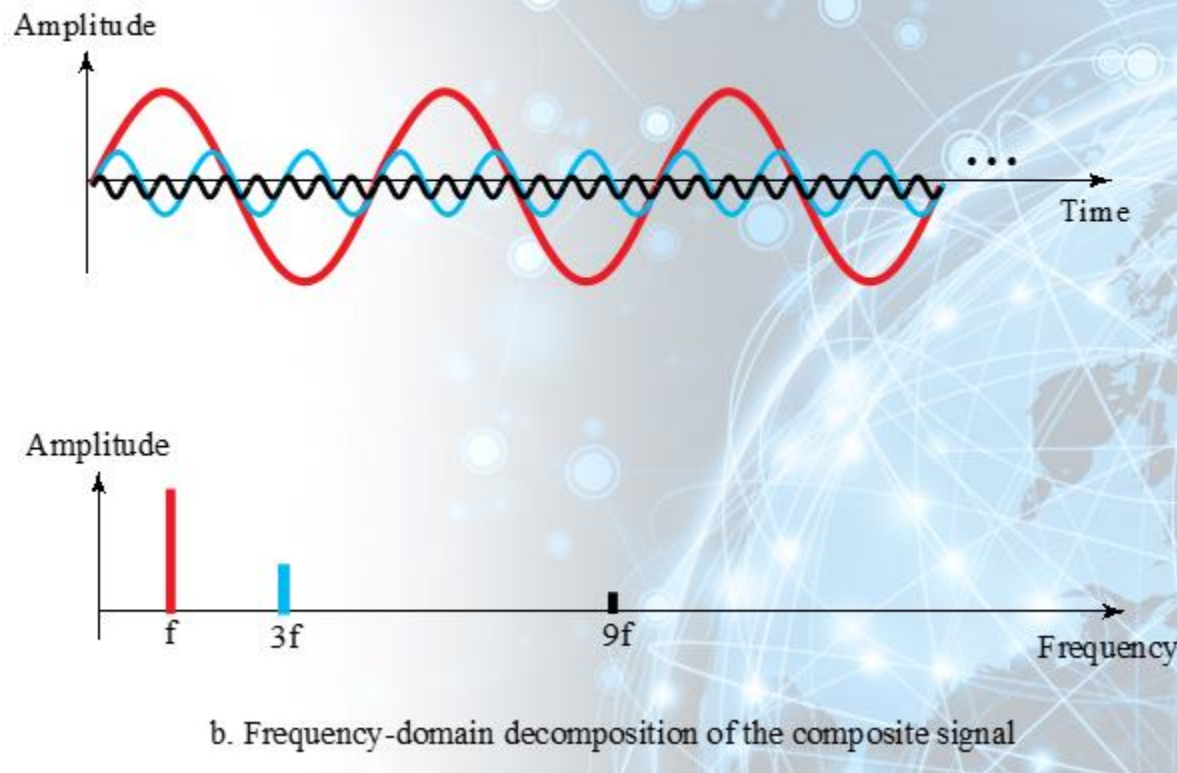
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### Decomposition of Composite Periodic Signal



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## Bandwidth

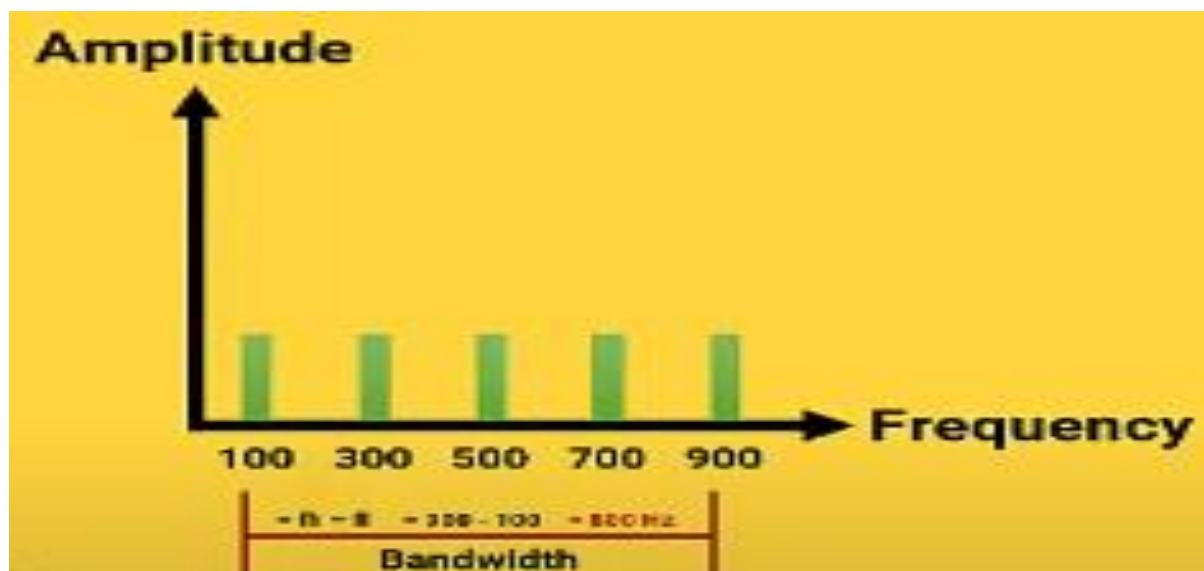
Bandwidth give us the range of frequencies contained in a composite signal. It give normally between two frequencies (High and Low).

### How to Calculate the Bandwidth:

To calculate Bandwidth, subtract the lowest frequency from the highest frequency.

### Example:

If a periodic signal is decomposed into five sine waves with frequencies 100, 300, 500, 700, and 900 Hz, what is the Bandwidth?





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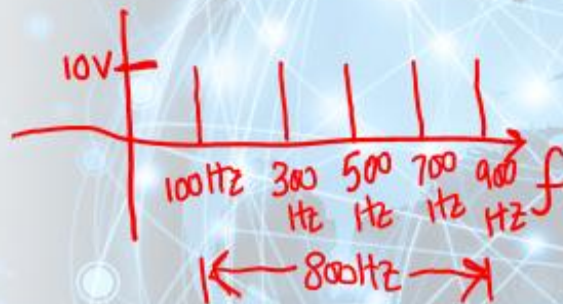


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### Example

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

$$B = f_h - f_l$$
$$= 900 - 100$$
$$B = 800 \text{ Hz}$$



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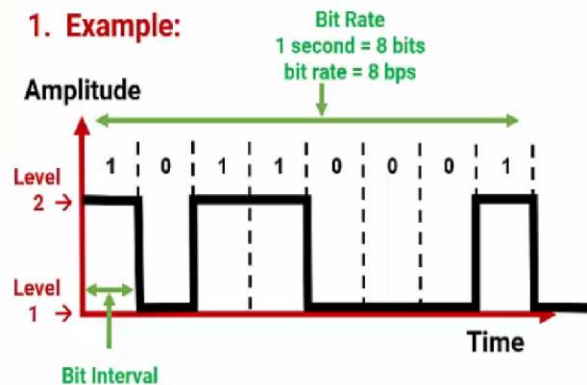


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## Digital Signal – Non-Periodic Signals

A signal in which the original information is converted into bits (0s Zero & 1s Positive) before being transmitted. And Digital Signal have more than 2 level.

### 1. Example:



**Bit Interval (Seconds):** The time required to send one signal bit.

**Bit Rate (bps):** Number of bits sent per second.

**Bit Length:** The distance one bit occupies on transmission medium.

**Signals:** It generate L level signals. If generate 1 level than  $\log^2 2$ , and if we have 4 level than  $\log^2 4$ .

**Example 1:** A digital signal has 8 Level. How many bits are needed?

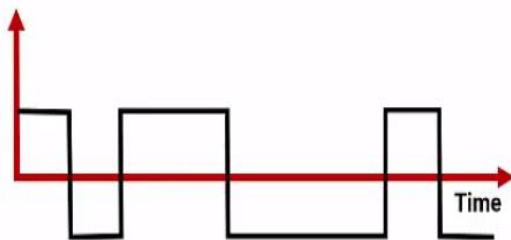
**Formula:** No. of Bits =  $\log_2 L = \log_2 8 = 3 \text{ bits}$

**Example 2:** Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel? 1 Page contained 24 line and 1 line contain 80 character and 1 character contain 8 bits.

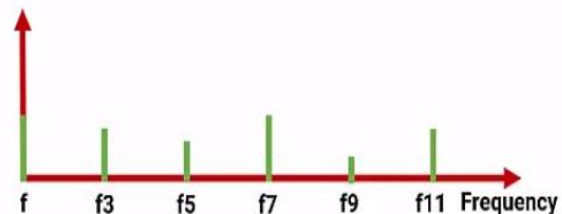
**Formula:** No. of Pages x No. of Lines x No. of Character x No. of Bits  
 $= 100 \times 24 \times 80 \times 8 = 1.536 \text{ Mbps}$

## Digital Signals as Composite Analog Signal

- A method is used for converting the digital signals into composite analog signals that is **Fourier Analysis**. It has infinite Bandwidth.
- Vertical line in the time domain: Frequency of infinity
- Horizontal line in the time domain: Frequency of zero



Time Domain Plot



Frequency Domain Plot

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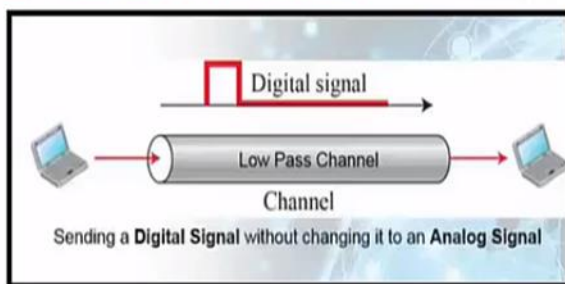


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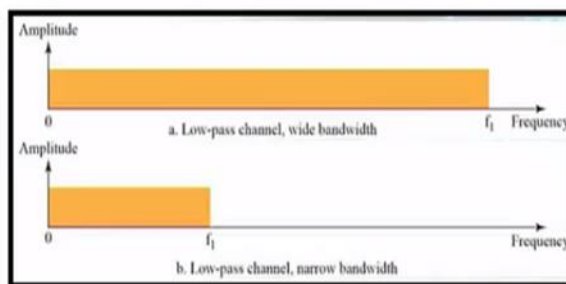
## Transmission of Digital Signals

We have already study that when the digital signal seen as composite signal it has infinite bandwidth and it has frequency from ZERO to ONE. To Transmit such signals we need two methods.

### ❑ Baseband Transmission Digital Signal to Digital Signals

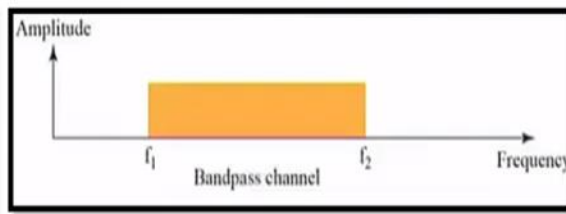
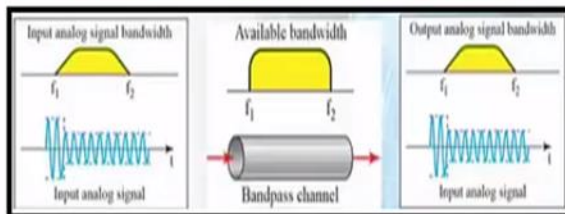


The output of the digital signal is NOT changed.



The low Pass Channel started from Zero to Infinite Frequency  
For this Transmission we need dedicated link (**Bus Topology**)

### ❑ Broadband Transmission (Modulation) Digital Signal to Analog Signals



The band Pass Chanel NOT started from Zero Frequency

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## Transmission of Digital Signals

- Digital signal, periodic or non-periodic, is a composite analog signal with frequencies between zero and infinity (Infinite Bandwidth)
- Two approaches for transmission:
  - ✓ Baseband Transmission
  - ✓ Broadband Transmission

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## Transmission of Digital Signals

A vertical line in the time domain means a frequency of infinity: a horizontal line in the time domain means a frequency of zero. Going from a frequency of zero to a frequency of infinity implies all frequencies in between are part of the domain.

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## Transmission Impairments

When we are sending signals from one point to another point the signals has lost some thing. These Losses can be heat, external factors etc. These in perfect signals are known as Transmission Impairments.

### Causes of Transmission Impairments

Following factors affect on Transmission Impairments.

1. Attenuation
2. Distortion
3. Noise



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## Attenuation | Distortion | Noise

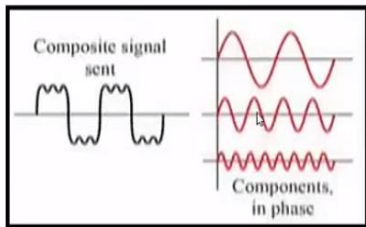
**1. Attenuation** – Loss of Energy. It cannot be recovered. (Heat)

The signals Strength can measured in terms of **Decible (dB)**. We can measure it at two Pints (Point 1 and Point 2).

**Formula:**  $10 \log_{10} P_2 / P_1$

If Decible is Negative its mean Signals has Attenuated and Lost Strength.

**2. Distortion** – is change in the form or shape of signal. (Delay)



### Attenuation Example

Suppose a signal travels through a transmission medium and its power is reduced to one half. Calculate the attenuation?

**Formula:**  $10 \log_{10} P_2 / P_1$

$$= 10 \log_{10} P_2 / P_1$$

$$= 10 \log_{10} (0.5 P_1) / P_1$$

$$= 10 \log_{10} (0.5)$$

$$= -3 \text{ dB}$$

= Signal has lost one half of the Power.

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### Noise Example

The signal power is 10 mW (Milli Wats) and noise power is 1 uW (Micro Wats). What is SNR?

**Formula = Avg. Signal Power / Avg. Noise Power**

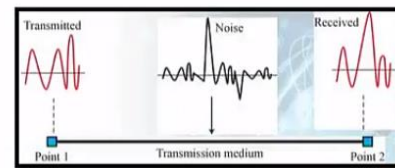
$$\begin{aligned}\text{SNR} &= 10 \text{ mW} / 1 \text{ uW} \\ &= 10,000 \text{ Milli Wats}\end{aligned}$$

$$\begin{aligned}\text{SNR}_{\text{dB}} &= 10 \log_{10} 10,000 \\ &= 40_{\text{dB}}\end{aligned}$$

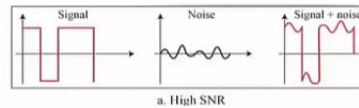
### Attenuation | Distortion | Noise

#### 3. Noise – Noise Corrupt Signal

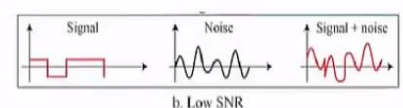
1. Thermal Noise
2. Induced Noise
3. Crosstalk Noise



### Signal to Noise Ratio (SNR)



1. High SNR mean Low Corrupt Signal



2. Low SNR mean High Corrupt Signal



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**Both Formulas are used to measure Data Rate Limits. Shannon's formula gives us the upper limit while the Nyquist formula gives us the signal levels.**

## Data Rate Limits

❑ Data Rate Limits mean how much data can be transfer over transmission medium or channel. It can be depend on three factors:

❑ Band Width Availability

❑ Level of Signals

❑ Level of Noise (Quality of Channel)

❑ The Data Rate Limits can be calculate by two methods:

1. One is provided by Nyquist (Noise = 0)

**Formula:** Bit Rate =  $2 \times \text{Bandwidth} \times \log_2 L$

**Example:** A Noiseless channel with a bandwidth of 3,000 Hz transmitting a signal with four signal levels. What is the maximum bit rate of the said channel?

$$\begin{aligned}\text{Bit Rate} &= 2 \times 3,000 \times \log_2 4 \\ &= 6,000 \times \log_2 4 \\ &= 6,000 \times 2\end{aligned}$$

**Bit Rate = 12,000 bps**

❑ Second is provided by Shanon (Noise > 0)

**Formula:** Capacity =  $\text{Bandwidth} \times \log_2 (1 + \text{SNR})$

**Example:** Theoretical highest bit rate of a Telephone line Bandwidth of 3000 Hz assigned for data communication. SNR is usually 3162. What is capacity?

$$\begin{aligned}\text{Capacity} &= \text{Bandwidth} \times \log_2 (1 + \text{SNR}) \\ &= 3,000 \times \log_2 (1+3162) \\ &= 3,000 \times \log_2 (3,163)\end{aligned}$$

**Capacity = 34,881 bps**

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## Network Performance

The network performance is based on three factors. Which includes:

### □ Bandwidth

Bandwidth can be measure in two Terms:

1. Bandwidth in Hertz (Range of Frequencies in a Composite Signals)
2. Bandwidth in Bits Per Second (No. of Bits Transmits on Channel or Network)

### □ Throughput

The throughput is how fast we can actually send data through a network.

1. Throughput T bps is always less than Bandwidth B.

### □ Latency - Delay

The time period to sent a signal from source to destination. Its depend on

Propagation Time + Transmission Time + Queuing Time + Processing Time

$$\frac{\text{Distance}}{\text{Propagation Speed}} + \frac{\text{Message Size}}{\text{Bandwidth}} + \text{Message Time Hold}$$



### Throughput Example

A network with bandwidth of 8 Mbps can pass only an average of 10,000 frames per minute with each frame carrying an average of 8,000 bits. What is the throughput of this network?

**Formula: No. of Frames in per Minute x No. of Bits per Second**

$$\text{Throughput} = 10,000 \times 8,000$$

$$= 80,000,000 / 60$$

$$= 1,333,333.33 \text{ bit per second}$$

**Throughput = 1.33 Mbps**  
**Bandwidth = 8 Mbps**

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## Example Propagation Time

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be  $2.4 \times 10^8$  m/s in cable.

### SOLUTION

Propagation Time + Transmission Time + Queuing Time + Processing Time

$$\frac{\text{Distance}}{\text{Propagation Speed}} + \frac{\text{Message Size}}{\text{Bandwidth}} + \text{Message Time Hold}$$

**Propagation Time = Distance / Propagation Speed**

$$= 12,000 \times 10,000 / 2.4 \times 10^8$$

$$= 120,000,000 / 2.4 \times 100,000,000$$

$$= 120,000,000 / 240,000,000$$

$$= 0.5$$

$$= 50 \text{ m/s}$$

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## Exercise

### Question No. 1:

A network with bandwidth of 15 Mbps can pass only an average of 20,000 frames per minute with each frame carrying an average of 14,000 bits. What is the throughput of this network?

### Question No. 2:

What is the propagation time if the distance between the two points is 15,000 km? Assume the propagation speed to be  $3.4 \times 10^{10}$  m/s in cable.

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## Digital Transmission

There are two types of signals that can be transmitted. These transmitted signals are known as Digital Transmission. These two digital transmission are:

Digital to Digital Conversion      Analog to Digital Conversion

First of all we discussed Digital to Digital Conversion. Three methods are used for this conversion:

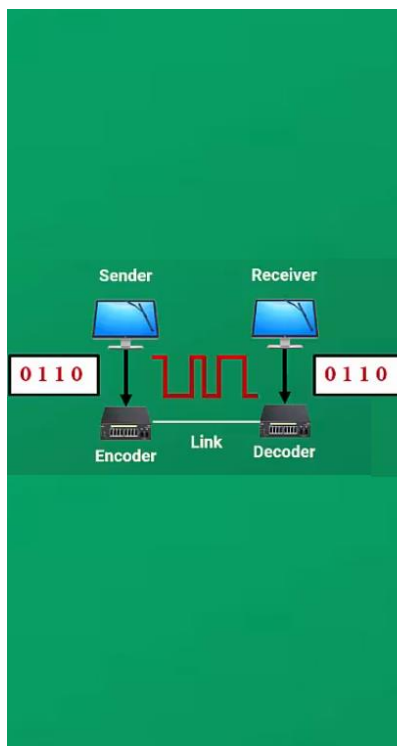
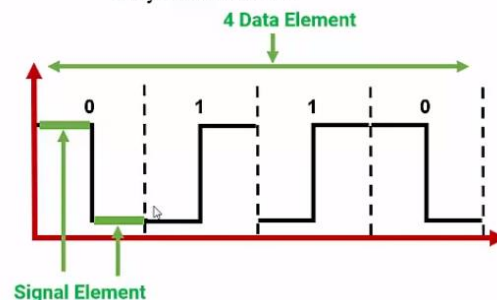
Line Coding  
Always Needed

Block Coding  
May NOT needed

Scrambling  
May NOT needed

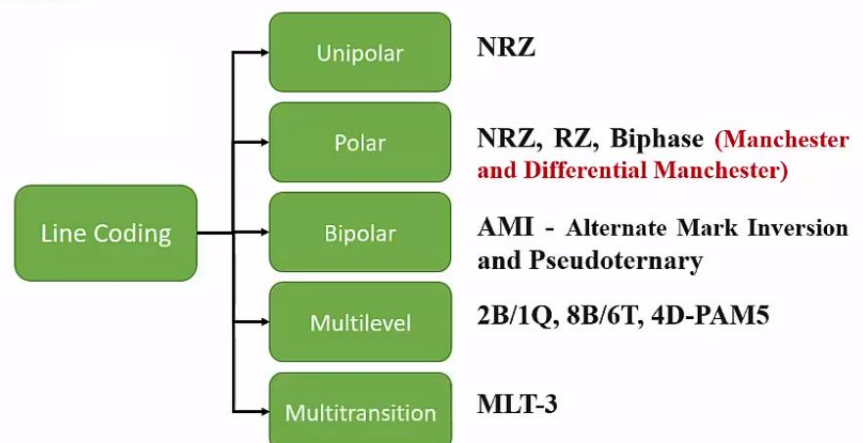
**Data Element:** A smallest unit which carry some piece of Information (Bits) → Carried

**Signal Element:** A shortest unit in terms of time of the digital signal. → Carriers



## Line Coding

Data can be Text, Numbers, Images, Audio or Video store into the computer in forms of bits in a sequence. In Line Coding these bits converted into Digital Signals. There are multiple techniques for line coding these are five in numbers.



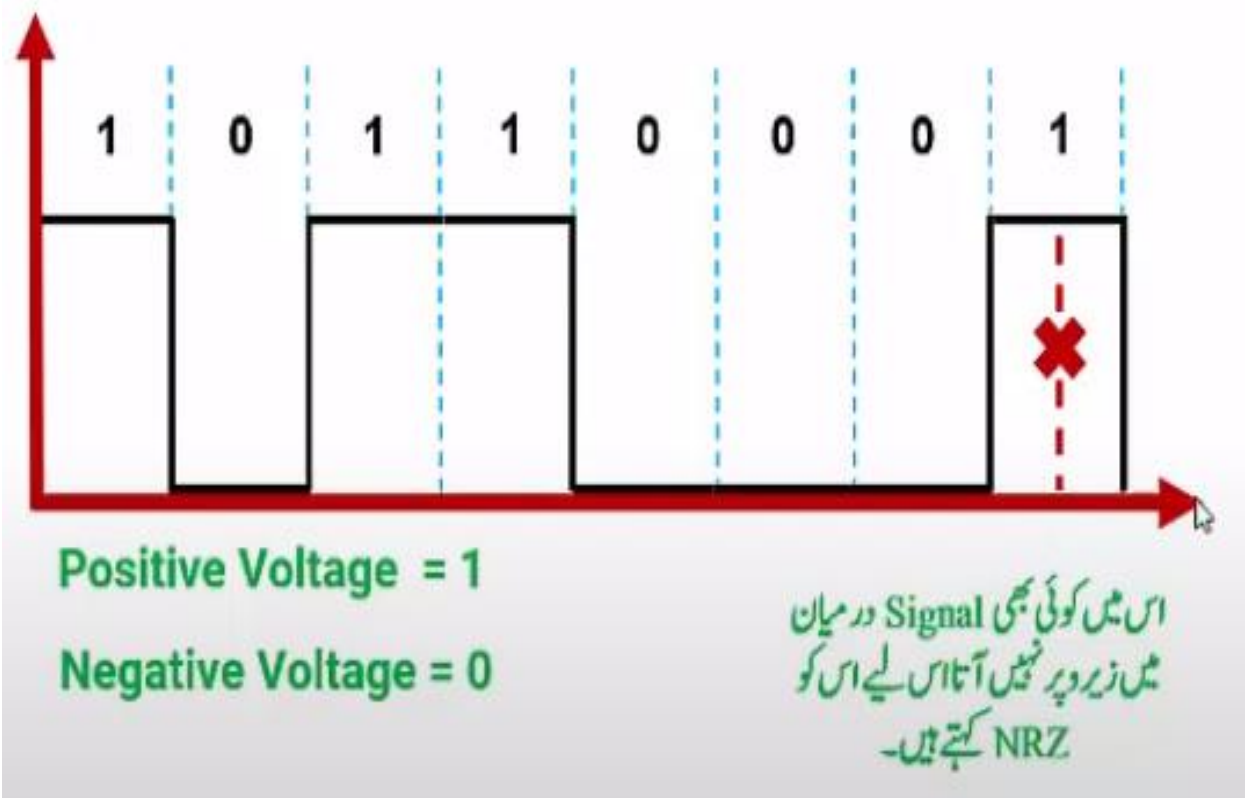
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# Unipolar - NRZ



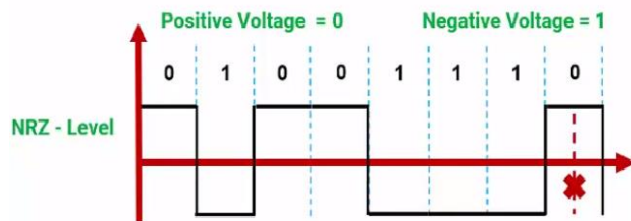
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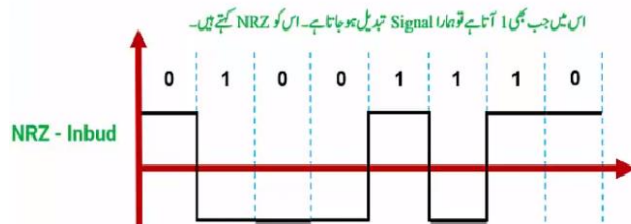


<https://www.youtube.com/watch?v=-0kr2sG8JUQ&list=PLAUocDaOe9K6nDHAXKa-7fHsIFAKBPE8q&index=9>

### Polar - NRZ

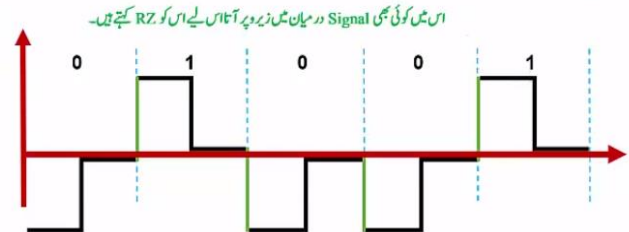


اس میں کوئی بھی Signal درمیان میں زیر و بر آتا اس لیے اس کو NRZ کہتے ہیں۔

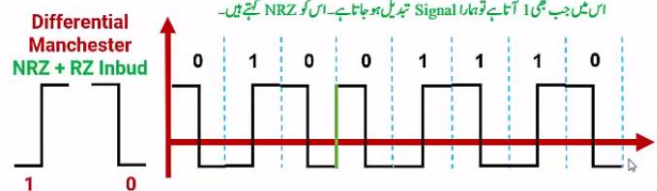
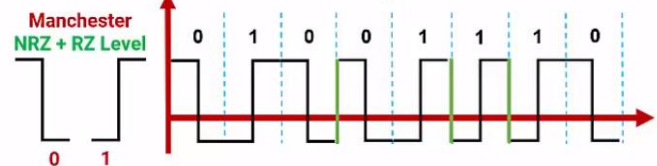


اس میں جب بھی 1 آتا ہے تو ہمارا Signal تبدیل ہو جاتا ہے۔ اس کو NRZ کہتے ہیں۔

### Polar - RZ



### Polar - Biphas



اس میں جب بھی 1 آتا ہے تو ہمارا Signal تبدیل ہو جاتا ہے۔ اس کو NRZ کہتے ہیں۔

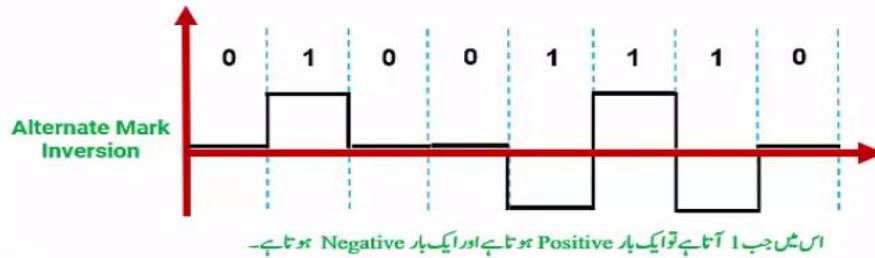
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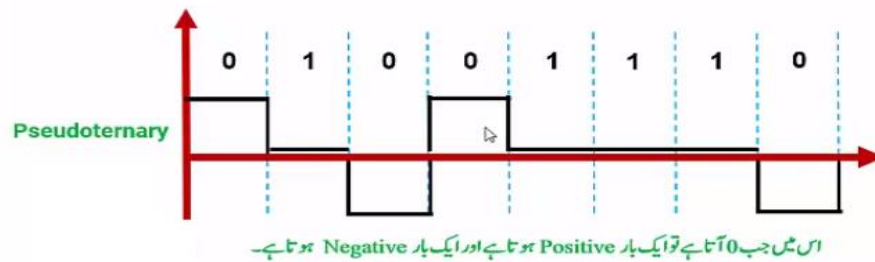


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## Bipolar - AMI



## Bipolar - Pseudoternary



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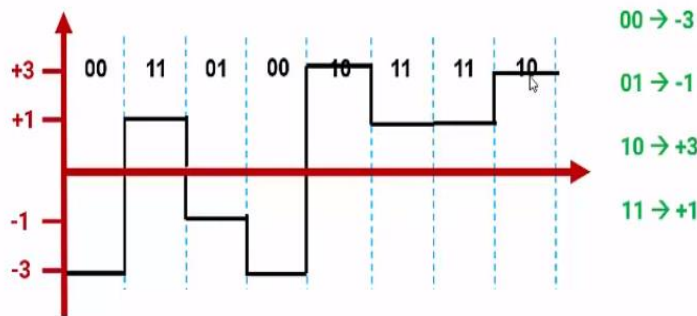


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## Multilevel - 2B/1Q (Two Binary and One Quaternary)

The objective of this method is to increase the data rate and decrease the required bandwidth.

اس Technique میں ڈیٹا کی 2 Bit ٹرانسمفر ہوتی ہیں۔



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# Today's Topic

## 1. Block Coding

- a) Division
- b) Substitution
- c) Combination

## 2. Scrambling

- a) Types of Scrambling
  - i) B8ZS
  - ii) High Density HDB3

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### Block Coding (Error Detection)

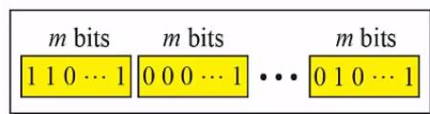
Block coding refers to the technique of adding extra bits to a digital word in order to improve the reliability of transmission.

When we are using Line Coding we not know where to start long ZEROs "0s" and where from start long Ones "1s". This problem is known as synchronization. To over the said problem we use second technique is known as Block Coding.

In this method we convert the " $m$ " bits into " $n$ " bits. These  $n$  bits is less than the  $m$   $n < m$ . Its tell about where is long series of ZEROs "0s" and where from start long series of Ones "1s". We add some extra bit with original bits known as redundancy bits.

Three Steps of Block Coding:

❑ Division



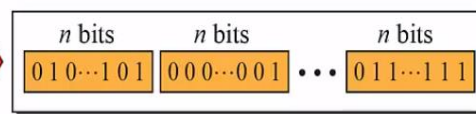
Division of a stream into  $m$ -bit groups

❑ Substitution

$mB$ -to- $nB$   
substitution



❑ Combination



Combining  $n$ -bit groups into a stream

Redundancy  
 $n > m$

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### Block Coding with Line Coding

We have select 4B / 5B scheme.

Example:

We need to send data at a 1 Mbps rate. What is the minimum required bandwidth, using a combination of 4B/5B and NRZ-I or Manchester coding?

Block Code Resolve:

Synchronization → ✓

Bandwidth  $\leftarrow$  Bandwidth  
NRZ-I Manchester ✓

DC Components → ✗

### SOLUTION

$$M = 4$$

$$N = 5$$

$$n > m$$

**Data Rate** 4 Bit = 1Mbps

5 Bit = 1.25 Mbps

↑ Signal Rate Grow Up

**NRZ-I** (Minimum Bandwidth is half of the Data Rate)

Formula =  $N / 2$

$$= 1.25 / 2 \text{ Mbps}$$

$$= 62.5 \text{ Mbps}$$

$$= 625 \text{ KHZ}$$

**Manchester** (1 MHZ)

**8B / 10B Scheme**

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## DC components

- When voltage level in a digital signal is constant for a while, spectrum creates very low frequencies (around zero), called **DC components**
- *Creates problems for a system that cannot pass low frequencies*
- a telephone line cannot pass frequencies below 200 Hz.

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## Scrambling

✓ In **Scrambling** we increase the **No. of Pulses** instead of **No. of Bits**.

Because in Block Coding when we increase the No. of Bits it will increase the data rate as well as it increase bandwidth. But in Scrambling when we increase Pulses it will increase data rate but not increase the bandwidth.

- ✓ Biphase schemes suitable for Dedicated Link (LAN) but not for Long Distance
- ✓ Block Coding + NRZ-I solves Synchronization issue but has DC Component
- ✓ Bipolar AMI has a narrow bandwidth (no DC Component) but synch issue (long series of 0s)

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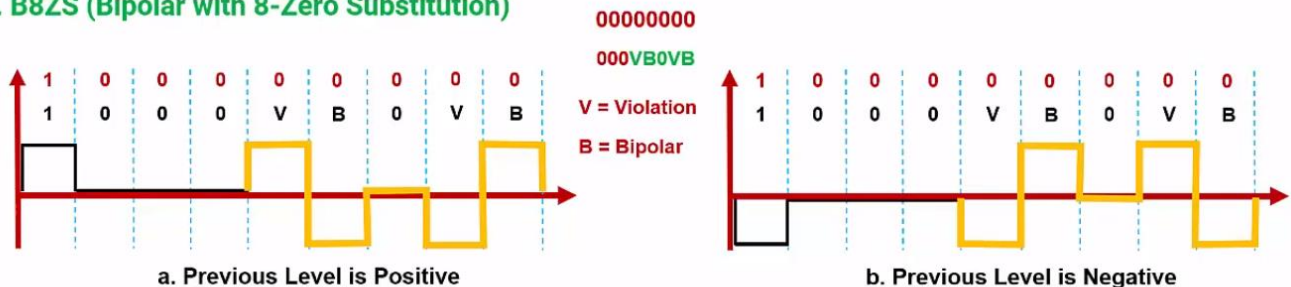
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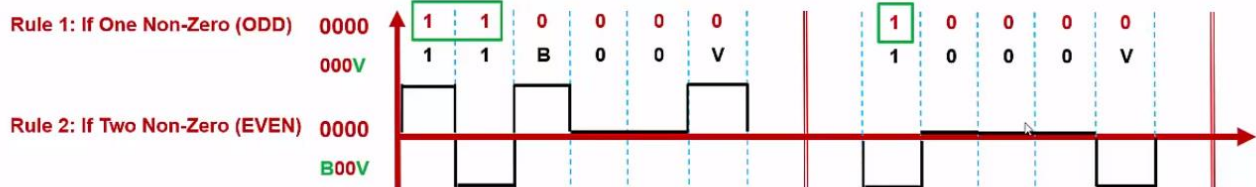
<https://www.youtube.com/watch?v=-0kr2sG8JUQ&list=PLAUocDaOe9K6nDHAXKa-7fHsIFAKBPE8q&index=9>

## Types of Scrambling Techniques

### 1. B8ZS (Bipolar with 8-Zero Substitution)



### 2. High Density HDB3 (Bipolar with 3-Zero Substitution)





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# Today's Topic

## Analog to Digital Conversion

### 1. (PCM) - Pulse Code Modulation

- i) Sampling
- ii) Quantization
- iii) Encoding

### 2. (DM) - Delta Modulation

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## Analog to Digital Conversion

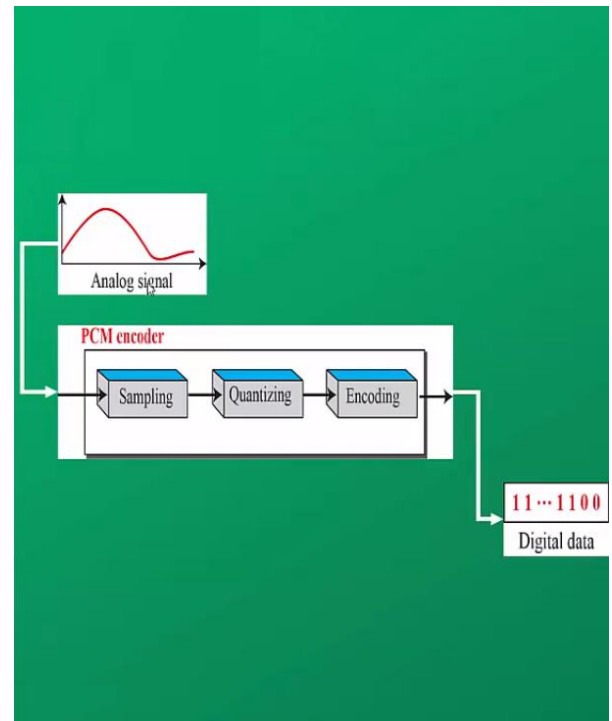
In this method we convert the analog signals to the digital signals. In this method first of all we convert the data into Digitization than convert it into Digital Signals. Its mean that two process are involved in it.

□ PCM consists of three steps to digitize an analog signal:

### 1. (PCM) - Pulse Code Modulation

- Sampling (Pulse Amplitude Modulation)
- Quantization
- Encoding

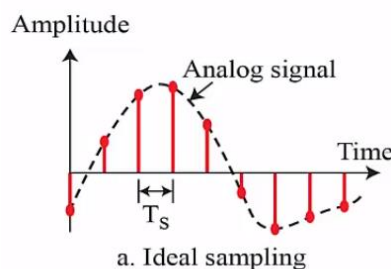
### 2. (DM) - Delta Modulation



## Sampling

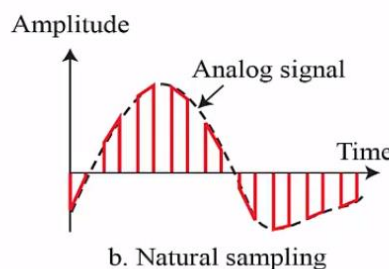
Series of pulses with amplitude values between min and max signal amplitude.

## Three Different Sampling Methods for PCM

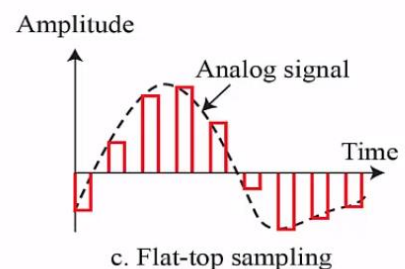


$T_s \rightarrow$  Sampled Interval or Sample Period

It has different pluses at different Amplitude. It is very difficult to deploy in Hardware.



It has high speed Switch. It has very small time interval



It is also called sample and hold technique.

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Sample is a piece of data taken from the whole data which is continuous in the time domain. When a source generates an analog signal and if that has to be digitized, having 1s and 0s i.e., High or Low, the signal has to be discretized in time. This discretization of analog signal is called as Sampling.

نمونہ پورے ڈیٹا سے لیا گیا ڈیٹا کا ایک ٹکڑا ہے جو ٹائم ڈومین میں مسلسل ہوتا ہے۔ جب کوئی ذریعہ ایک یعنی ہائی یا لو ہو، تو سگنل کو 1 اور 0 یا نیلا لگ سگنل تیار کرتا ہے اور اگر اسے ڈیجیٹائز کرنا ہو، جس میں 1 وقت کے ساتھ الگ کرنا ہوگا۔ نیلا لگ سگنل کی اس ڈسکریٹائزیشن کو سیمپلنگ کہا جاتا ہے۔

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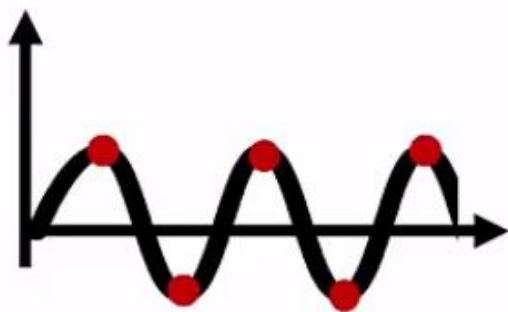
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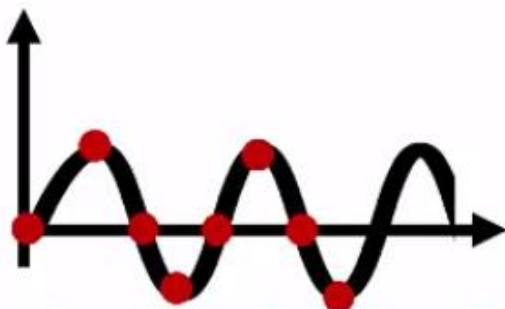
<https://www.youtube.com/watch?v=-0kr2sG8JUQ&list=PLAUocDaOe9K6nDHAXKa-7fHsIFAKBPE8q&index=9>

# Nyquist Sampling Rate

Nyquist  $\rightarrow f_s$  (Frequency of Sampling) =  $2f$  (Twice Highest Frequency)



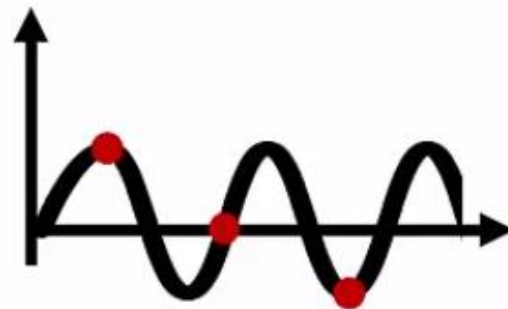
**Ideal Sampling  $f_s = 2f$**



**Over Sampling  $f_s = 4f$**

**(Double Nyquist Rate)**

Extra on Line Sample has no Value



**Under Sampling  $f_s = f$**

**(Half Nyquist Rate)**

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## Quantization

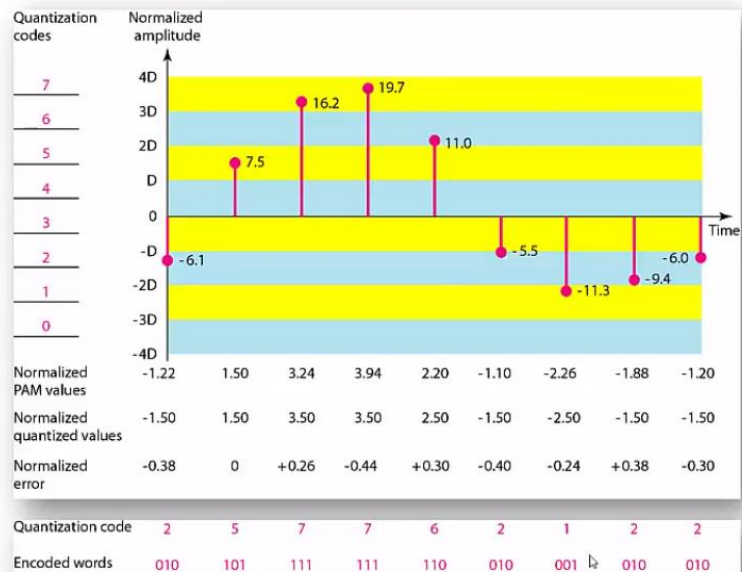
Quantization is normalization of sample data according to certain level. The no. of level chose on two factors.

-- Range of Amplitude

-- Accuracy (If we acquire higher accuracy than we need more Quantization Level and Voice Versa).

## Encoding

The digitization of the analog signal is done by the encoder. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n bit code. Encoding also minimizes the bandwidth used.





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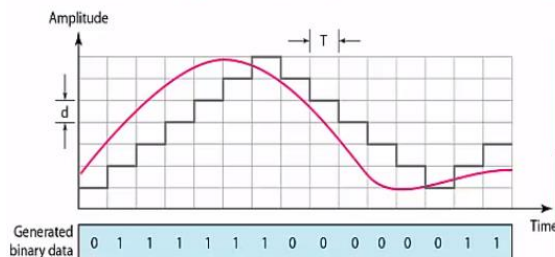


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## Delta Modulation

This scheme sends only the difference between pulses, if the pulse at time  $t_{n+1}$  is higher in amplitude value than the pulse at time  $t_n$ ,

- ☐ Then a single bit, say a “1”, is used to indicate the positive value.
- ☐ If the pulse is lower in value, resulting in a negative value, a “0” is used.
- ☐ This scheme works well for small changes in signal values between samples.
- ☐ If changes in amplitude are large, this will result in large errors.



## Delta Modulation

- ☐ Delta modulation is a simpler technique
- ☐ DM finds the change from the previous sample
- ☐ No code words

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### 1. Transmission Mode

a) Parallel Transmission

b) Serial Transmission

i) Asynchronous Transmission

ii) Synchronous Transmission

iii) Isochronous Transmission

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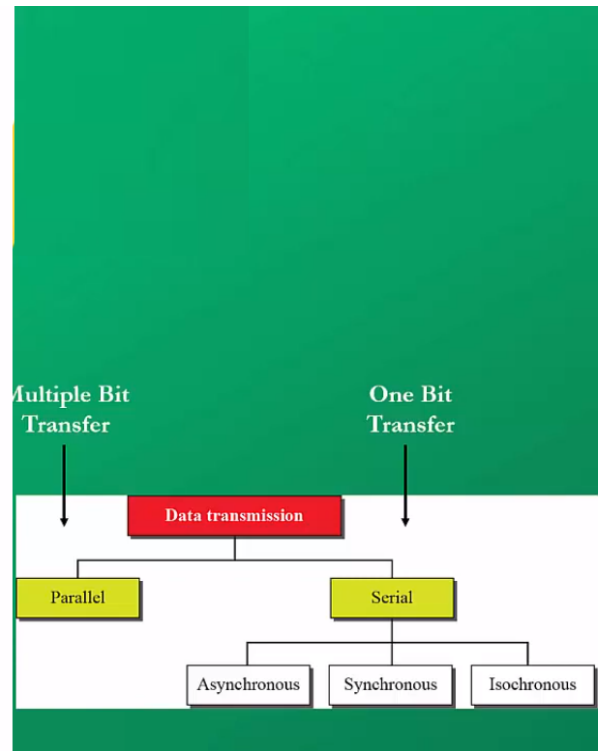
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## Transmission Mode

We have convert the data into digital data. Now, the process is to transmission that from source to destination. This data is transmitted through Transmission Mode. One problem is arises before transmitting data that is “WIRING”. We have used different types of wires to transmit the data. Its depends on no. of bits to transmit. We have one bit or multiple bits to transmit.

□ We have two method to transmit data.

1. Parallel Transmission
2. Serial Transmission

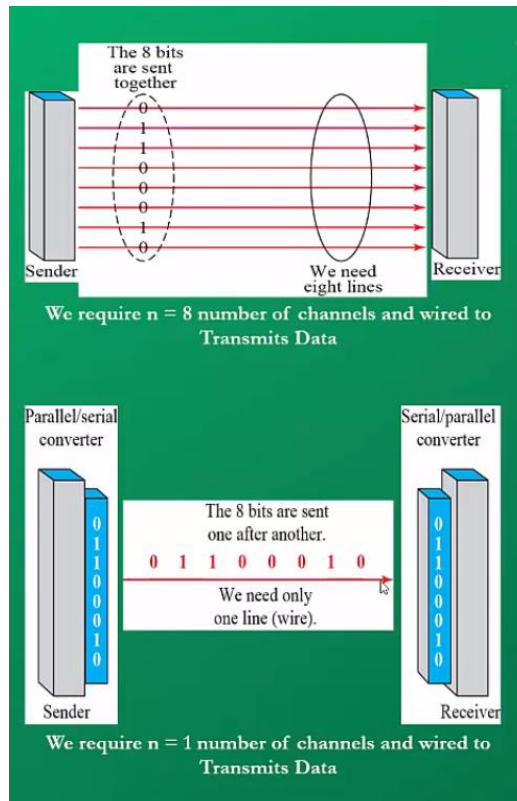


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## Parallel Transmission

In this type of Transmission Mode we have Transmit Multiple Bit to the destination at the same time.

**Advantages** → Speed (Data Transmit Quickly)

**Disadvantages** → Cost (We Require  $n[8]$  Wire)

## Serial Transmission

In this type of Transmission Mode we have Transmit Only One Bit to the destination.

**Advantages** → Cost (We Require  $n[1]$  Wire)

**Disadvantages** → Speed (Data Transmit Slowly)

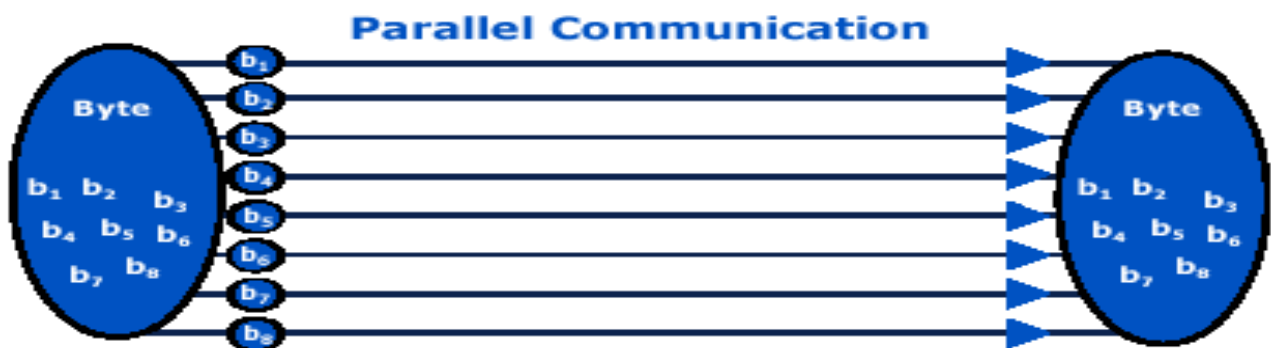
**Serial Transmission:** In this type of transmission a single link is used to transmit data bits and only one bit is transferred at a time. **Parallel Transmission:** In this transmission, data bits are transmitted simultaneously through multiple links, which are placed parallel to each other.

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VS





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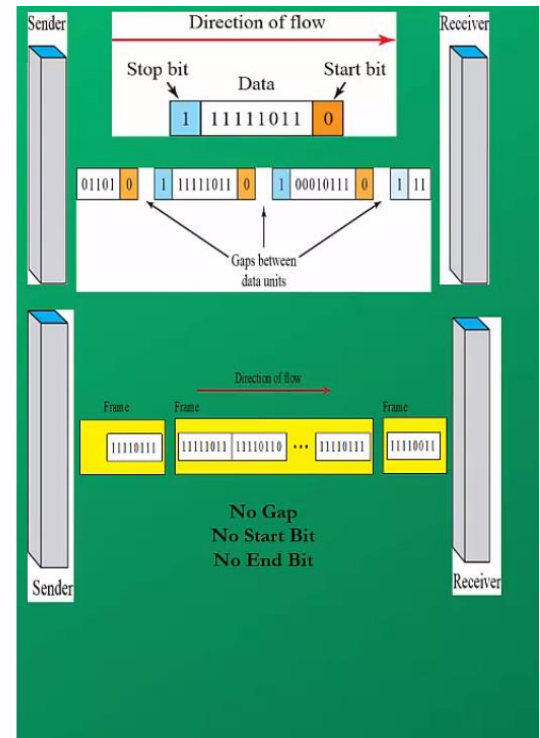
### Asynchronous Transmission

In this type of Transmission Mode we have Transmit Bits on a specific pattern.

**Advantages** → Cost (Cheap)

**Disadvantages** → Speed (Slow)

**Example** → Keyboard input to PC



### Synchronous Transmission

In this type of Transmission Mode we have Transmit data into large units called frame.

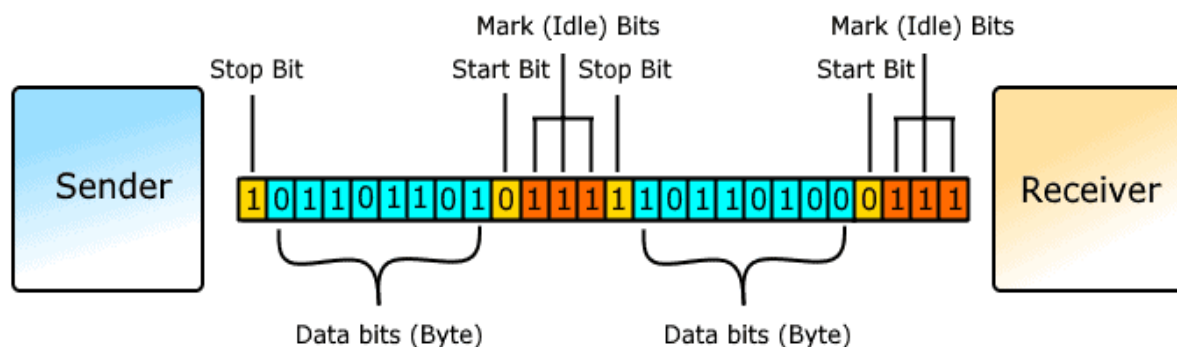
**Advantages** → Speed (No Gap)

**Example** → High Speed Application

### Isochronous Transmission

In this type of Transmission Mode we have synchronized Entire bit of stream and then Transmit it.

The **Isochronous Transmission** Format is also known as **START-STOP mode** or **CHARACTER mode**. Each character or byte is framed as a separate and independent unit of DATA that may be transmitted and received at irregular and independent time intervals.



### Asynchronous Transmission

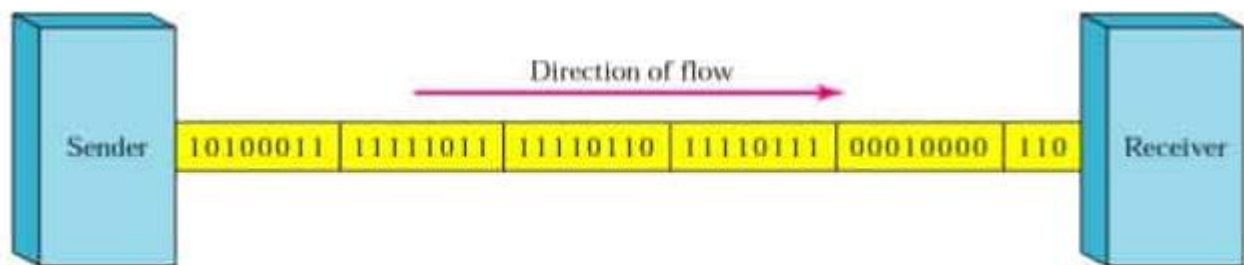
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## Synchronous Transmission



| Synchronous Transmission                                         | Asynchronous Transmission               |
|------------------------------------------------------------------|-----------------------------------------|
| Transmits data in the form of chunks or frames or blocks         | Transmits 1 byte or character at a time |
| Transmission speed is faster                                     | Transmission speed is slower            |
| Cost is expensive                                                | Cost is economical                      |
| Time interval is constant                                        | Time interval is random                 |
| There are gaps between the data                                  | There are no gaps between the data      |
| Used in Chat Rooms, Telephonic Conversations, Video Conferencing | Used in Email, Forums, Letters          |

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# Today's Topic

## 1. Transmission Mode of Analog Data

a) Amplitude Shift Keying(ASK)

b) Frequency Shift Keying(FSK)

c) Phase Shift Keying (PSK)

i) Bit Rate

ii) Data Rate

iii) Baud Rate

iv) Signal Rate

v) Carrier Signal

vi) No. of Data Element

vii) High Frequency Signals

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## Digital-To-Analog Conversion

Digital-To-Analog Conversion (DAC), **Process by which digital signals (which have a binary state) are converted to analog signals** (which theoretically have an infinite number of states). For example, a modem converts computer digital data to analog audio-frequency signals that can be transmitted over telephone lines.

A sine wave is defined by 3 characteristics:

Amplitude

Frequency

Phase

**How do your data convert from digital-to-analog?**

**There are three kinds of digital-to-analog conversions:**

1. Amplitude Shift Keying. In this conversion technique, the amplitude of analog carrier signal is modified to reflect binary data. ...
2. Frequency Shift Keying. ...
3. Phase Shift Keying. ...
4. Quadrature Phase Shift Keying.

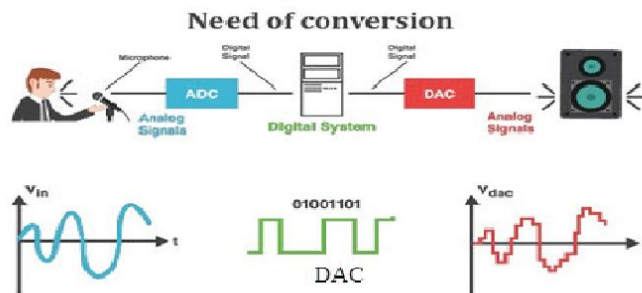
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### Digital to Analog Converter (DAC) and Its Applications

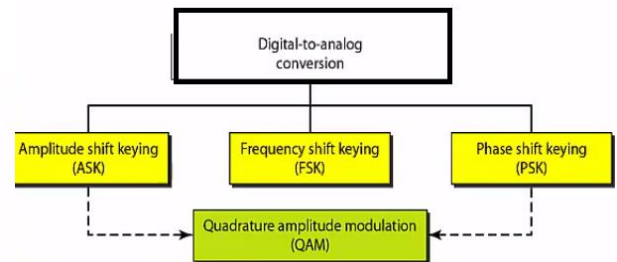
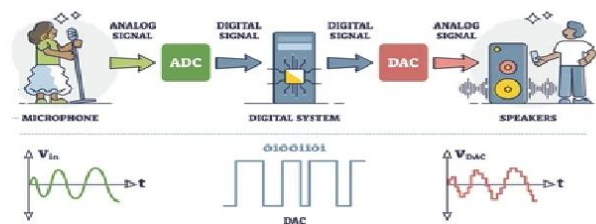


□ We have two method to transmit data.

1. Amplitude Shift Keying(ASK)
2. Frequency Shift Keying(FSK)
3. Phase Shift Keying (PSK)

NOTE: If one technique vary other remains constant.

### DIGITAL TO ANALOG CONVERTER (DAC) AND ITS APPLICATIONS



In addition, there is a fourth and better mechanism that combines changes in both amplitude and phase called Quadrature Amplitude Modulation(QAM)



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# DAC

## DIGITAL TO ANALOG CONVERTER

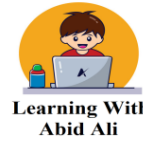
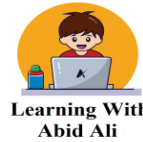
Types, Working, Block Diagram & Applications



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An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate?

### SOLUTION

Formula:  $S = N / r$

$S = 1000$     $R = 4$     $N = ?$

$S = N / r$     $N = r \times S$     $N = 4 \times 1000$    **4000 pbs**

An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

### SOLUTION

Formula:  $S = N / r$

$N = 8000$     $S = 1000$     $L = ?$     $r = ?$

$S = N / r$     $r = N / S$     $r = 8000 / 1000$    **8 bits**

$L = 2^r$     $L = 2^8$    **256 bits**

## Digital to Analog Conversion

In conversion from digital to analog signals we must review two issues.

1. **Bit Rate** → **Data Rate**  
No of Bits Transmitted in One Second
2. **Baud Rate** → **Signal Rate**  
No of Signal Units Per Second

Baud Rate is less than Bit Rate

## Calculation of Bit & Baud Rate

$(S = N / r)$  **Signal Rate** = (Data Rate / No. of Data Element)

$R = \log_2 L$

$L$  = No. of Different Signal Elements

**Bandwidth Required** → Its depend on Signal Rate as more signal required more bandwidth & vice versa. **FSK**

### 3. The Carrier Signal →

When we transmit the signals from sender to receiver. Than we transmit **High Frequency Signals**, these are called **Carrier Signal**.

## Exercise

### Example # 01

An analog signal carries 8 bits per signal element. If 2000 signal elements are sent per second, find the bit rate?

### Example # 02

An analog signal has a bit rate of 16000 bps and a baud rate of 2000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

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### Amplitude Shift Keying (ASK)

In this technique the amplitude is vary and it vary with the carrier signals as the carrier signal change the amplitude changed.

- ❑ This technique is also known as (OOK - On Off Key) or Binary Technique (0's and 1's)
- ❑ When Amplitude change than Frequency and Phase remain (Constant).

**Example:** We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with  $d = 1$ ?

**Bandwidth Requirements**  
**Formula:  $B = (1+d) S$**   
 (d is always between 0 and 1)  
 $d = 1 = \text{Maximum Bandwidth}$   
 $d = 0 = \text{Minimum Bandwidth}$

**Amplitude**

Same Amplitude as Carrier Signal      1 = Maintain Amplitude      0 = Zero Amplitude

Amplitude ↑      Bit rate: 5      Time →

1 signal element   1 signal element   1 signal element   1 signal element   1 signal element

1 s      Baud: 5

**SOLUTION:**     $B = 100$      $R = D = 1$      $f_c = 250$      $N = ?$

**Formula:**     $B = (1+d)S$        $S = N/r$

$= (1+1) S$        $B = 2(N/r)$        $B = 2(N/1)$   
 $= 2S$        $B = 2N$        $N = B/2$   
 $B = 100/2$        **$N = 50 \text{ kps}$**

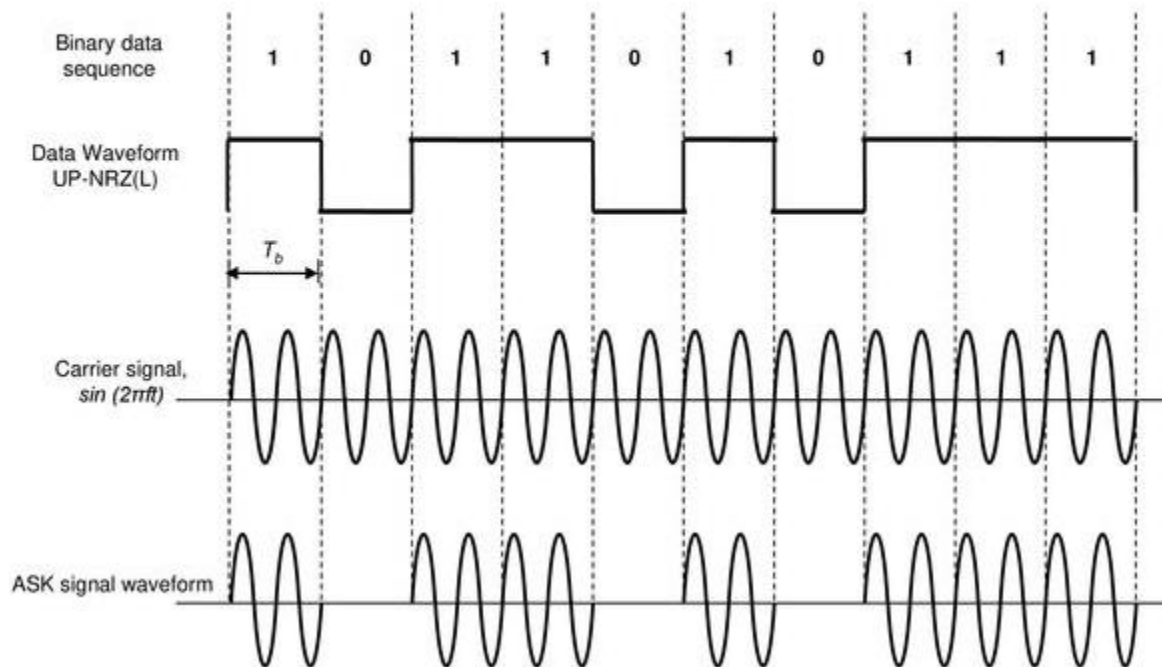
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### 3.1.1 Amplitude Shift Keying (ASK)



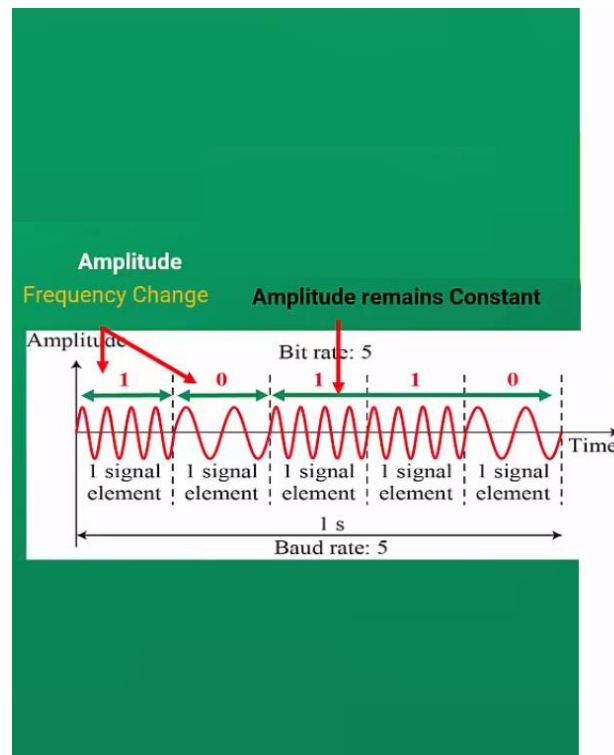


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### Frequency Shift Keying (FSK)

In this technique in which the frequency vary (change) but the Amplitude and Phase remain constant.

❑ Formula:  $B = (1+d)S + 2\Delta f$

**Example:** We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with  $d = 1$ ?

**SOLUTION:**  $B = 100$     $R = D = 1$     $\Delta f = 50$     $N = ?$

$$B = (1+d)S + 2\Delta f \quad 100 = 2S + 50$$

$$B = (1+1)S + 50 \quad 100 - 50 = 2S$$

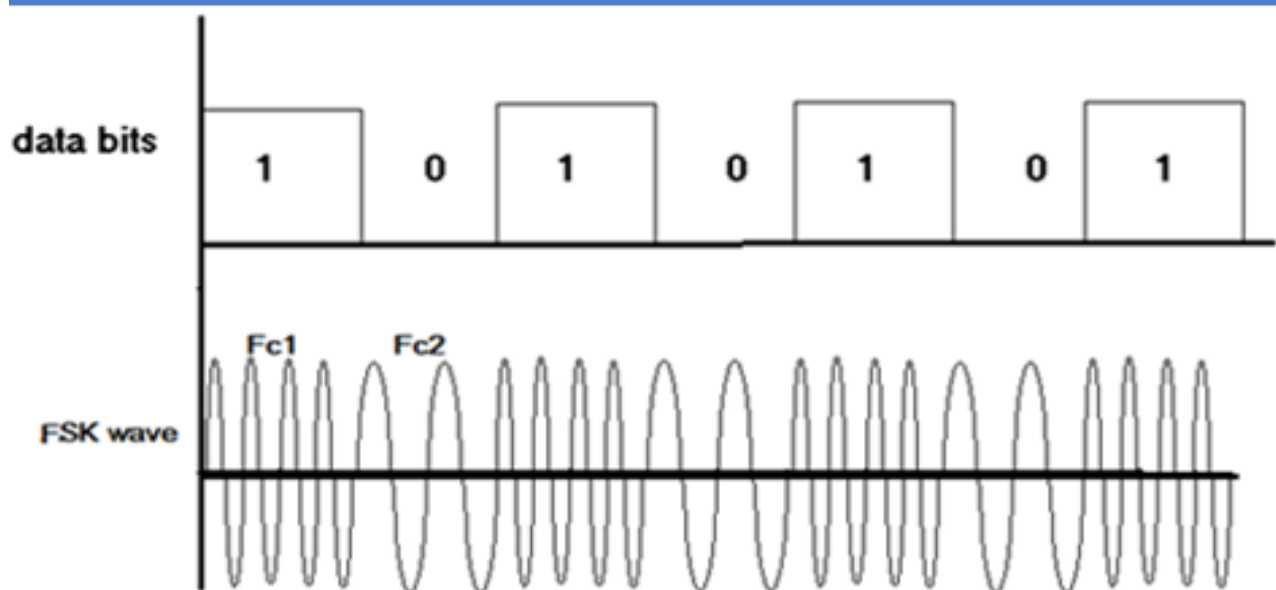
$$B = 2S + 50 \quad 2S = 50$$

$$100 = 2S + 50 \quad S = 25 \text{ kHz}$$

Activate Windows  
Go to Settings to activate Windows.

**N = 25 kHz**  
Because Data Rate = 1

### FSK Modulation and Demodulation







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### Phase Shift Keying (PSK)

In this technique in which the Phase vary (change), but the Amplitude and Frequency remain constant.

PSK is relatively common than ASK or FSK

**Example:** Find the bandwidth for a signal transmitting at 12 Mbps for QPSK. The value of  $d = 0$ .

#### SOLUTION:

$$N = 12 \text{ Mbps} \quad R = 2 = 1 \quad S = ?$$

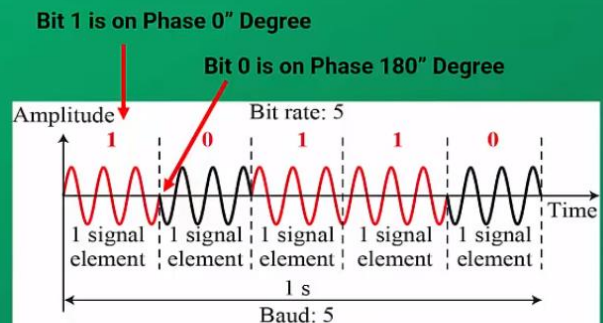
$$S = N/r \\ = 12/2$$

$$S = 6 \text{ Mbps}$$

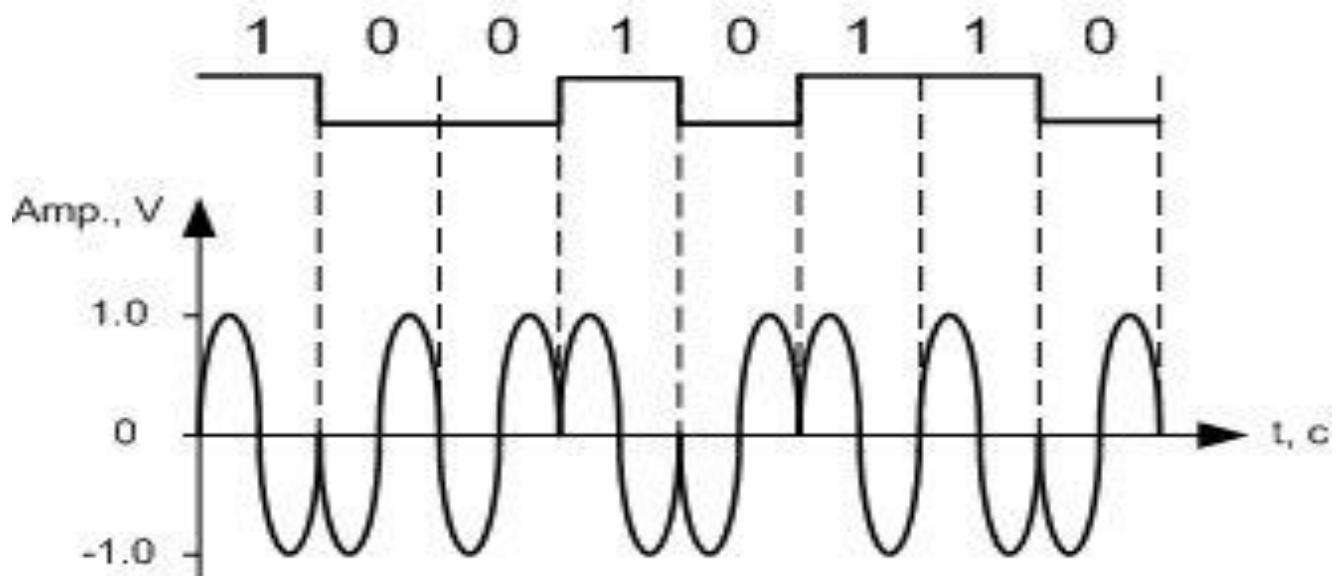
#### Bandwidth Requirements

Formula:  $B = (1+d) S$

(d is always between 0 and 1)



🔍🔍🔍🔍🔍



### Phase shift keying (PSK)

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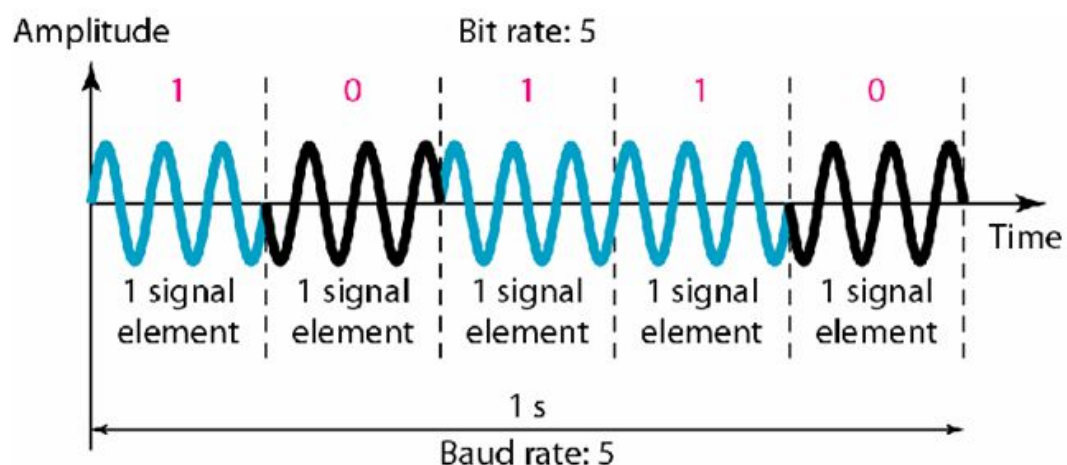
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## Phase Shift Keying (PSK)

- In PSK, the phase of the carrier is shifted to represent data.
- Two-Level PSK(Binary PSK) - BPSK
  - Uses two phases (0 and  $180^\circ$ ) to represent the two binary digits.
  - The resulting transmitted signal for one bit time is:



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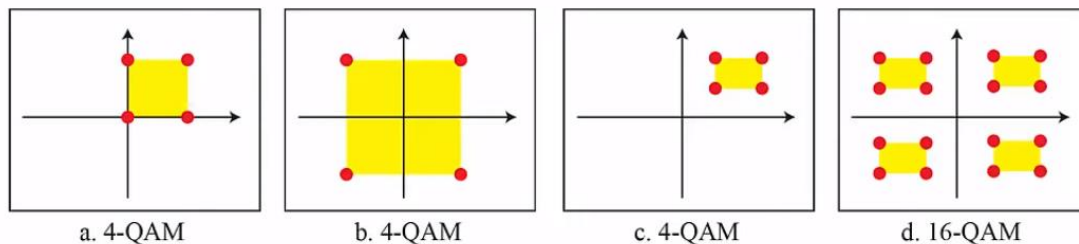


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## (QAM) Quadrature Amplitude Modulation

We have studied about ASK, FSK, and PSK. In these three techniques the **PSK is most effective** technique.

We have **combine two ASK and PSK** to make more effective that become QAM. It has **four variation**.



### Bandwidth Requirements:

The QAM requires Bandwidth as ASK and PSK requires.

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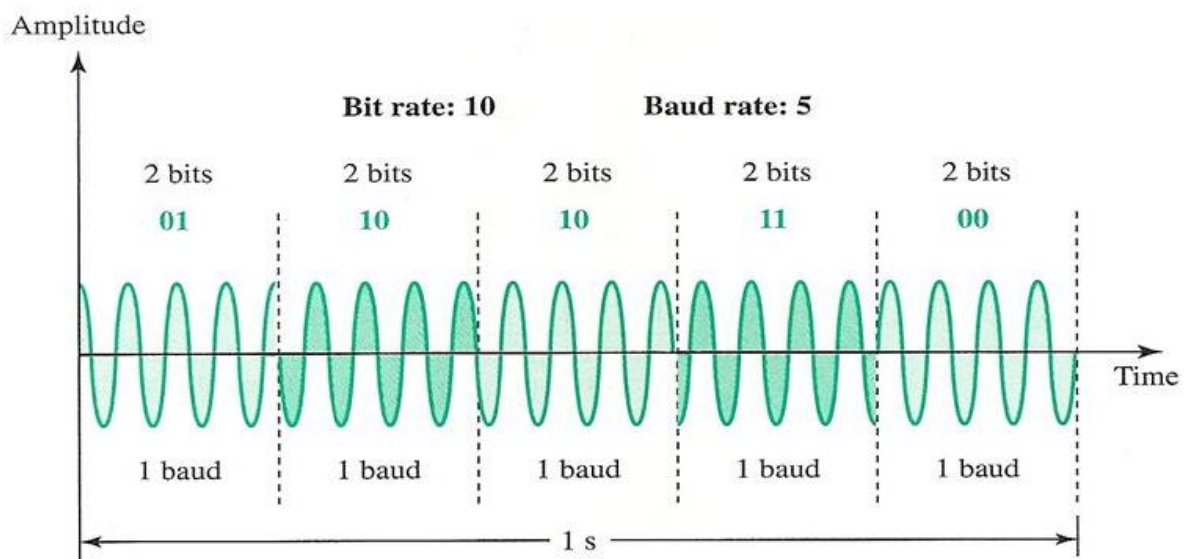
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## Quadrature PSK (QPSK)

- The term “quadrature” implies that there are four possible phases (4-PSK) which the carrier can have at a given time.
- The pair of bits represented by each phase is called **dibit**.
- The rate of change (baud) in this signal determines the signal bandwidth.
- BUT the throughput or bit rate for QPSK is twice the baud rate.





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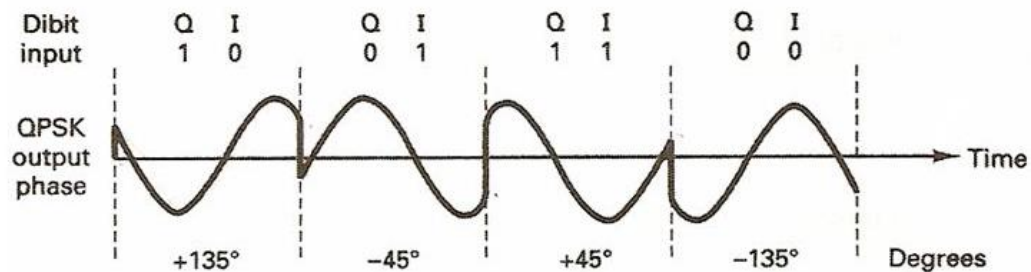
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## QPSK = 4-PSK

PSK that uses phase shifts of  $90^\circ = \pi/2$  rad  $\Rightarrow$  4 different signals are generated, each representing 2 bits



Assumption

| Binary input |   | QPSK output phase |
|--------------|---|-------------------|
| Q            | I |                   |
| 0            | 0 | -135°             |
| 0            | 1 | -45°              |
| 1            | 0 | +135°             |
| 1            | 1 | +45°              |



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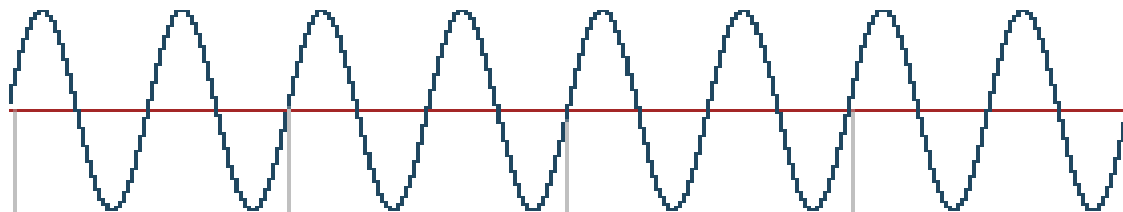
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### QPSK

Carrier / Channel



Modulating value from two bits.

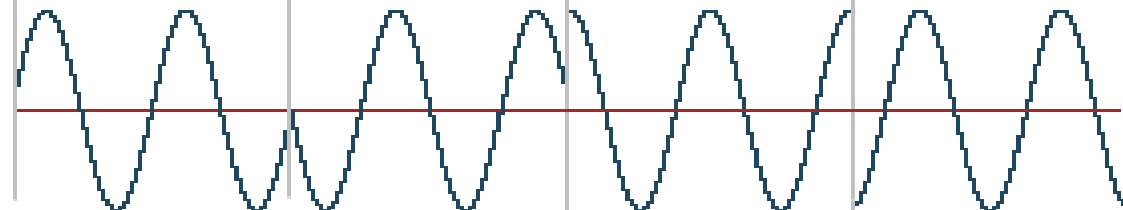
0  
(00)

2  
(10)

1  
(01)

3  
(11)

Modulated Result



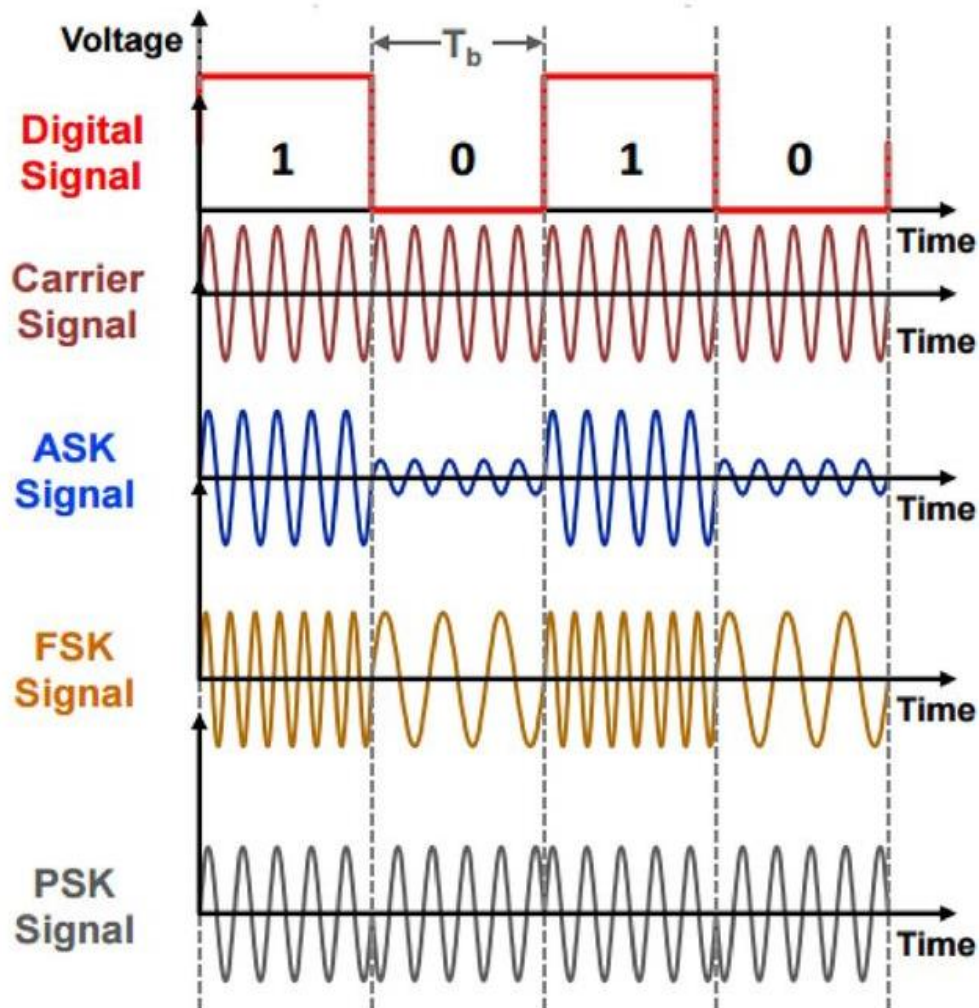
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## Comparison between ASK, FSK and PSK



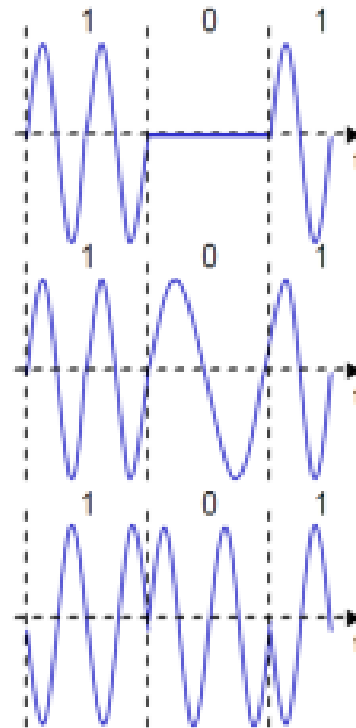
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- **Amplitude Shift Keying (ASK):**
  - change amplitude with each symbol
  - frequency constant
  - low bandwidth requirements
  - very susceptible to interference
- **Frequency Shift Keying (FSK):**
  - change frequency with each symbol
  - needs larger bandwidth
- **Phase Shift Keying (PSK):**
  - Change phase with each symbol
  - More complex
  - robust against interference



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CS601 Lecture No 18 Topic 86 to 89

### Analog to Analog Conversion

Analog-to-analog conversion, or analog modulation, is the representation of analog information by an analog signal. Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us.

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**An example is radio.** The government assigns a narrow bandwidth to each radio station. The analog signal produced by each station is a low-pass signal, all in the same range. To be able to listen to different stations, the low-pass signals need to be shifted, each to a different range.

## Analog To Analog Conversion



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**Que.** The example of an analog to analog conversion is

- a. radio
- b. Video
- c. Television
- d. internet

**Answer:** radio

**What are the 3 ways of conversion analog to analog?**

**Analog to Analog conversion can be done in three ways:**

- Amplitude Modulation.
- Frequency Modulation.
- Phase Modulation.

**What is difference between modulated Signal Carrier Signal and modulating signal?**

We take the help of such high frequency signal which is called as a carrier signal to transmit our message signal. Such a process is simply called as Modulation. Modulation is the process of changing the parameters of the carrier signal, in accordance with the instantaneous values of the modulating signal.



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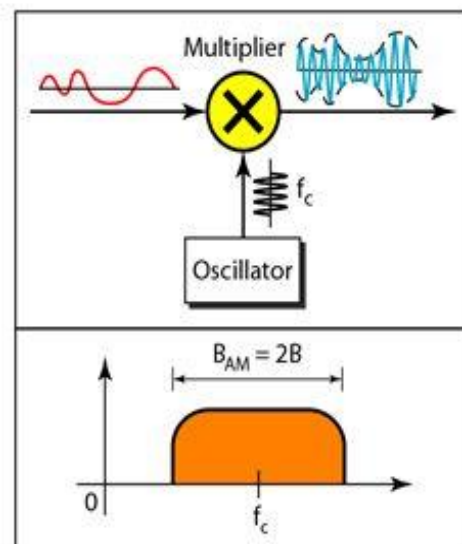
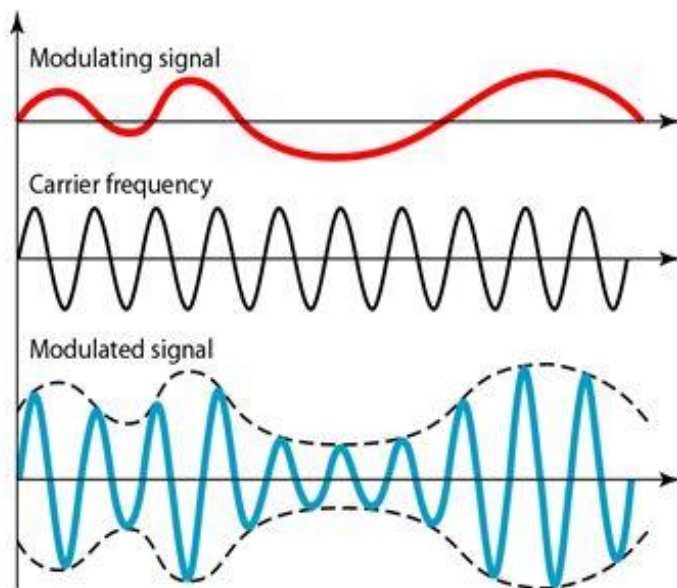
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### 1. Amplitude Modulation:

In AM transmission, the carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal. The frequency and phase of the carrier remain the same. Only the amplitude changes to follow variations in the information. The following figure shows how this concept works. The modulating signal is the envelope of the carrier.



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## AMPLITUDE MODULATION

- ▶ Band width of AM signal = twice the bandwidth of modulating signal.
- ▶  $B_{AM} = 2 B$
- ▶ If the modulation signal frequency increases, the amplitude of the carrier changes at a greater rate.

AM is normally implemented by using a simple multiplier because the amplitude of the carrier signal needs to be changed according to the amplitude of the modulating signal.

## 2. Frequency Modulation

In FM transmission, the frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and phase of the carrier signal remain constant, but as the amplitude of the information signal changes, the frequency of the carrier changes correspondingly.

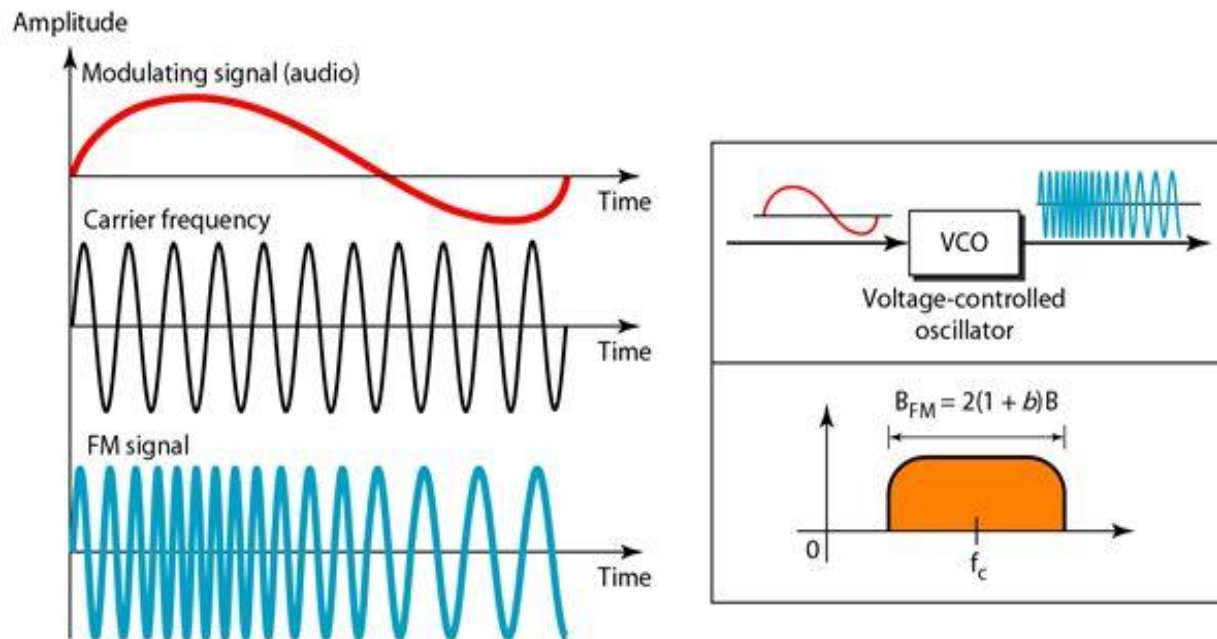
The following figure shows the relationships of the modulating signal, the carrier signal, and the resultant FM signal. FM is normally implemented by using a voltage-controlled oscillator as with FSK. The frequency of the oscillator changes according to the input voltage which is the amplitude of the modulating signal.

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### 3. Phase Modulation:

In PM transmission, the phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and frequency of the carrier signal remain constant, but as the amplitude of the information signal changes, the phase of the carrier changes correspondingly. It is proved mathematically that PM is the same as FM with one difference.

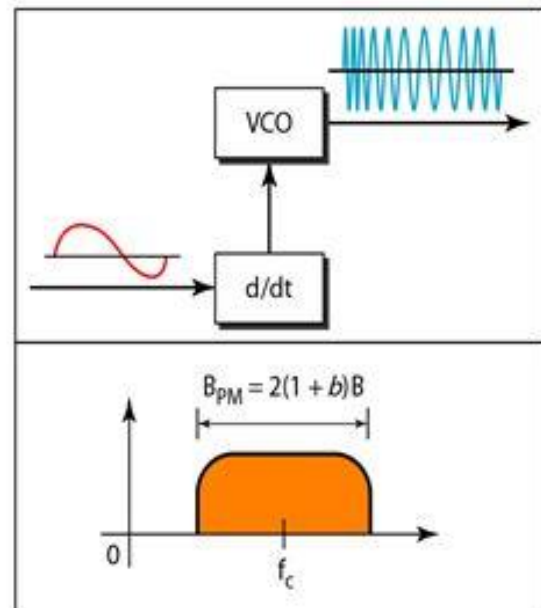
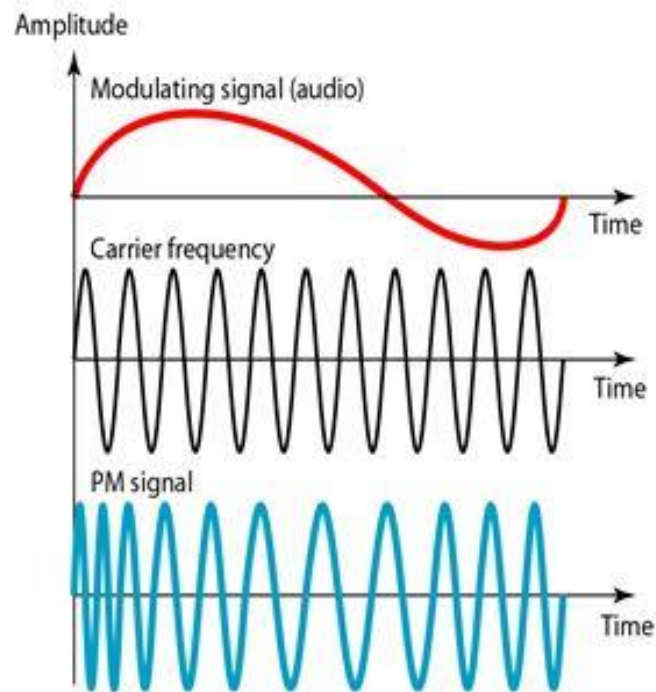
In FM, the instantaneous change in the carrier frequency is proportional to the amplitude of the modulating signal; in PM the instantaneous change in the carrier frequency is proportional to the derivative of the amplitude of the modulating signal. The following figure shows the relationships of the modulating signal, the carrier signal, and the resultant PM signal.

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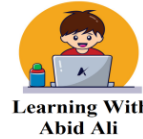


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## CS601 Lecture No 19 Topic 89 to 100

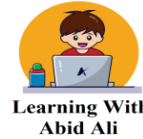
### Multiplexing

Simultaneous transmission of multiple signals across a single data link **MCQS**

In telecommunications and computer networking, multiplexing (sometimes contracted to muxing) is a method by which multiple analog or digital signals are combined into one signal over a shared medium.

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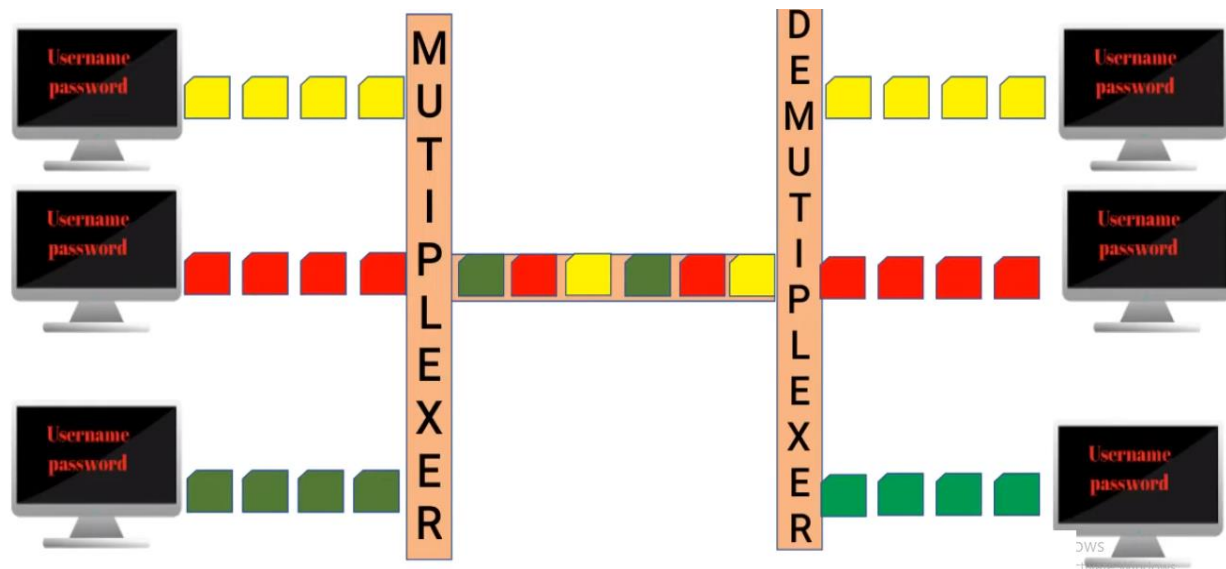
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The aim is to share a scarce resource. For example, in telecommunications, several telephone calls may be carried using one wire. Multiplexing originated in telegraphy in the 1870s, and is now widely applied in communications. In telephony, George Owen Squier is credited with the development of telephone carrier multiplexing in 1910.

The multiplexed signal is transmitted over a communication channel such as a cable. The multiplexing divides the capacity of the communication channel into several logical channels, one for each message signal or data stream to be transferred. A reverse process, known as demultiplexing, extracts the original channels on the receiver end.

A device that performs the multiplexing is called a multiplexer (MUX), and a device that performs the reverse process is called a demultiplexer (DEMUX or DMX).

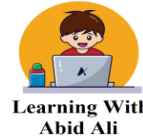
Inverse multiplexing (IMUX) has the opposite aim as multiplexing, namely to break one data stream into several streams, transfer them simultaneously over several communication channels, and recreate the original data stream.



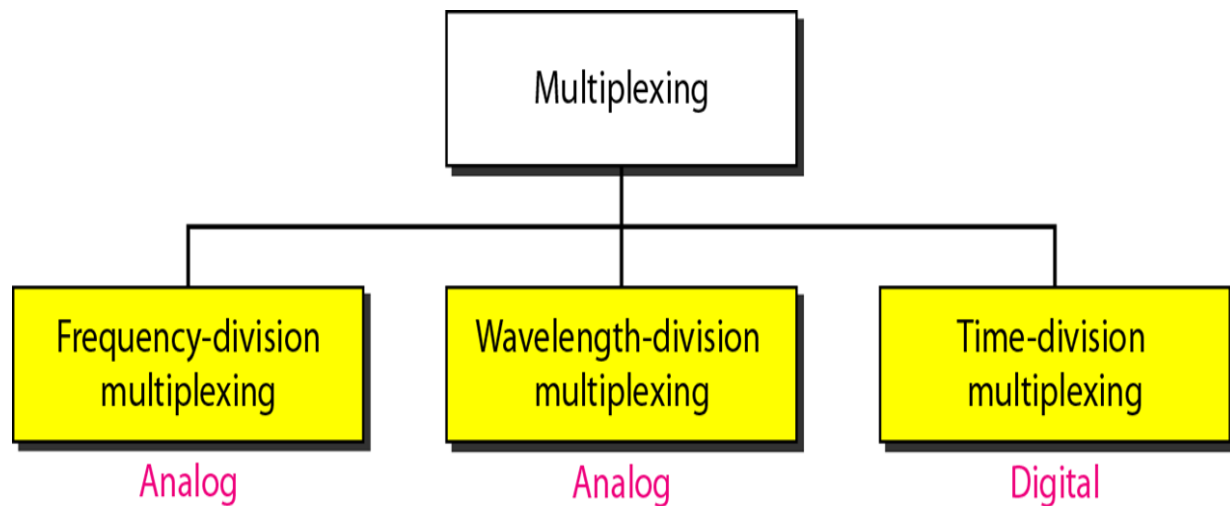


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## Overview

### TDM

Different users' digital signals over one link by dividing time into slots or intervals and assigning them equally among these users

### STDM

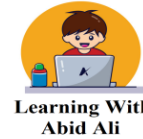
Analyzes statistics related to the workload, need or priority and determines on-the-fly how much time each user should get

### FDM

Divides the bandwidth of the communication line so that each terminal can transmit data. Individual terminals are separated from guard bands to ensure the signals do not interfere with each other

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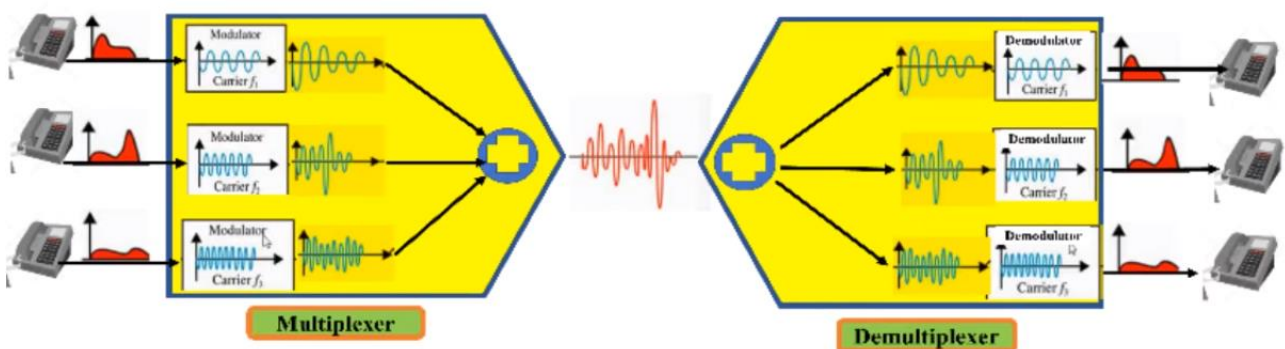
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## Frequency-Division Multiplexing

An analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.

### Frequency-division multiplexing

is an analog technology that supports the transmission of multiple communications over a single carrier by shifting each communication to a unique frequency range. A related technology for fiber channels is known as wavelength division multiplexing. This technology requires that each communication be transmitted as unique wavelength (i.e. color) of light along the shared fiber optic channel.



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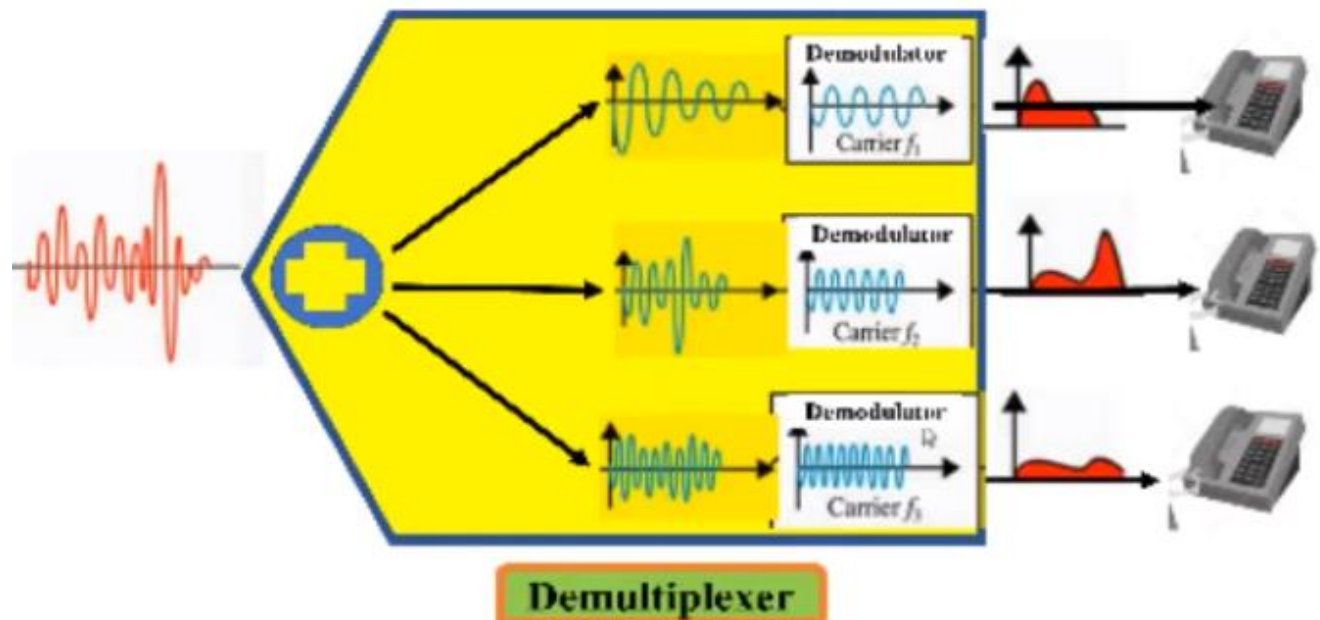
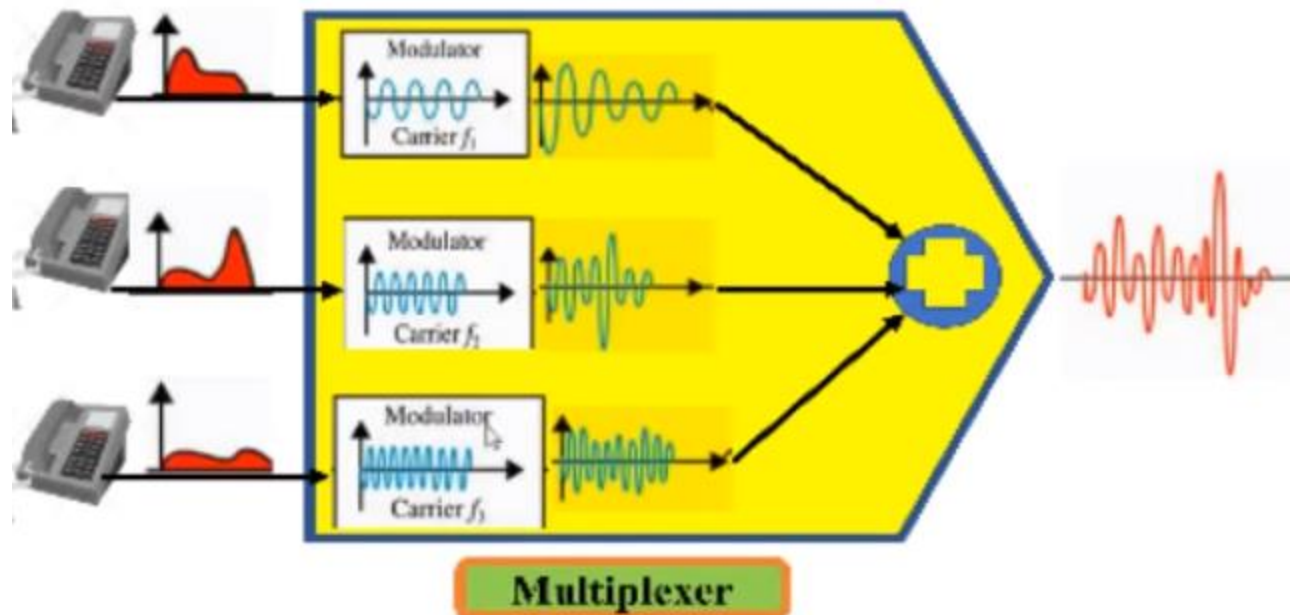


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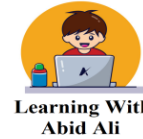
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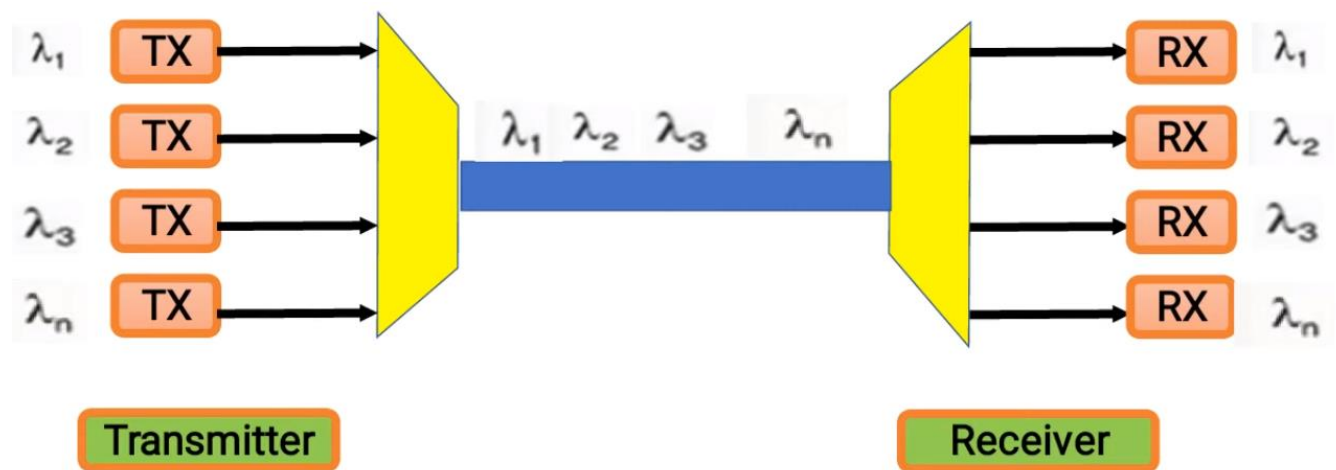
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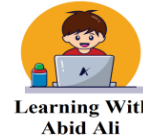
### Wavelength-Division Multiplexing

Wavelength division multiplexing (WDM) is a technique of multiplexing multiple optical carrier signals through a single optical fiber channel by varying the wavelengths of laser lights. WDM allows communication in both the directions in the fiber cable.



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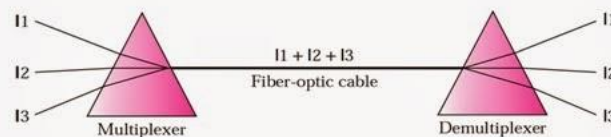
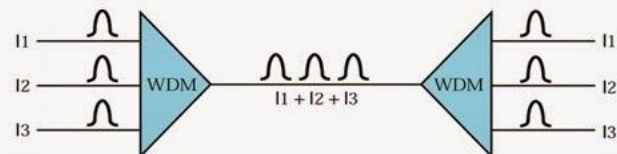
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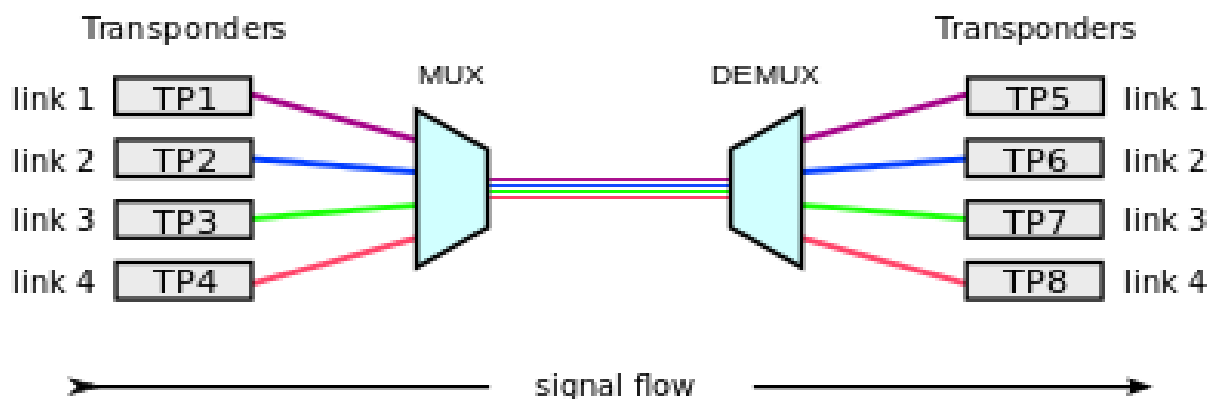
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### WAVE DIVISION MULTIPLEXING (WDM)

- An analog multiplexing technique to combine optical signals
- Multiple beams of light at different frequency
- Carried by optical fiber
- A form of FDM
- Each color of light (wavelength) carries separate data channel
- Commercial systems of 160 channels of 10 Gbps now available



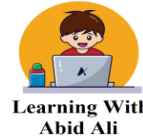
### wavelength-division multiplexing (WDM)





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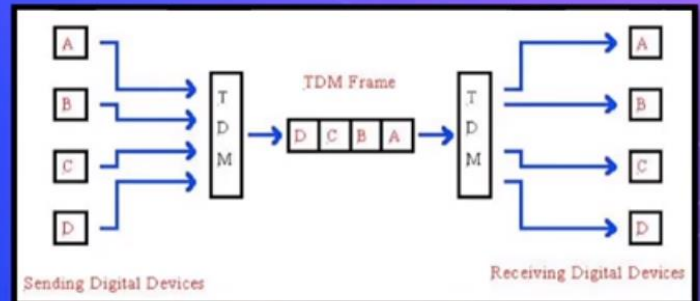
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## Time-Division Multiplexing

Digital process that allows several connections to share the high bandwidth of a link

### TDM

- Baseband Transmission
  - One at a time
  - Virtual Signals Only
- Time Slots
- Sends signals equally
  - At the same time
- Not the most efficient



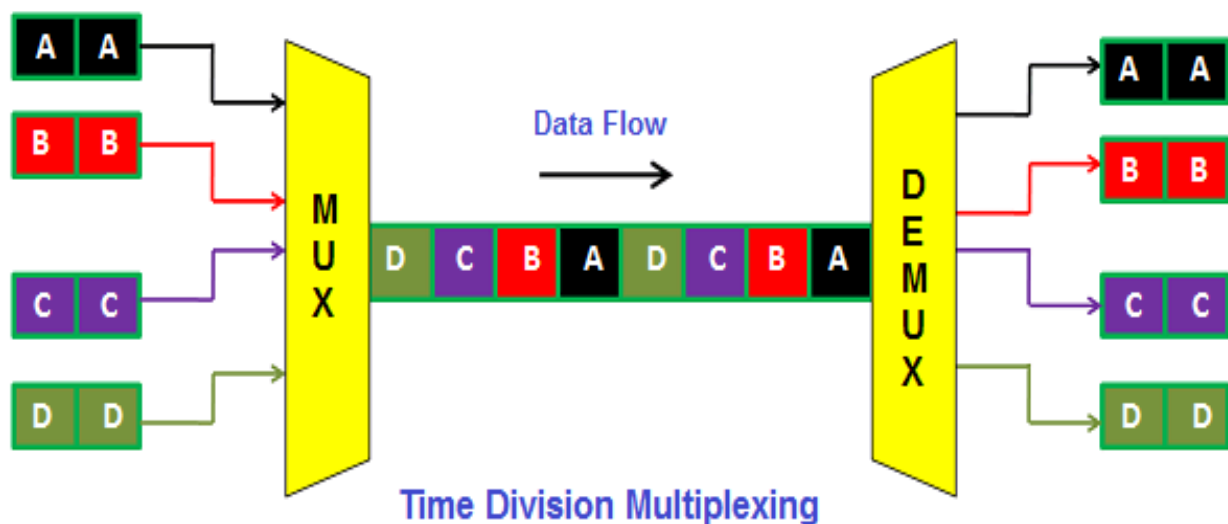
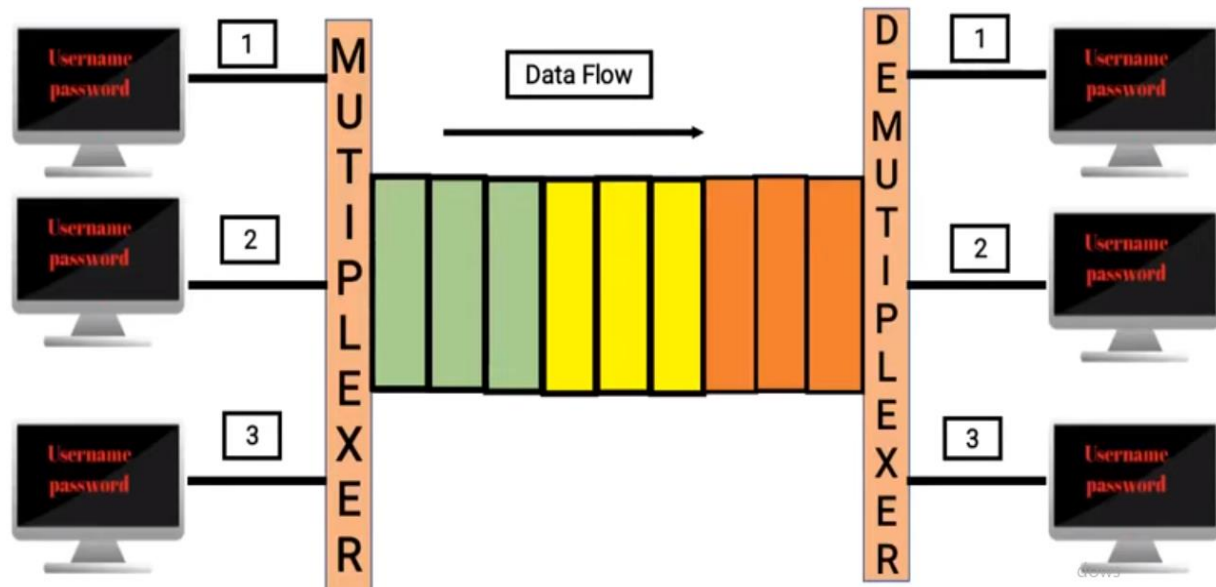


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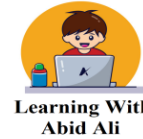


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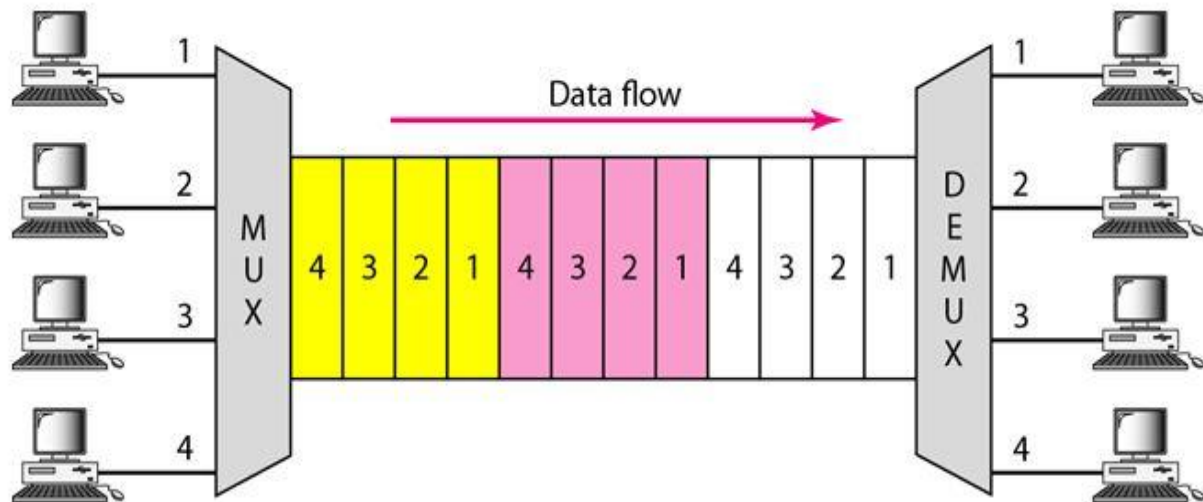


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### Time-Division Multiplexing

- Synchronous TDM
- Statistical TDM

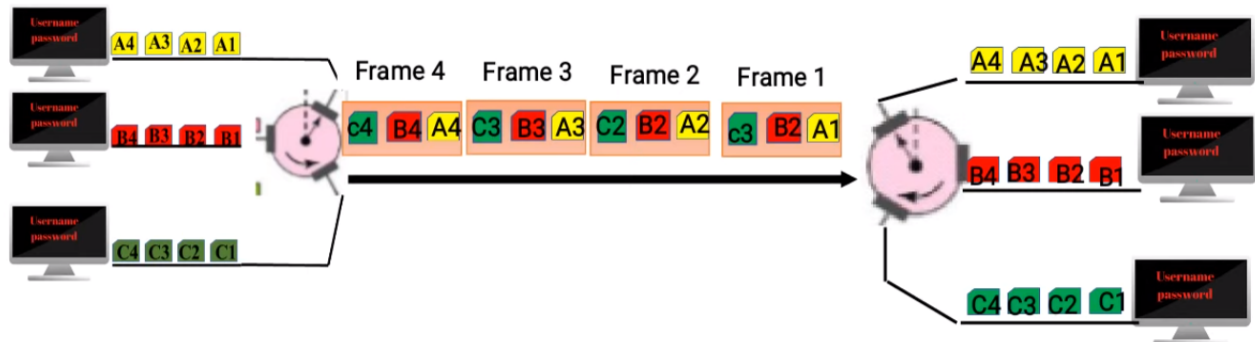
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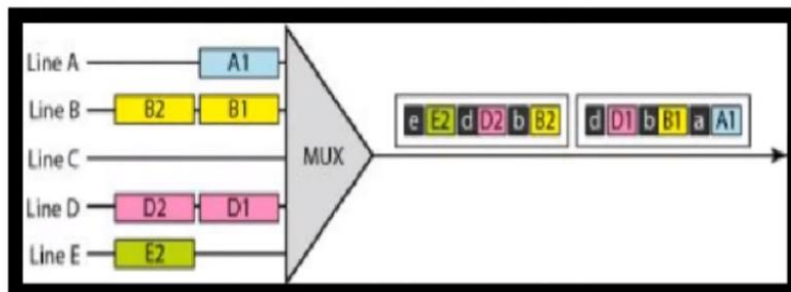
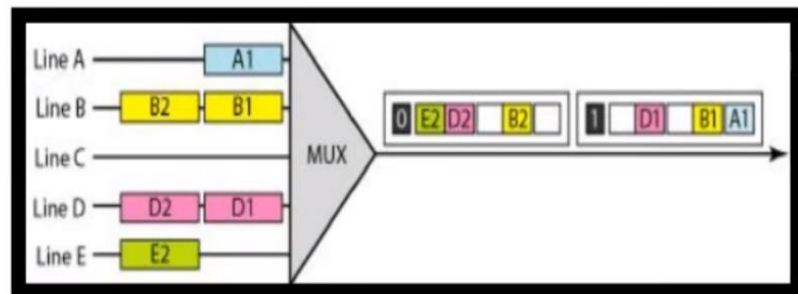
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### Synchronous TDM



Empty slots If a source does not have data to send, the corresponding slot in the output frame is empty.

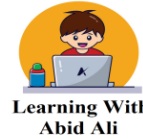
### Synchronous TDM



### Statistical TDM

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## SPREAD SPECTRUM

Spread spectrum is **a form of wireless communications in which the frequency of the transmitted signal is deliberately varied**. This results in a much greater bandwidth than the signal would have if its frequency were not varied.

### Spread Spectrum :-

1. It is a SIGNAL STRUCTURING TECHNIQUE.
2. Spread-spectrum techniques are methods by which any signal generated with a particular bandwidth is deliberately spread in the frequency domain, results in signal with wide bandwidth.

Here, signals maybe electrical, electromagnetic or any other signal.

## What is spread spectrum & Types?

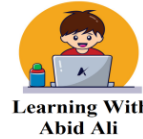
Spread Spectrum can be categorized into two types:

Frequency Hopping Spread Spectrum (FHSS)

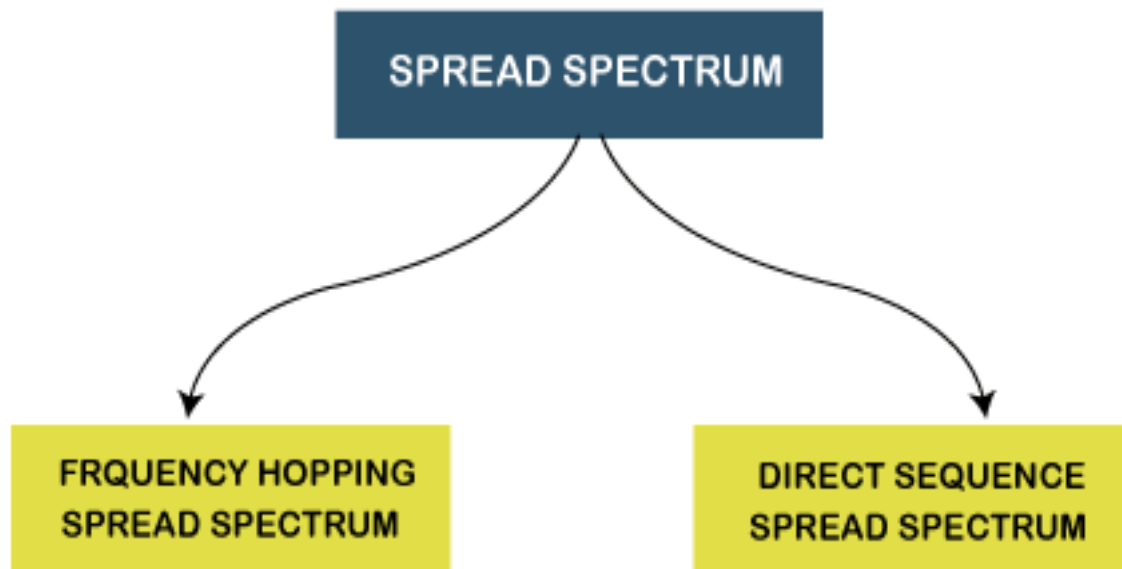
Direct Sequence Spread Spectrum (DSSS)

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### DSSS

- DSSS also expands the bandwidth of the original signal, but the process is different
- We replace each data bit with 'n' bits using a spreading code
- Each bit is assigned a code of 'n' bits, called chips, where the chip rate is 'n' times that of the data bit

### What is the difference between FHSS and DSSS?

FHSS and DSSS, which stand for Frequency Hopping Spread Spectrum and Direct Sequence Spread Spectrum, are two spread spectrum techniques. The main difference is in how they spread the data into the wider bandwidth. **FHSS utilizes frequency hopping while DSSS utilizes pseudo noise to modify the phase of the signal.**

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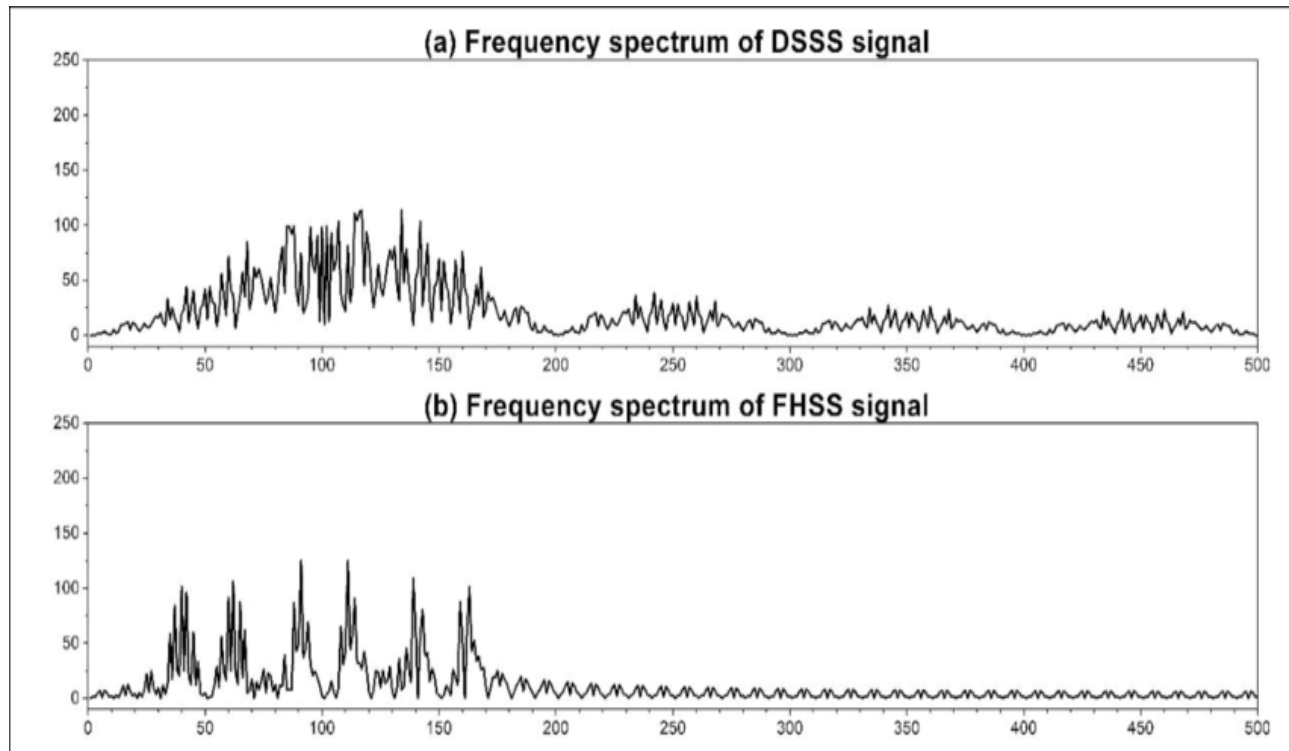


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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

## Data Communication CS601 Lecture No 20

Lecture Prepared By  
Muhammad Abid Ali

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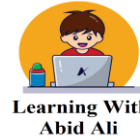
CS601 Lecture No 20 Topic 100 to 113

What do you mean by transmission medium?

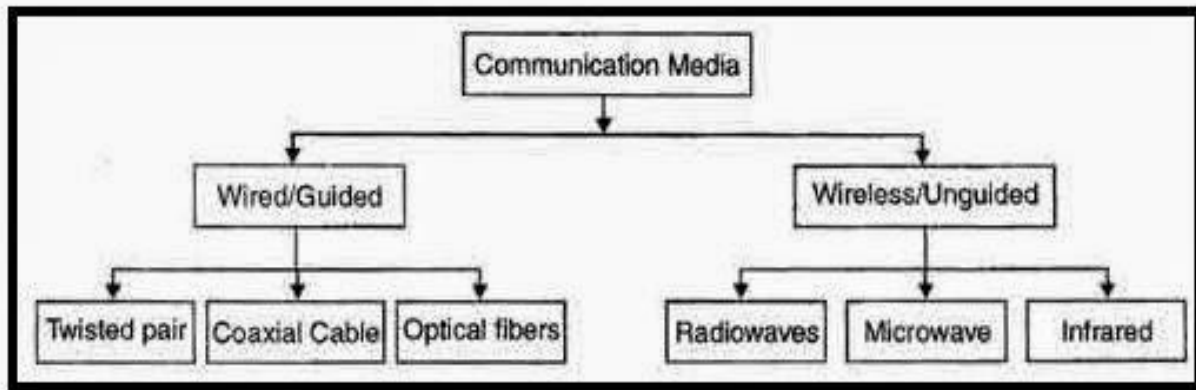
The transmission medium can be defined as a **pathway that can transmit information from a sender to a receiver**. Transmission media are located below the physical layer and are controlled by the physical layer. Transmission media are also called communication channels.

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In transmission media is the way the systems are connected to route data signals in a network. Unguided media (wireless).

...

Differences.

### Guided media

It is called wired communication or bounded transmission media.

### Unguided media

It is called wireless communication or unbounded transmission media.

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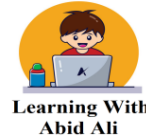
| BASIS FOR COMPARISON | GUIDED MEDIA                                                    | UNGUIDED MEDIA                                                       |
|----------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
| Basic                | The signal requires a physical path for transmission.           | The signal is broadcasted through air or sometimes water.            |
| Alternative name     | It is called wired communication or bounded transmission media. | It is called wireless communication or unbounded transmission media. |
| Direction            | It provides direction to signal for travelling.                 | It does not provide any direction.                                   |
| Types                | Twisted pair cable, coaxial cable and fibre optic cable.        | Radio wave, microwave and infrared.                                  |

### Definition of Guided Media

**Guided transmission media** are more commonly known as the **wired communication** or **bounded transmission media**. The electromagnetic signals travel between the communicating devices through a physical medium/conductor. As the medium

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for transmission is a physical conductor, it also provides **direction** to the signal. But there are physical limitations of the conductor in the guided media. Like the length of the conductor, its installation cost, its maintenance, etc.

The guided media is categorized further into three categories that are **twisted-pair cable**, **coaxial cable** and **fiber-optic cable**. The twisted pair cable has two conductors' wires wound around each other and each surrounded by an insulating material. The twisted pair cable is flexible and easy to install. But it has *low bandwidth* and provide *less protection from interference*. Twisted pair cable is also of two types **shielded and unshielded twisted pair cable**.

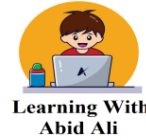
The **coaxial cable** has a central core conductor (usually copper) enclosed in an insulating sheath, which is further encased in an outer metallic braid, it serves as both protection against noise and as a second conductor which completes the circuit. Now, the outer metallic covering is also covered by an insulating sheath. The coaxial cable carries signals of *higher frequency* than the twisted pair cable.

The third category is the **optical fiber** which is made of glass or plastic, and it transmits signals in the form of light. The optical fiber is *noise resistance*, *has less signal attenuation and has a higher bandwidth* in comparison to twisted pair cable and coaxial cable. But it also has some drawbacks like; it is *very expensive*; it requires a lot of *installation and maintenance charge* as any defect in the cable can diffuse light and alter the signals. As the optical fiber is made of glass, it is very *fragile*.

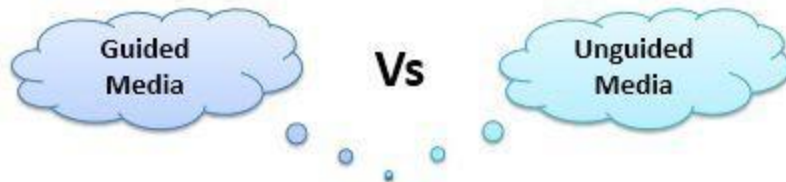
## Difference Between Guided and Unguided Media

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The computer and other communicating devices represent data in the form of signals. The signals transmit between communicating devices in the form of electromagnetic energy, and hence the signals are called **electromagnetic signals**. Electromagnetic signals are the combination of electric and magnetic fields that vibrates in relation to each other. The electromagnetic signals can travel through various transmission media. The transmission media is broadly classified into two categories that are **guided** and **unguided media**.

The basic difference between guided and unguided media is that in the **guided media**, the signal travels through a physical medium whereas, in **unguided media**, the signal travel through the air. There are some more differences between guided and unguided media which I have discussed with the help of comparison chart shown below.

### Content: Guided Vs Unguided Media

1. [Comparison Chart](#)
2. [Definition](#)
3. [Key Differences](#)
4. [Conclusion](#)

Comparison Chart

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| BASIS<br>COMPARISON | FOR                                                             |                                                                      |
|---------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
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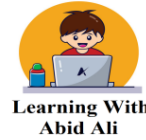
### Definition of Guided Media

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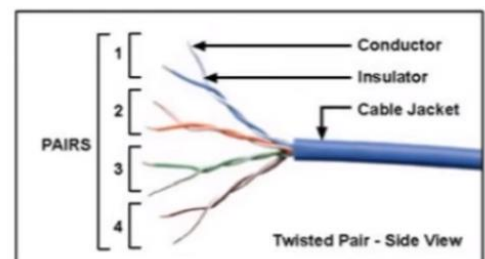
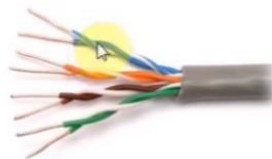
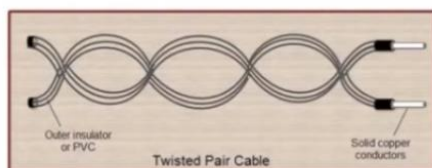
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## Twisted Pair Cable

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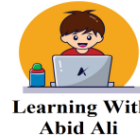
## Twisted Pair Cable



- A twisted pair cable is a pair of **copper wires**.
- Copper wires are the most common wires used for transmitting signals because of good performance at low costs.
- A twisted pair consists of **two conductors** (normally copper), each with its own **plastic insulation**, twisted together to form a single media.

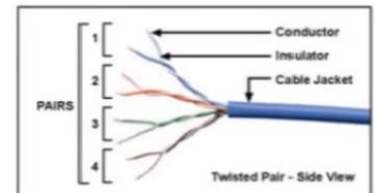
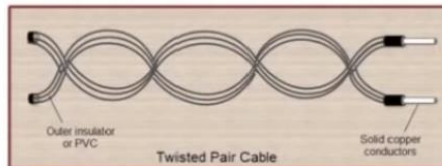
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## Twisted Pair Cable



- Out of these two wires, only one carries **actual signal** and another is used for ground reference.
- To identify every cable, these cables are **colour coated**.
- The twists between wires are helpful in **reducing noise** (electro-magnetic interference) and **crosstalk**.
- This type of cable is used in **telephone lines** to provide voice and data channels.

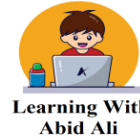
## There Are Two Main Types of Twisted Pair Cables

**Unshielded Twisted Pair (UTP)**

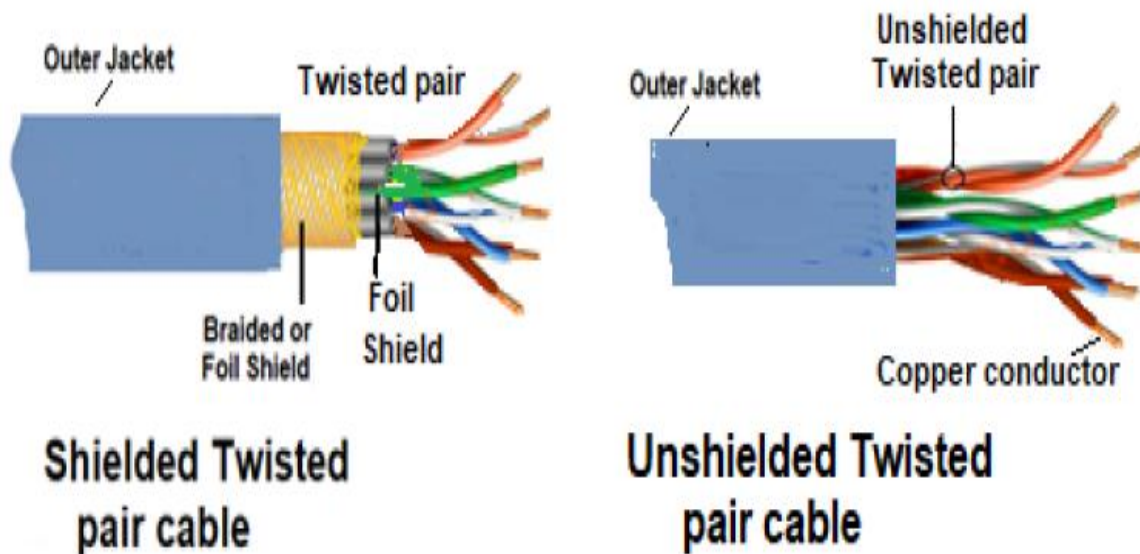
**Shielded Twisted Pair (STP)**

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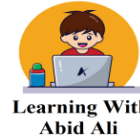


**What is the difference between UTP and STP When would you use STP?**

Both UTP and STP are widely used in transmitting information across distance. UTP is a cable with wires that are twisted together to reduce noise generated by an external source. On the other hand, STP is a twisted pair cable with an additional mesh shield that guards the cable against electromagnetic interference.

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### **Distinguish between STP and UTP:**

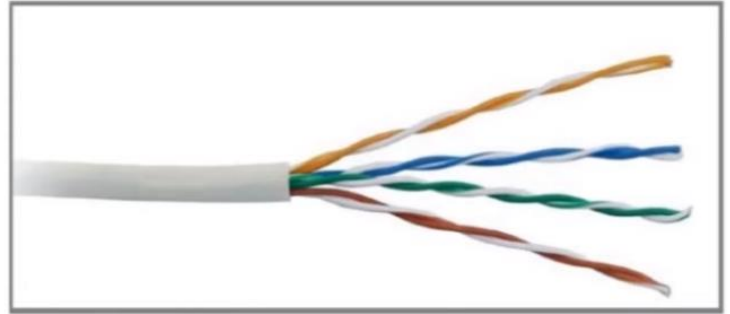
| Shielded Twisted Pair                                                              | Un-shielded Twisted Pair                                                                      |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1. STP has a metal foil covering.                                                  | 1. UTP does not have a metal foil Covering.                                                   |
| 2. STP gives better resistance to electromagnetic interference as compared to UTP. | 2. UTP does not provide better resistance to electromagnetic interference as compared to STP. |
| 3. STP is little expensive then UTP.                                               | 3. UTP is less expensive than STP.                                                            |
| 4. Grounding is possible.                                                          | 4. Grounding is not possible.                                                                 |
| 5. Possibility of crosstalk is less as it resist to EMI.                           | 5. Possibility of crosstalk is more as it provide less resistance to EMI.                     |
| 6. Distance travelled is large and can be used in MAN.                             | 6. Distance travelled is less and used in LAN.                                                |

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- **Unshielded twisted pair (UTP):-**

- UTP is more common.
- UTP cost less than STP and easily available due to its many use.
- Due to its low cost, UTP cabling is used extensively for local-area networks (LANs) and telephone connections.
- UTP cables consist of 2 or 4 pairs of twisted cable.
- Cable with 2 pair use **RJ-11** connector and 4 pair cable use **RJ-45** connector.

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- RJ-45 connectors is used with Ethernet cables in computer networking.
- RJ-11 connectors is used in connecting telephone units.



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- **Shielded twisted pair (STP):-**

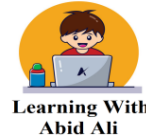
- This type of cable has a **metal foil covering** which encases each pair of insulator conductors.
- Electromagnetic noise penetration is prevented by metal casing. Shielding also eliminates crosstalk.
- It is similar to UTP but has a mesh shielding that's protects it from EMI which allows for higher transmission rate.
- It is more expensive than coaxial and unshielded twisted pair.

## Coaxial Cable

The **coaxial cable** has a central core conductor (usually copper) enclosed in an insulating sheath, which is further encased in an outer metallic braid, it serves as both protection against noise and as a second conductor which completes the circuit. Now, the outer metallic covering is also covered by an insulating sheath. The coaxial cable carries signals of *higher frequency* than the twisted pair cable.

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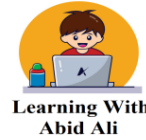
## Coaxial Cable



- **Coaxial cables** are copper cables with better shielding than twisted pair cables, so that transmitted signals may travel longer distances at higher speeds.
- The shield minimizes electrical and radio frequency interference.
- Coaxial cabling is the primary type of cabling used by the **cable television industry** and is also widely used for **computer networks** such as Ethernet.

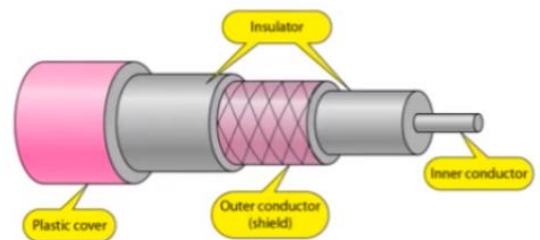
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## Coaxial Cable



- Coaxial cable has two wires of copper.
- The core/inner copper wire in centre and is made of solid conductor. It is enclosed in an insulating sheath.
- The second/outer copper wire is wrapped around, and is used to protect from external electromagnetic interference(Noise).
- This all is covered by plastic cover used to protect the inner layers from physical damage such as fire or water.

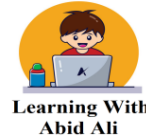
## Coaxial Cable Standards

- Coaxial cables are categorized by their Radio Government (RG) ratings. Each RG number denotes a unique set of physical specifications

- |          |               |                              |
|----------|---------------|------------------------------|
| • 50-Ohm | RG-7 or RG-11 | : used with thick Ethernet.  |
| • 50-Ohm | RG-58         | : used with thin Ethernet    |
| • 75-Ohm | RG-59         | : used with cable television |

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50 Ohm



75 Ohm



### RG CABLE



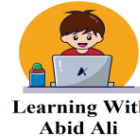
## Optical Fiber

The third category is the **optical fiber** which is made of glass or plastic, and it transmits signals in the form of light. The optical fiber is *noise resistance*, *has less signal attenuation* and *has a higher bandwidth* in comparison to twisted pair cable and coaxial cable. But it also has some drawbacks like; it is very expensive; it requires a lot of installation and maintenance charge as any defect in the cable can diffuse light and alter the signals. As the optical fiber is made of glass, it is very fragile.

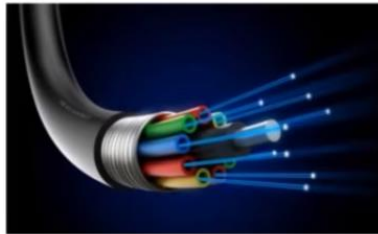


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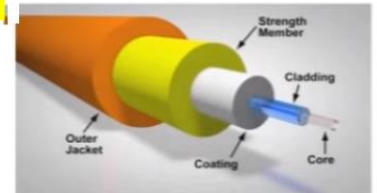
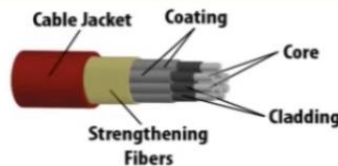
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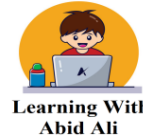
## Optical Fiber



- A fibre optic cable is made of high quality of **thin glass or plastic** and is used to transfer digital data signals in the form of **light** up to distance of thousand of miles.
  - Fibre optic cables are **not affected** by electromagnetic interference, so noise and distortion is very less.
  - Fibre optic cables carry communication signals using pulses of light generated by **small lasers or light-emitting diodes (LEDs)**.
  - The cable consists of one or more strands of glass, each only slightly thicker than a human hair. The centre of each strand is called the **core**, which provides the **pathway for light to travel**. The core is surrounded by a layer of glass called **cladding** that reflects light inward to avoid loss of signal and allow the light to pass through bends in the cable. No light escapes the glass core because of this reflective **cladding**.
- 
- **Advantages of Optical Fibre:-**
    - Fibre optic cables have a much High bandwidth than metal cables. This means that they can carry more data.
    - Smaller Size and Lighter weight.
    - low attenuation
    - Not affected electromagnetic interference (No EMI interference)
    - Signals carrying data can travel long distances without weakening
    - Suitable for industrial and noisy areas
  - **Disadvantages of Optical Fibre:-**
    - Optical fibre cables are expensive
    - Difficult to install
    - Maintenance is expensive and difficult

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### Definition of Unguided Media

The **unguided media** is also called **wireless communication**. It does not require any physical medium to transmit electromagnetic signals. In unguided media, the electromagnetic signals are broadcasted through air to everyone. These signals are available to one who has the device capable of receiving those signals.

The unguided media is also called unbounded media as it does not have any border limitation. The unguided media allows the user to connect all the time, as the communication is wireless the user can connect himself from anywhere to the network.

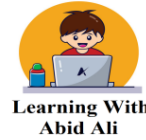
The unguided media is categorized into **radio waves, microwaves and infrared waves**.

## wireless communication Unguided Media

- A **wave** can be described as a disturbance that travels through a medium from one location to another location.
- A **wave** is a **transfer of energy**, usually through a form of matter called a **medium**.
- There are a special type of wave that can travel without a medium, called **electromagnetic waves** (also called **EM waves**), which are waves like radio waves and microwaves.
- Unlike sound waves and water waves, electromagnetic waves don't need a fluid, or a solid, or even air to help them travel from one place to another. EM waves can travel across the great vacuum of space, which is why we see light from distant stars and planets.



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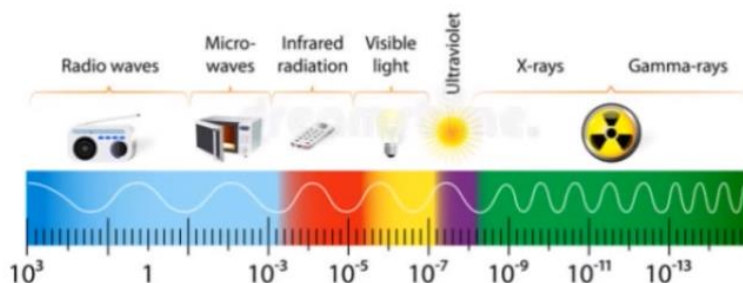


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## wireless communication **Unguided Media**

- **Electromagnetic waves** are formed when an **electric field** comes in contact with a **magnetic field**. They are hence known as '**electromagnetic**' waves.
- Electromagnetic (EM) radiation is a form of energy that is all around us and takes many forms, such as radio waves, microwaves, X-rays and gamma rays.
- **Sunlight** is also a form of EM energy. Electromagnetic energy from the sun comes to Earth in the form of radiation.
- The **Electromagnetic Spectrum** describes a wide range of different **electromagnetic waves**.
- A little part of **electromagnetic spectrum** can be used for wireless transmission.

### THE ELECTROMAGNETIC SPECTRUM

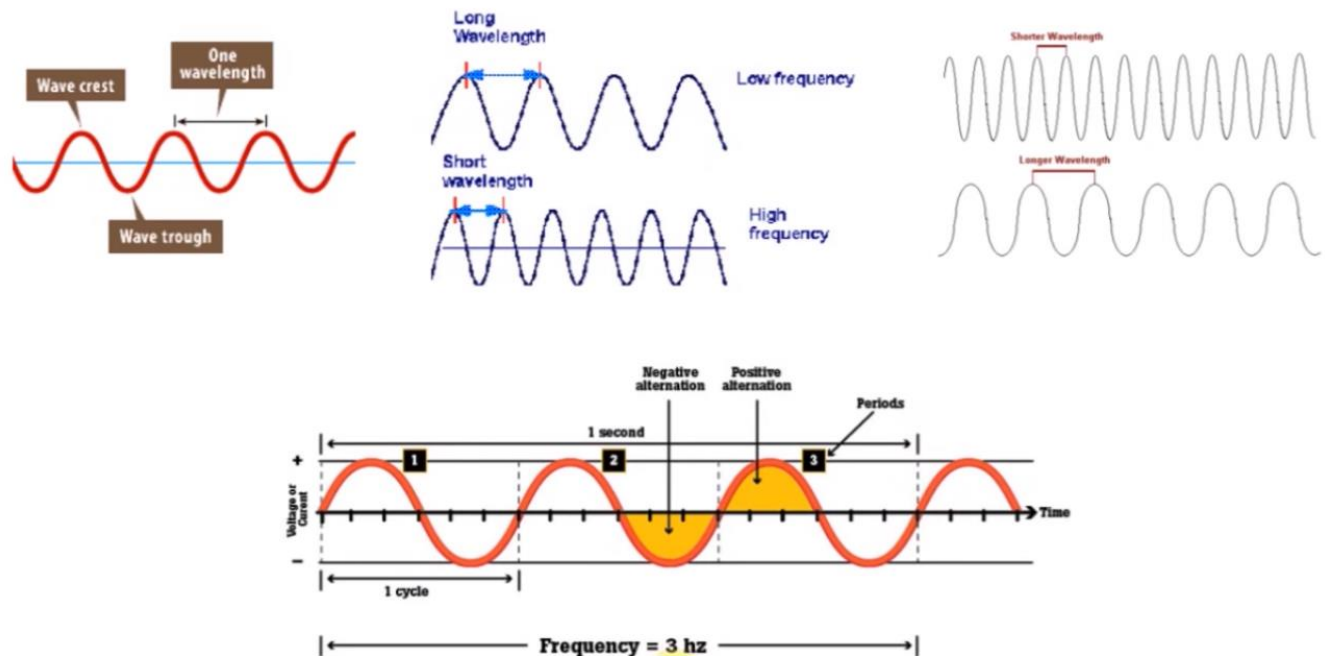


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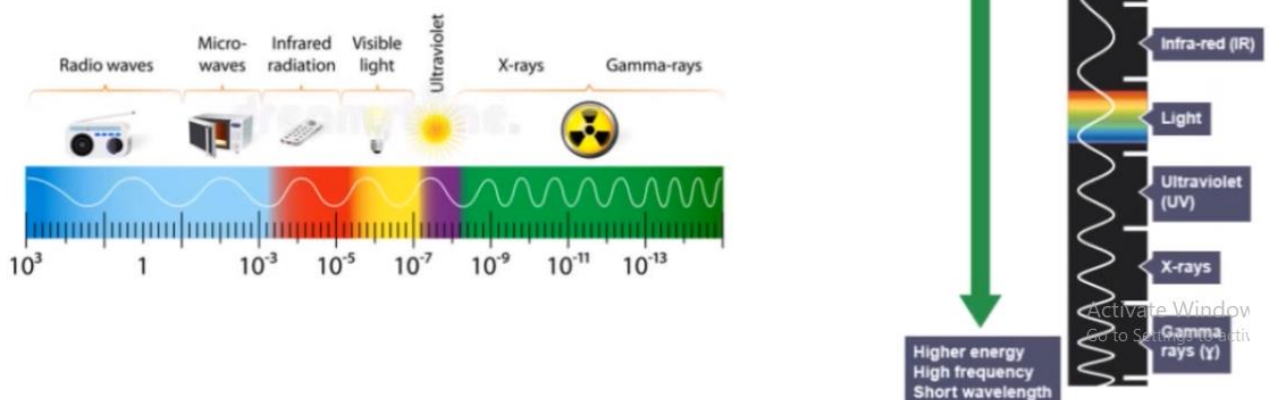


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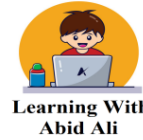
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### THE ELECTROMAGNETIC SPECTRUM



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## Radio Waves

The **radio waves** are generated easily; they are *low-frequency signals* and can travel a *long distance*. The radio waves can penetrate through the buildings.

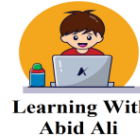
# Radio Waves



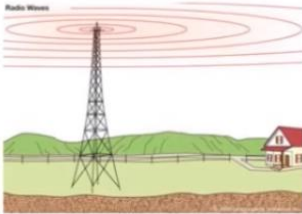
- Radio waves are **EM (Electromagnetic) waves** that have wavelengths between 1 millimetre and 100 kilometres (or 300 GHz and 3 kHz in frequency).
- Radio frequency is easy to generate because it has large wavelength and can travel long distance.
- Radio waves are generated by **radio transmitters** and received by **radio receivers**.
- Radio stations transmit radio waves using **transmitters**, which are received by the **receiver** installed in our devices. Both transmitters and receivers use antennas to radiate or capture radio signals.
- It can penetrate walls easily, so these waves are widely used for communication both indoors and outdoors.

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# Radio Waves



- Radio waves are **omnidirectional** means they travel in all the directions from the source.
- When an antenna transmits radio waves, they are propagated in all directions.
- A sending antenna send waves that can be received by **any receiving antenna**. The omnidirectional property has disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signal using the same frequency or band.
- It is Used Mobile, AM/FM radio, television



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# Radio Waves

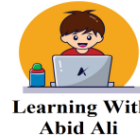


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## Radio Spectrum

| Radio Band               | Frequency         | Some Applications                                                                                       |
|--------------------------|-------------------|---------------------------------------------------------------------------------------------------------|
| Very Low Frequency VLF   | 3 KHz to 30 KHz   | Radio Navigation                                                                                        |
| Low Frequency LF         | 30 KHz to 300 KHz | Long Wave Radio                                                                                         |
| Medium Frequency MF      | 30 KHz to 3 MHz   | AM Radio                                                                                                |
| High Frequency HF        | 3 MHz to 30 MHz   | CB Radio (HAM)<br>Point to Point Radio<br>Search and Rescue Services                                    |
| Very High Frequency VHF  | 30 MHz to 300 MHz | FM radio 88-108 MHz<br>VHF Broadcast TV                                                                 |
| Ultra High Frequency UHF | 300 MHz to 3 GHz  | UHF Broadcast TV<br>Cellular Phones<br>Microwave Links<br>Wi-Fi in 2.4 Band<br>Satellite Communications |
| Super High Frequency SHF | 3 GHz to 30 GHz   | Microwave Links<br>Wi-Fi in 5 GHz Band<br>Satellite Communications                                      |
| Extra High Frequency EHF | 30 GHz to 300 GHz | Microwave Links                                                                                         |

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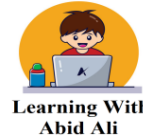
## Microwaves

The **microwaves** are transmitted in a straight line and hence require the **line-of-sight transmission**. The distance covered by the microwave signal depend on the height of the two antennas. More the taller are antennas longer is the distance covered by the signal. The microwave has a *frequency higher* than the radio waves. Microwave are used for telephone communication mobile phones, television distribution, etc.



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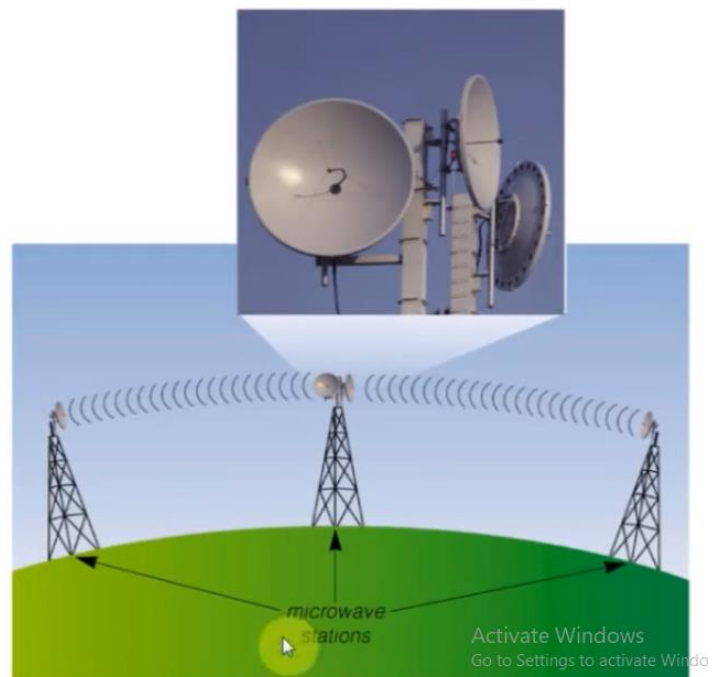
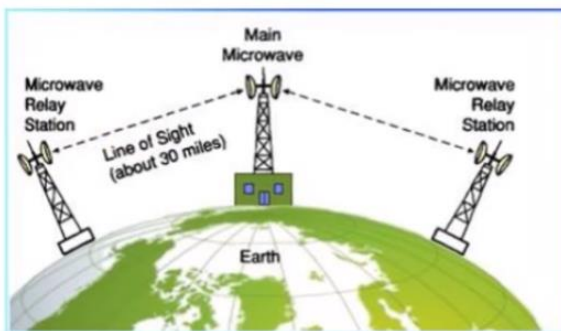
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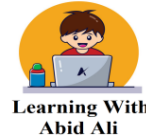
# Microwaves

- Microwaves are a type of radio waves with high frequencies. It can be classified as a subclass of radio waves. The frequency of microwaves lies in the 300 MHz to 300 GHz.
- Unlike radio waves, microwaves are unidirectional, in which the sending and receiving antennas need to be aligned.
- Microwaves are widely used for point-to-point communications because their small wavelength, which means that the signal is focused into a narrow beam. Additionally, each antenna must be within line of sight of the next antenna.
- Electromagnetic waves above 100 MHz tend to travel in a straight line and signals over them can be sent by beaming those waves towards one particular station. Because Microwaves travels in straight lines, both sender and receiver must be aligned to be strictly in line-of-sight.



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**Infrared waves** are used for short range communication. Like, the remote control for televisions, VCRs, etc. uses infrared waves. It can not penetrate through obstacles. The government license is not required, to operate an infrared system as it is more secure against eavesdropping.

# Infrared waves



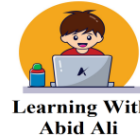
- Infrared signals have frequencies between 300 GHz to 400 THz. They are used for short-range communication.
- Infrared waves are used for very short distance communication like TV remote, wireless speakers, automatic doors, hand held devices etc.
- Infrared waves having high frequencies prevents interference b/w one system to another.
- Infrared signals have high frequencies and cannot penetrate walls. Due to its short-range communication system, the use of an infrared communication system in one room will not be affected by the use of another system in the next room. This is why using an infrared TV remote control in our home will not interfere with the use of our neighbour's infrared TV remote control.
- **THE DISADVANTAGES OF USING INFRARED**
  - Infrared signals cannot be used for long distance communication. In addition, we cannot use infrared waves outside a building because sun's rays contain infrared waves that can interfere with communication.

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### Key Differences Between Guided and Unguided Media

1. The key difference between guided and unguided media is that guided media uses a **physical path or conductor** to transmit the signals whereas, the unguided media **broadcast the signal** through the air.
2. The guided media is also called **wired communication** or **bounded transmission media**. However, the unguided media is also called **wireless communication** or **unbounded transmission media**.
3. The guided media provide **direction** to the signal whereas, the unguided media **does not direct** the signal.
4. Categories of guided media are **twisted pair cable, coaxial cable** and **optical fibre**. On the other hands, the categories of unguided media are **radio wave, microwave, and infrared signal**.